



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Università degli Studi di Padova
Master Degree in Astrophysics and Cosmology



Binary Stars Beyond the Main Sequence and the Enigma of UV-Dim Stars

Supervisor: Prof. Antonino Milone

Candidate: Giada Aggio

First topic

Binaries beyond the MS

How does the fraction of binaries vary outside the
Main Sequence?

Development of a framework to infer binary fractions in the
Horizontal Branch and in the Red Giant Branch.
Application to [BS 90](#), [47 Tucanae](#) and [NGC 1783](#)

UV-dim stars are peculiar objects discovered recently, 2022. They are
common in young clusters (~ 200 Myr).

Are they also present in very young, star-forming clusters?

Test case: NGC 346 (~ 3 Myr)

Second topic

UV-dim stars

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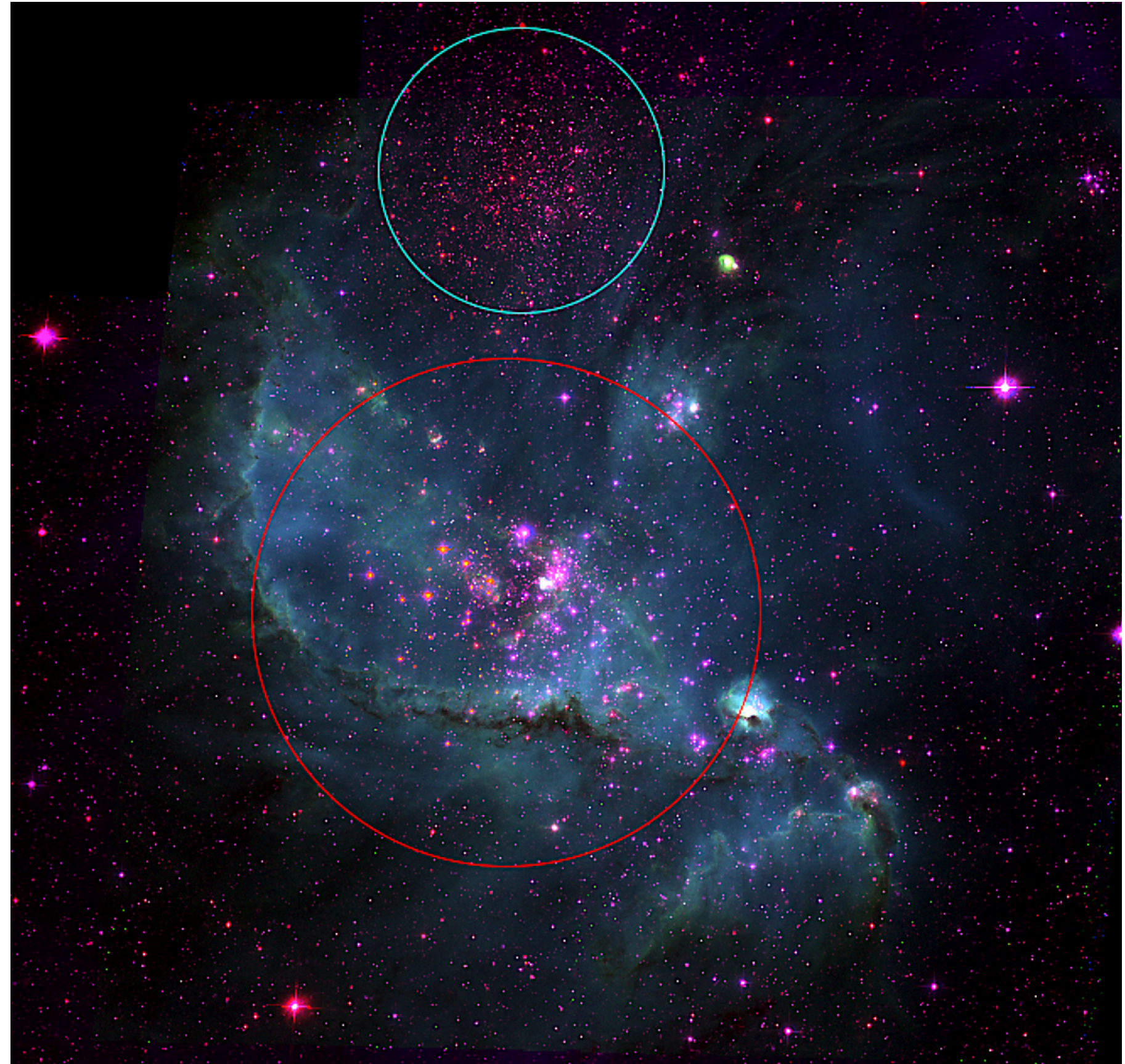
UV-dim stars

Data reduction

BS 90 and NGC 346

I derived high-precision multi-band photometry for the young, star-forming cluster NGC 346 and the intermediate-age cluster BS 90.

Stellar fluxes and positions were obtained through several FORTRAN codes (Anderson et al. 2008).

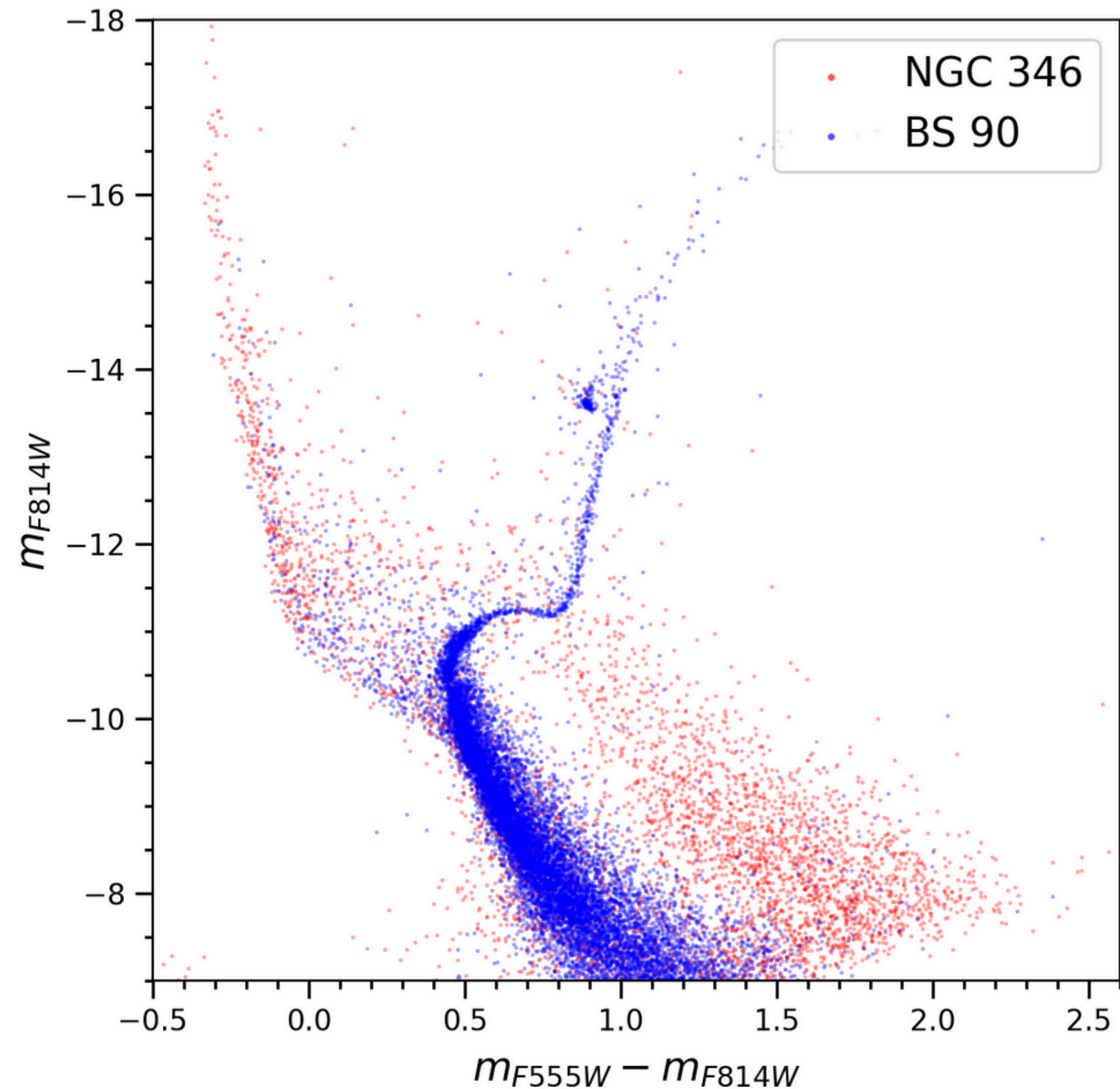


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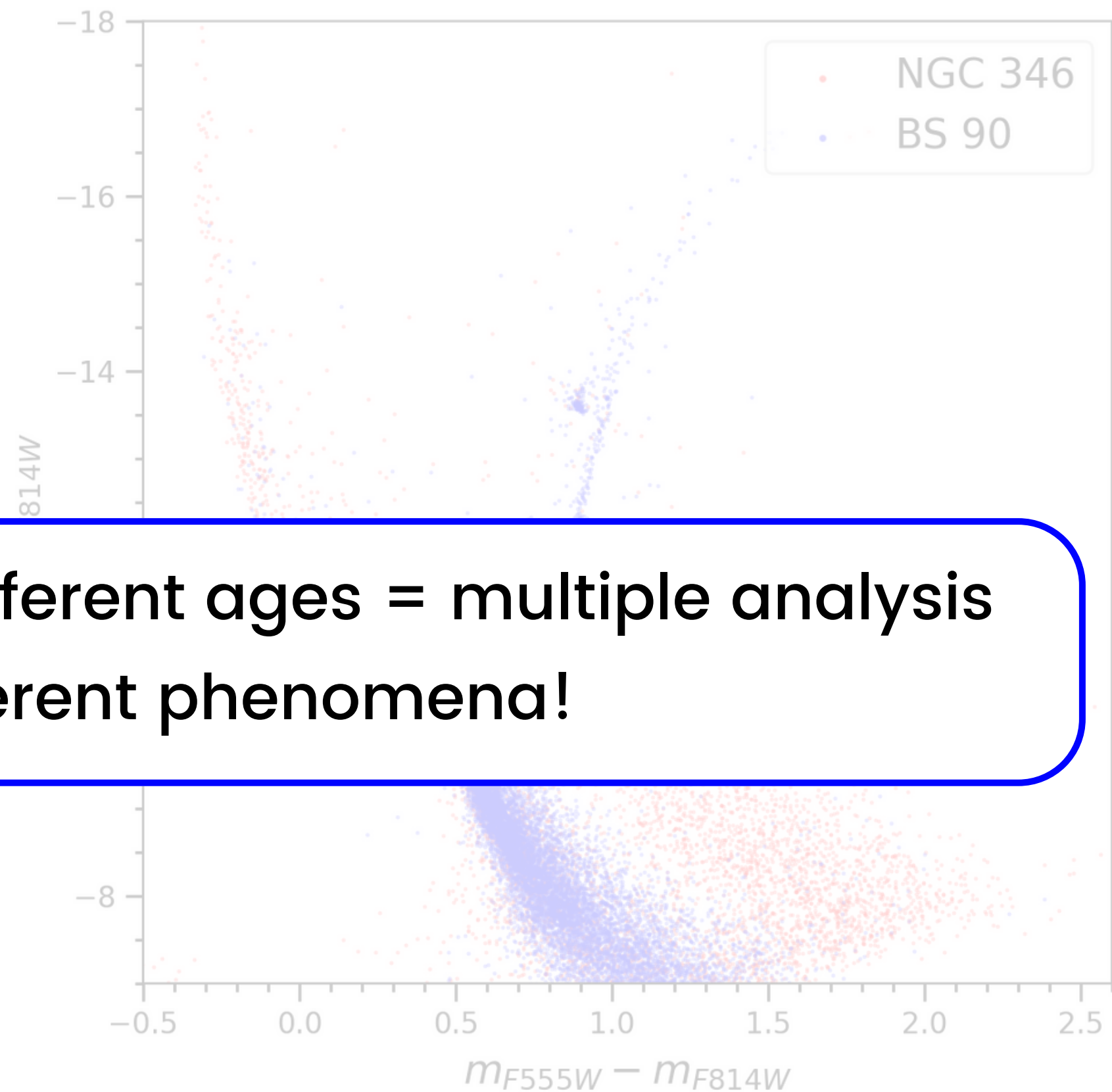
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Two clusters with different ages = multiple analysis on different phenomena!



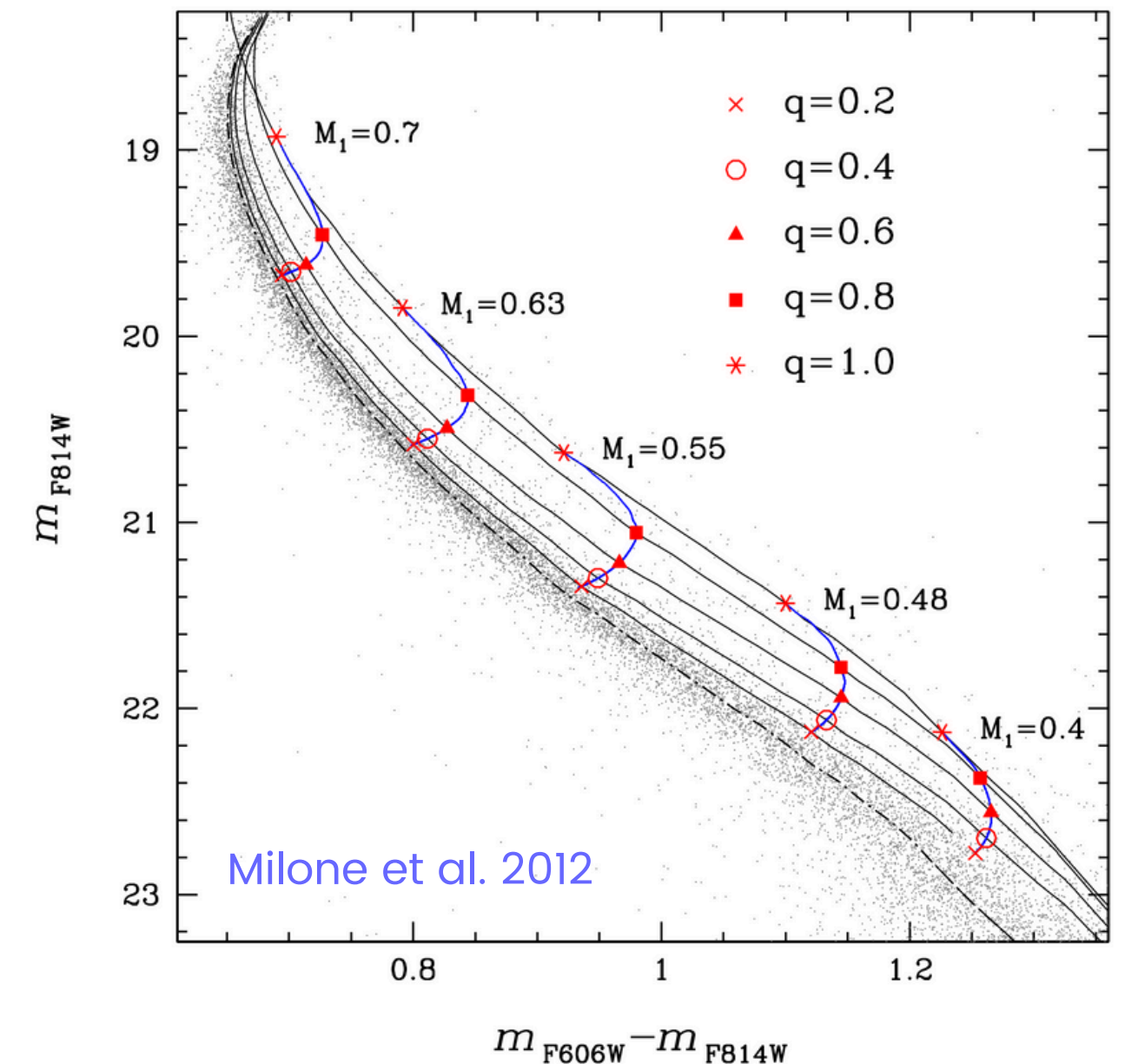
Binaries in the MS

Why are binaries important?

- Important role in cluster dynamical evolution
- Understanding of exotic objects like Blue Stragglers etc.

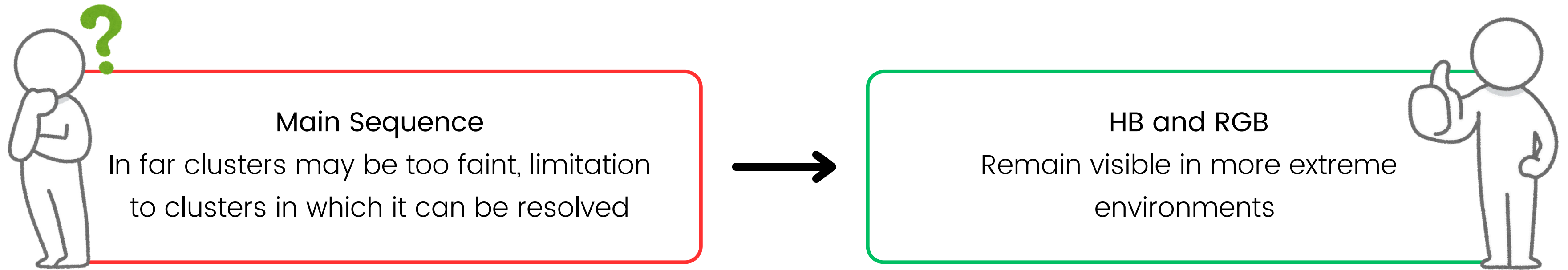
Lots of studies on binaries in the MS, e.g. [Milone et al. \(2012\)](#), [Cordoni et al. \(2023\)](#), [Mohandasan et al. \(2024\)](#) etc..

Beautiful studies but with some limitations!



Detecting Binaries

Why going outside the MS?



Analysis of the binary fraction in the Horizontal Branch and in the Red Giant Branch of:

47 Tucanae (~ 12 Gyr)

BS 90 (~ 4 Gyr)

NGC 1783 (~ 2 Gyr)

Binaries beyond the MS

Synthetic Binaries

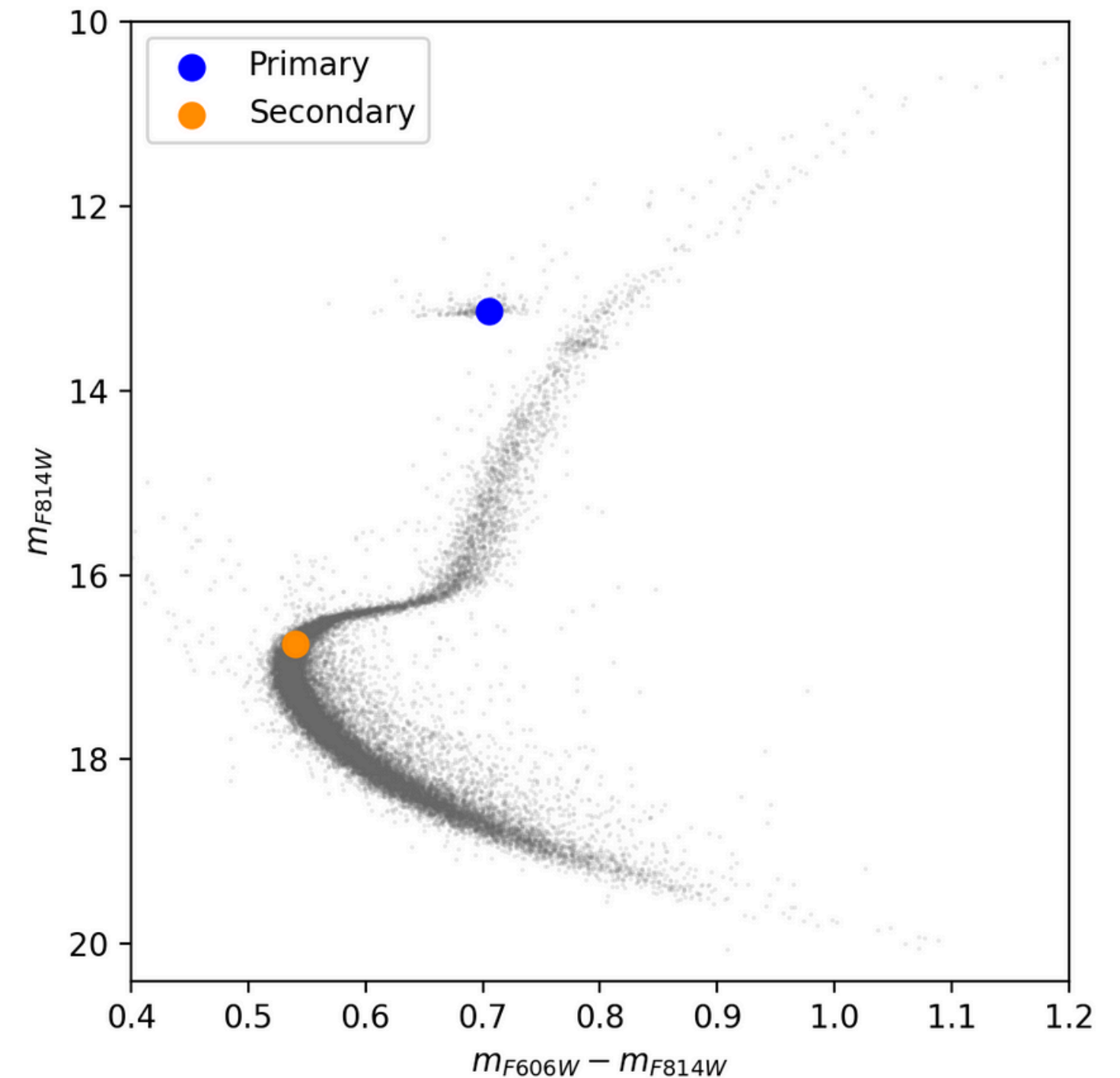
Magnitude of a binary system:

$$m_{\text{bin}} = -2.5 \log(F_{\text{bin}})$$

but the flux is given by the sum of the fluxes of the members of the binary:

$$m_{\text{bin}} = -2.5 \log(10^{-\frac{m_1}{2.5}} + 10^{-\frac{m_2}{2.5}})$$

47 Tucanae - HB binaries



Binaries beyond the MS

Synthetic Binaries

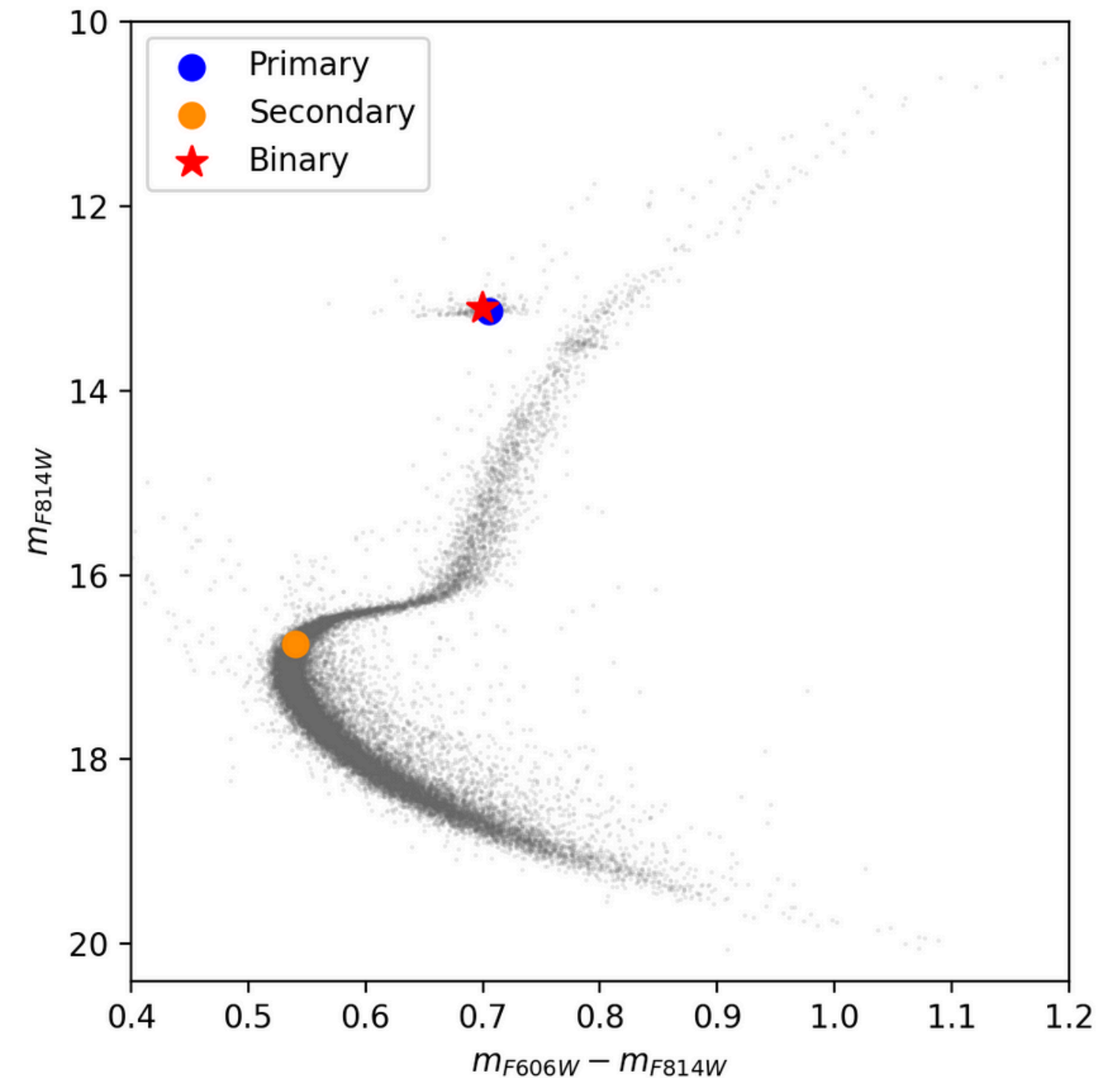
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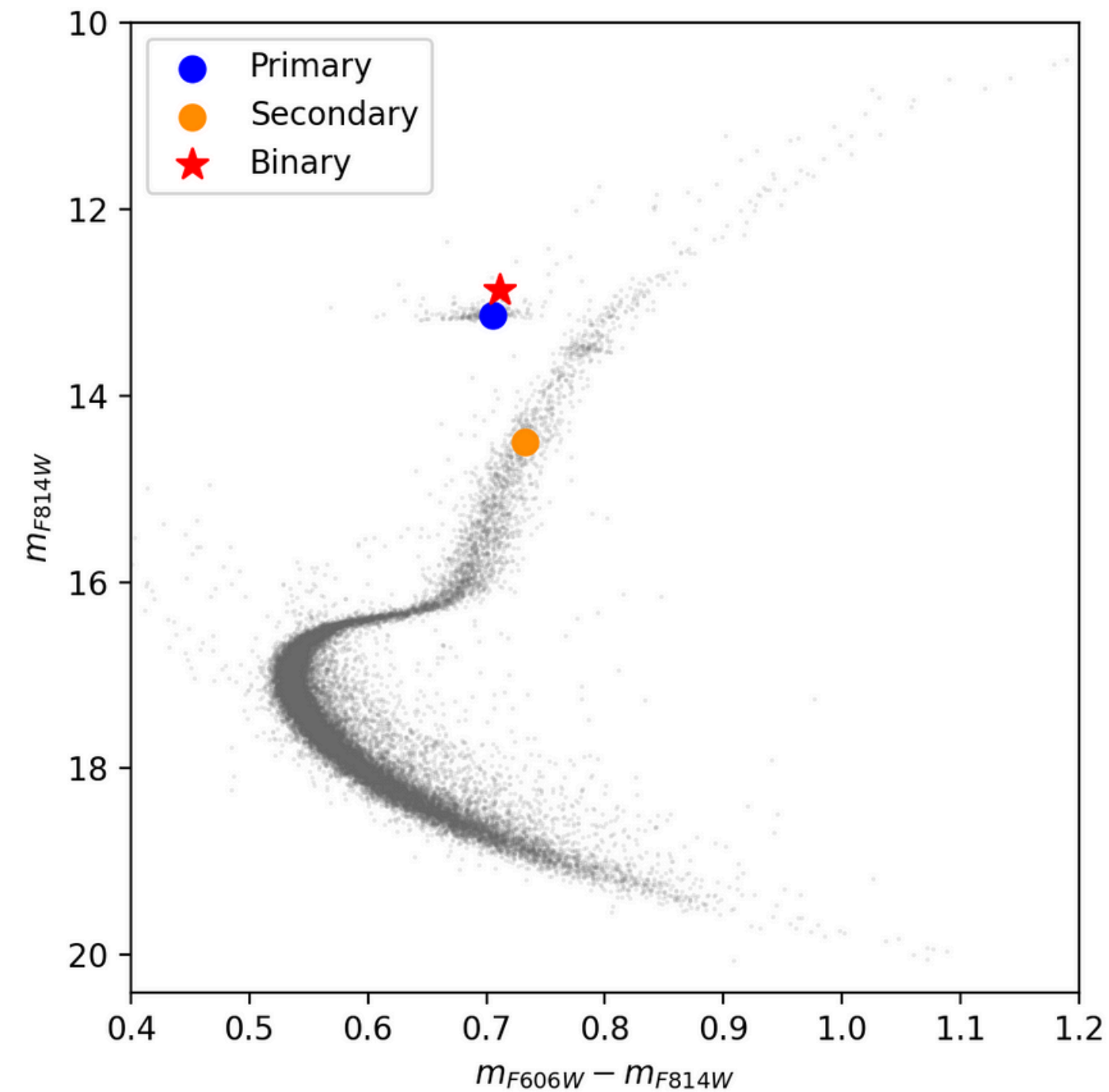
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Binaries beyond the MS

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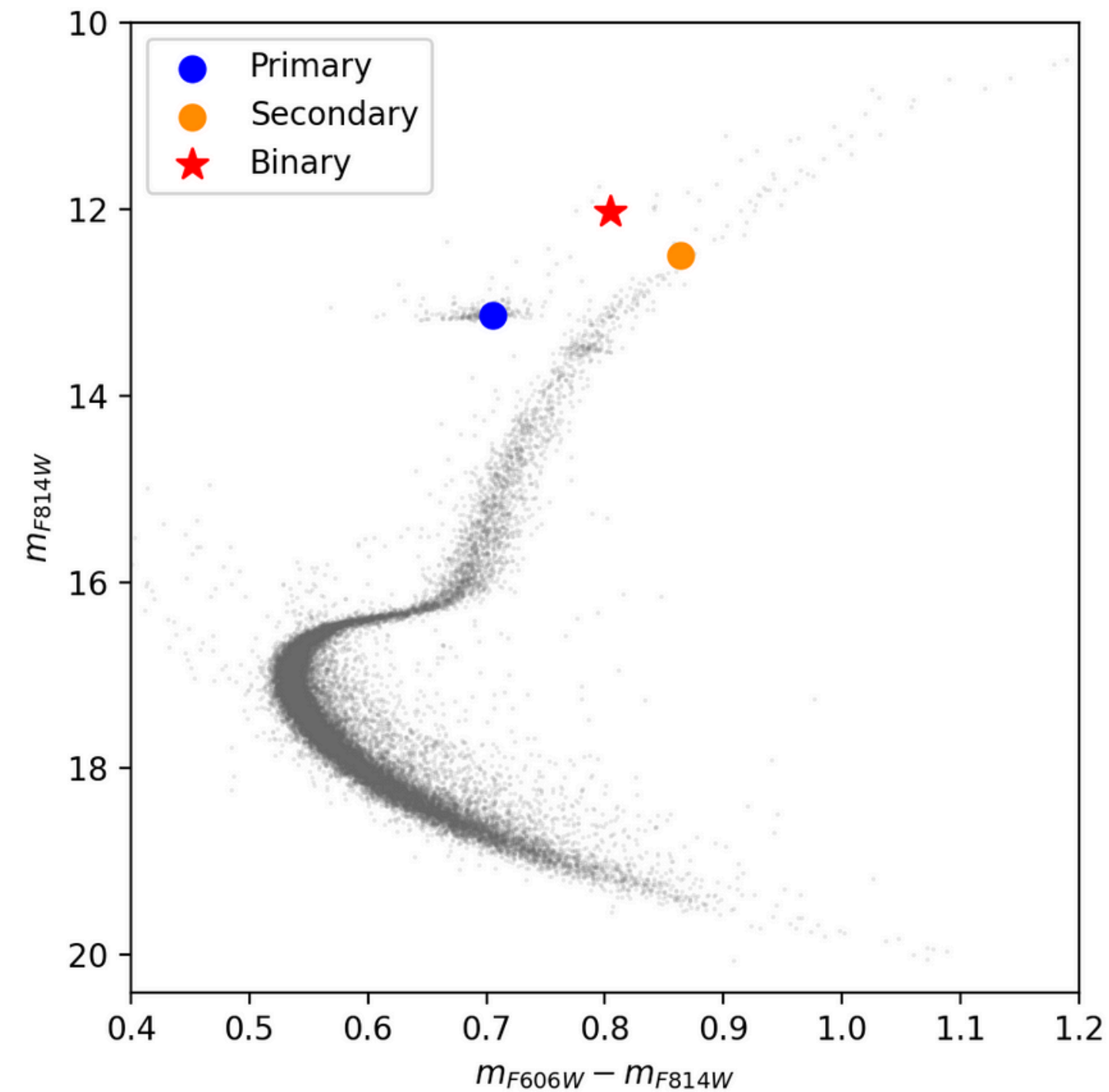
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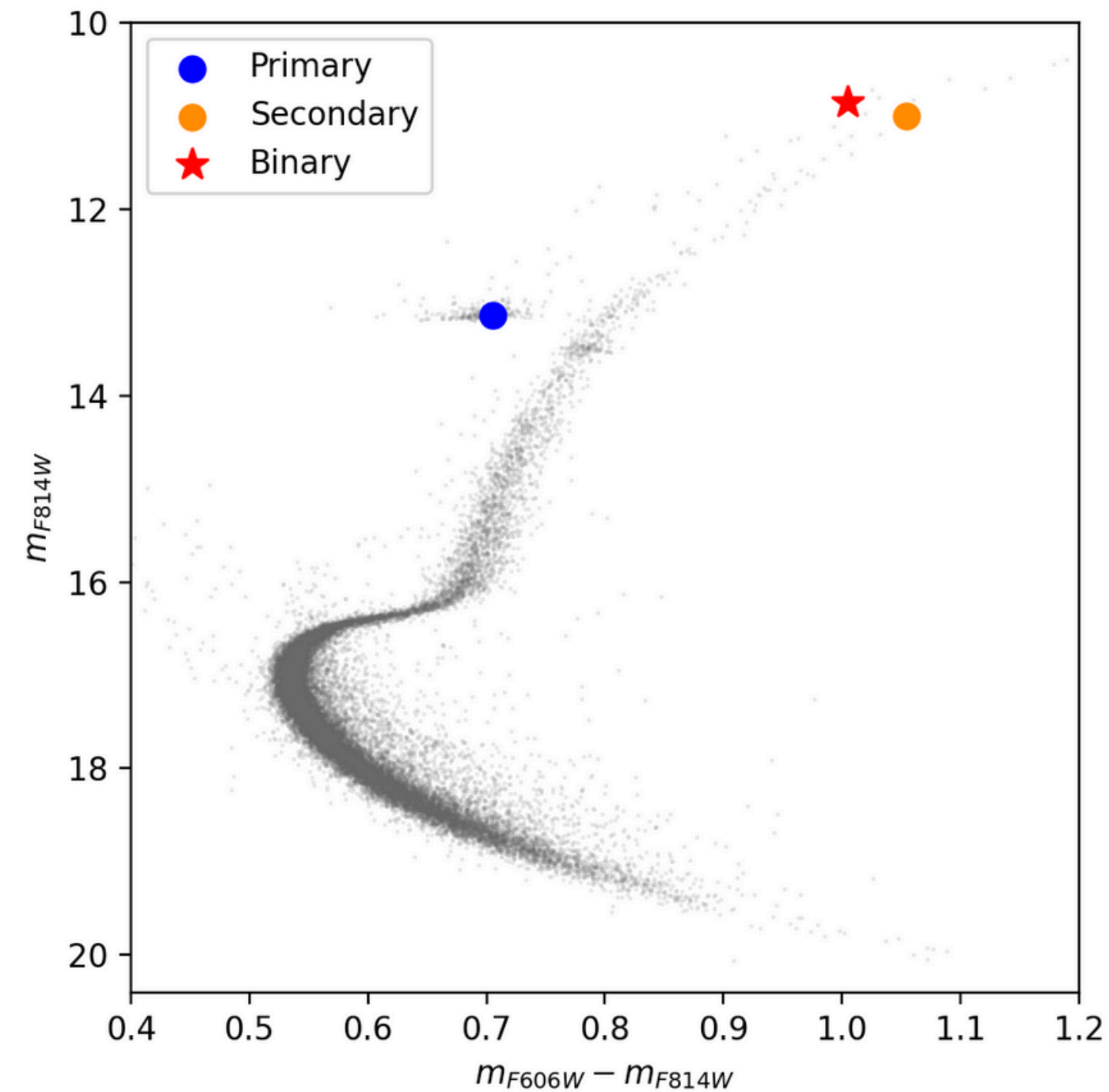
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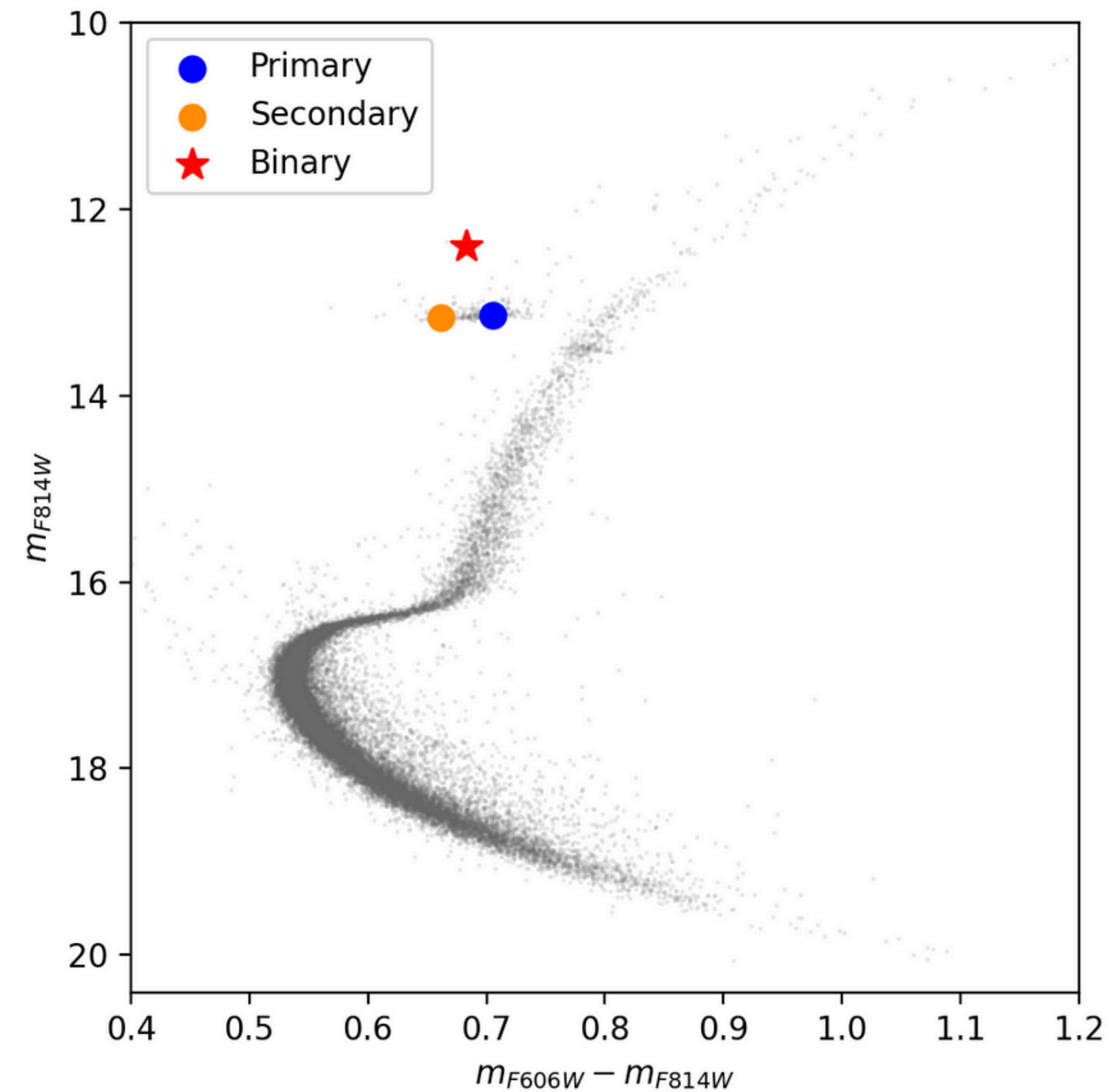
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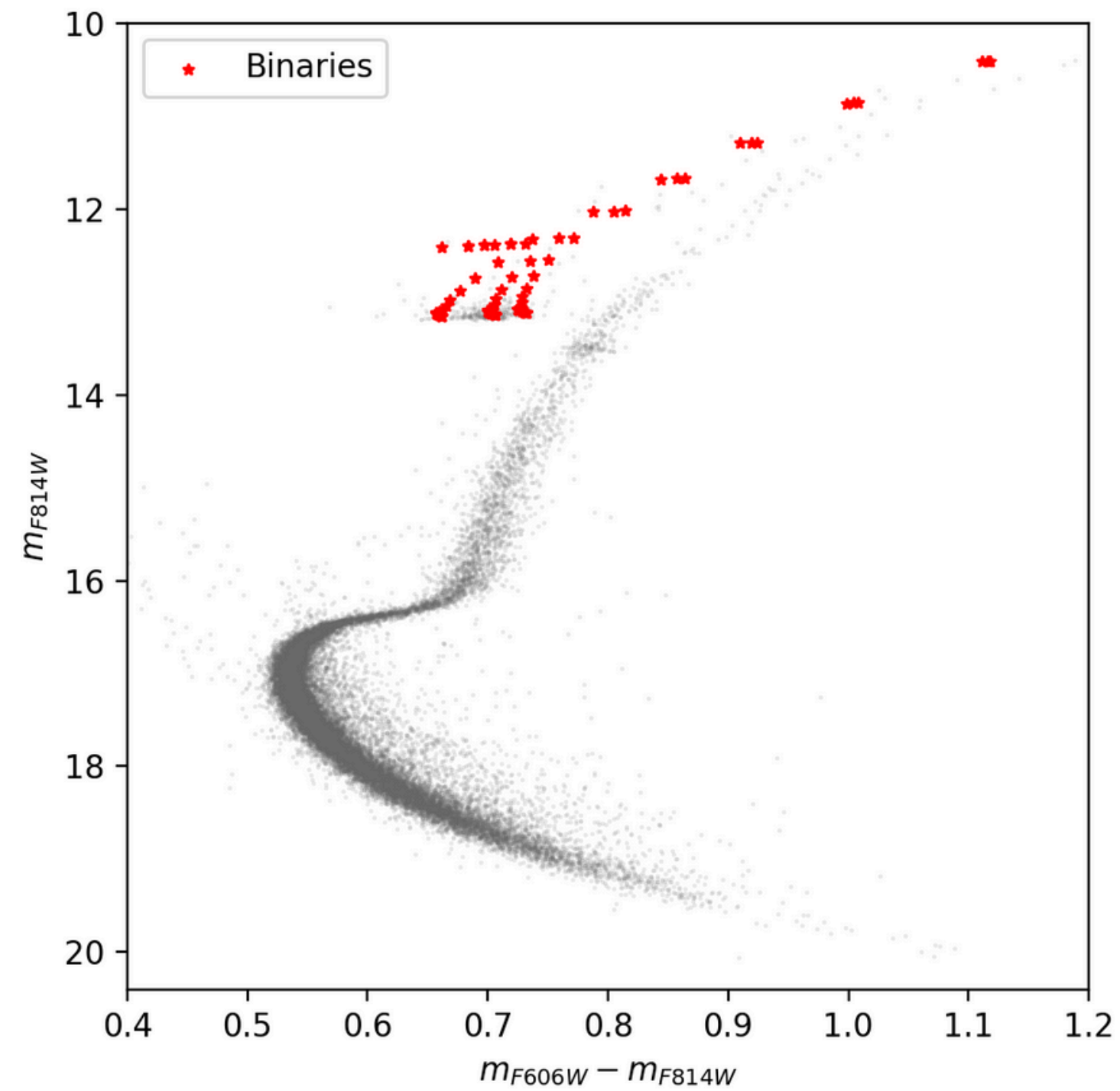
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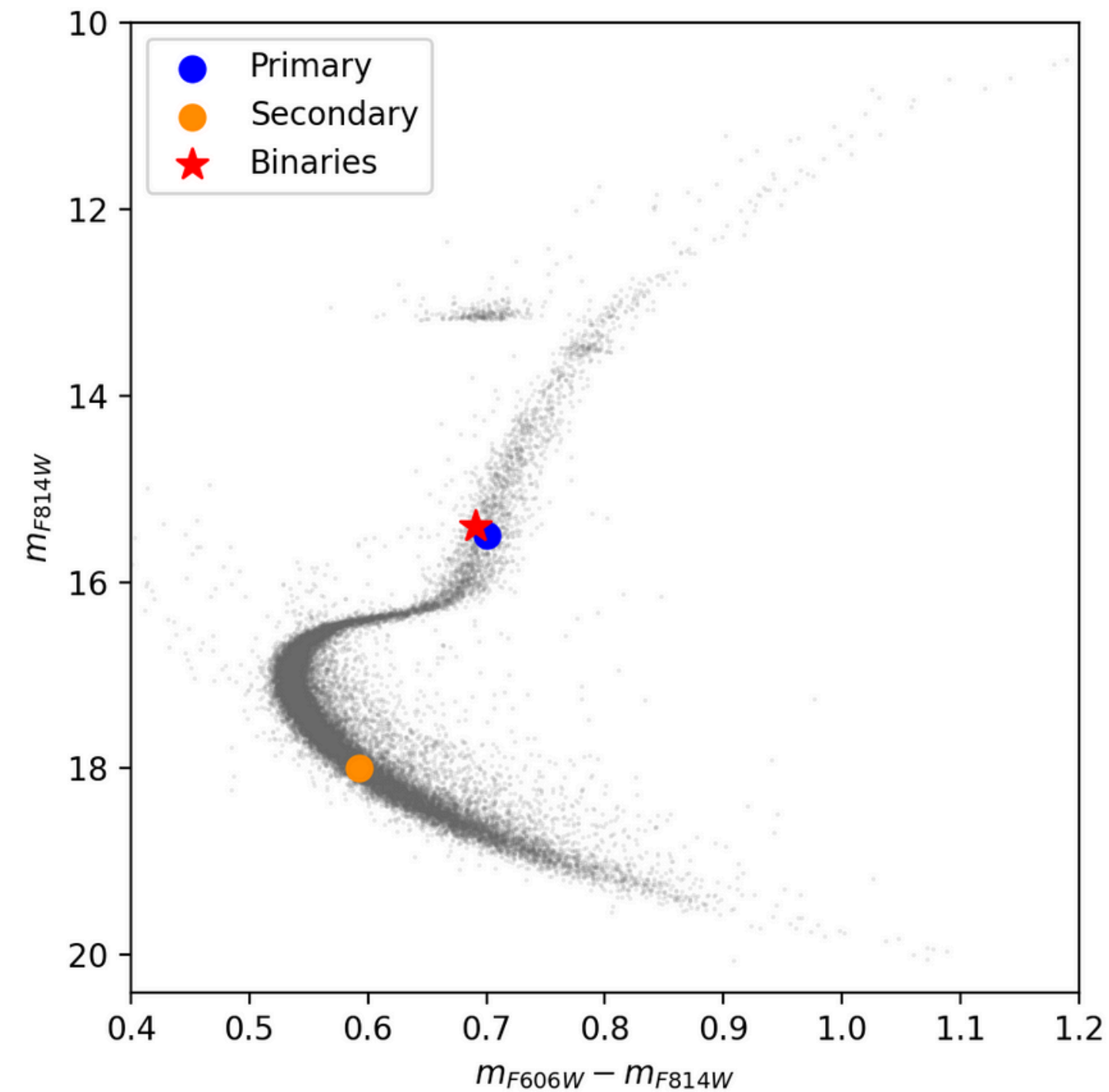
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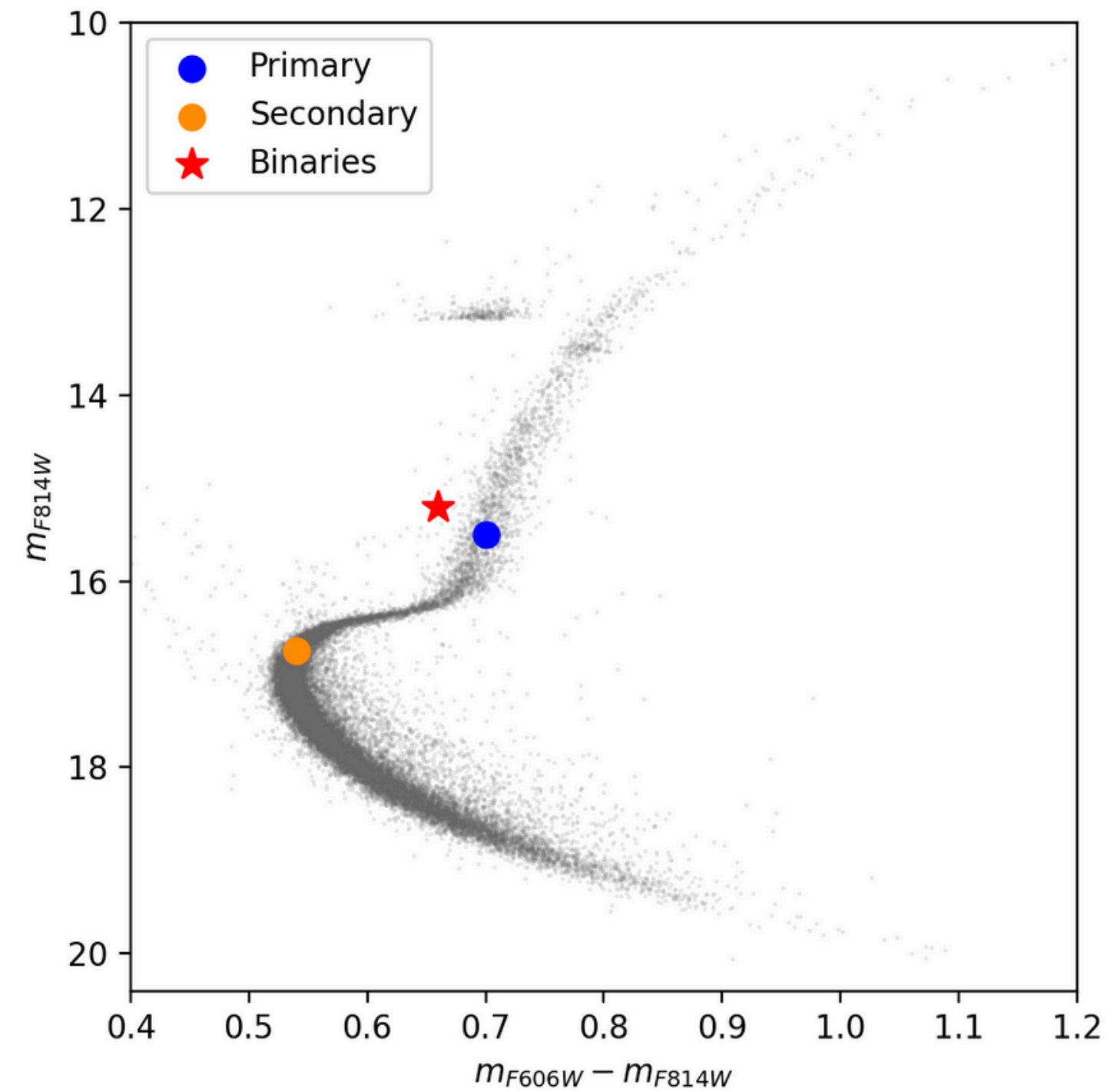
47 Tucanae – RGB binaries



Binaries beyond the MS

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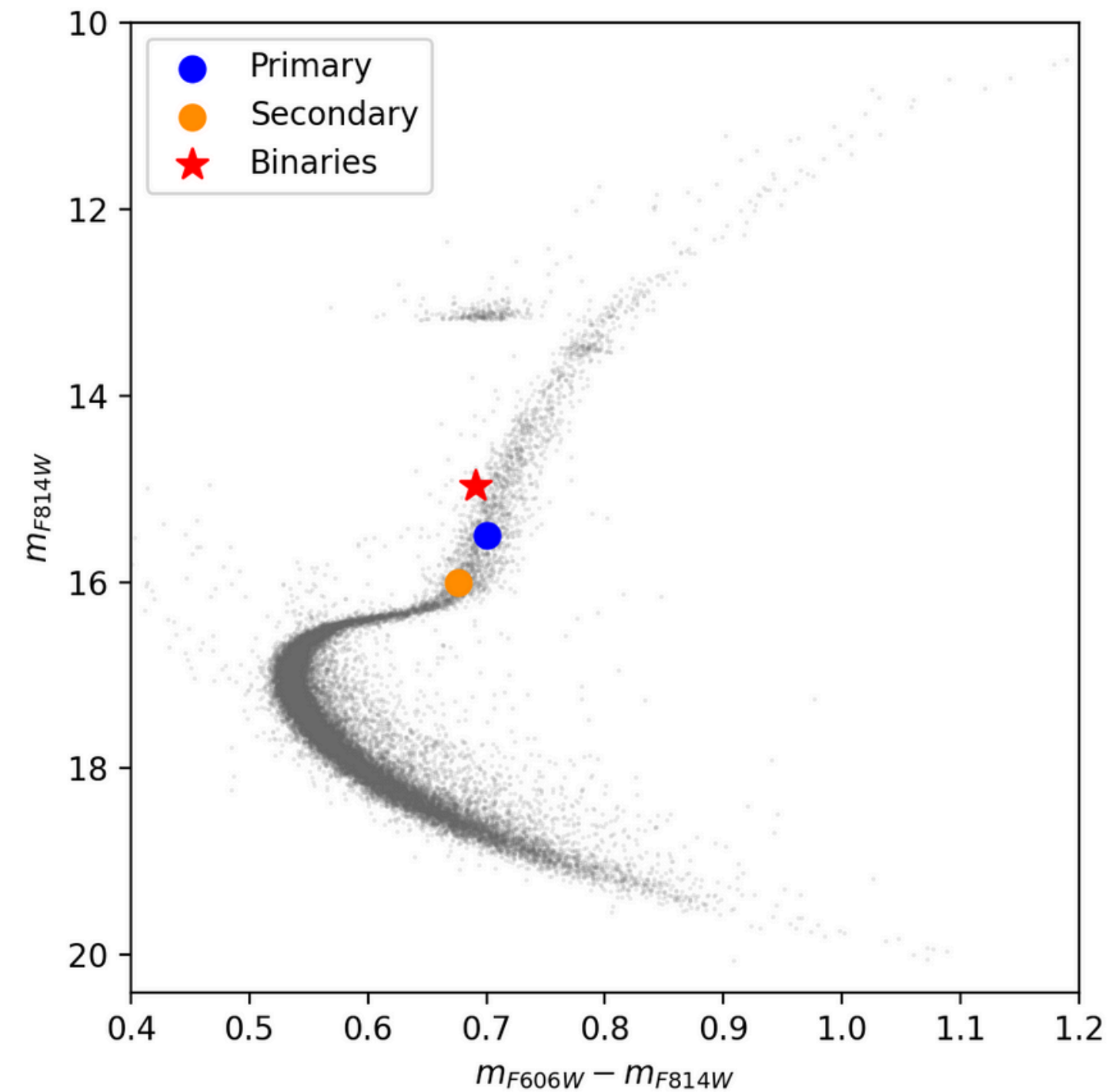
47 Tucanae – RGB binaries



Binaries beyond the MS

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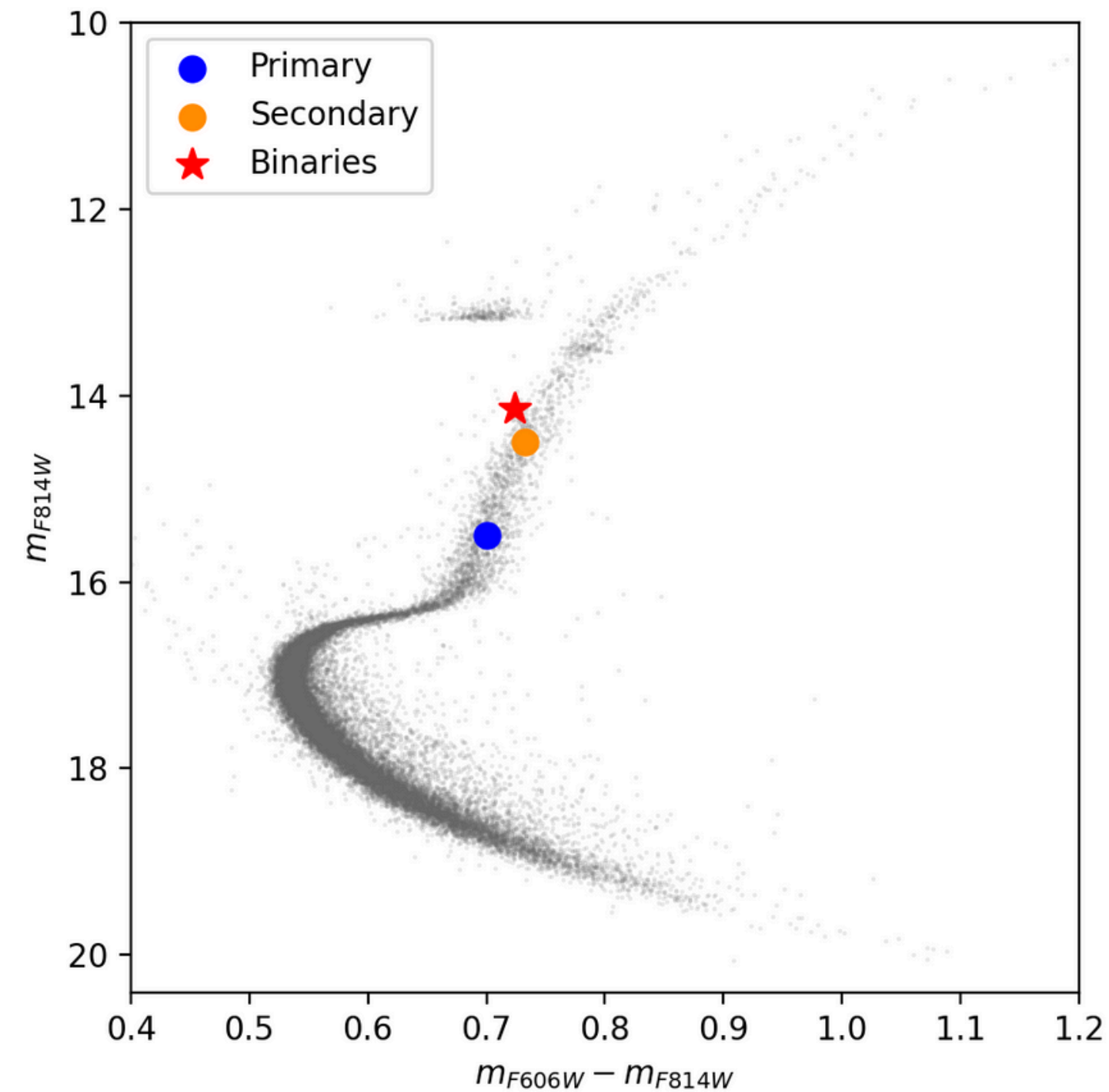
47 Tucanae – RGB binaries



Binaries beyond the MS

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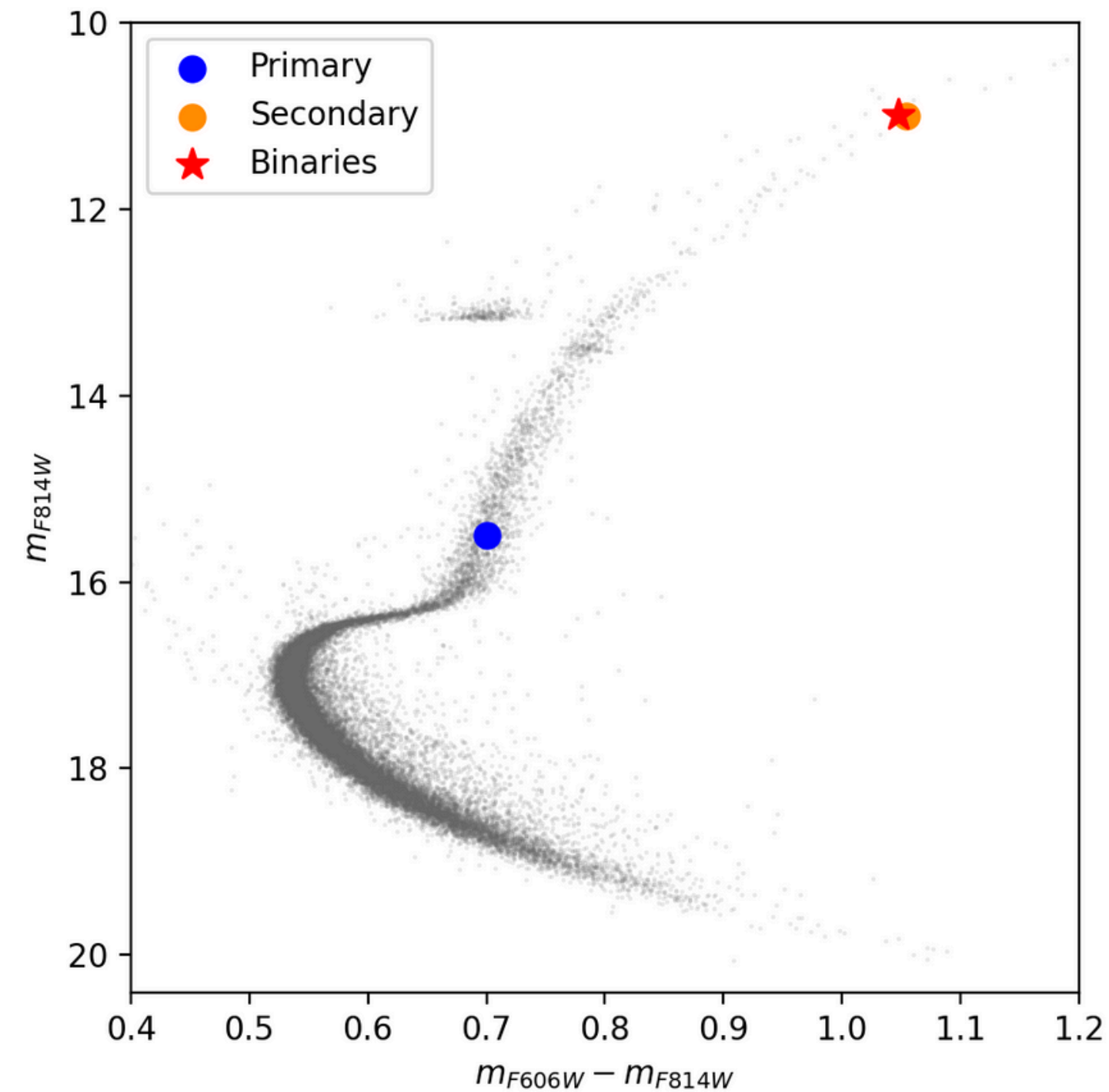
47 Tucanae – RGB binaries



Binaries beyond the MS

Synthetic Binaries

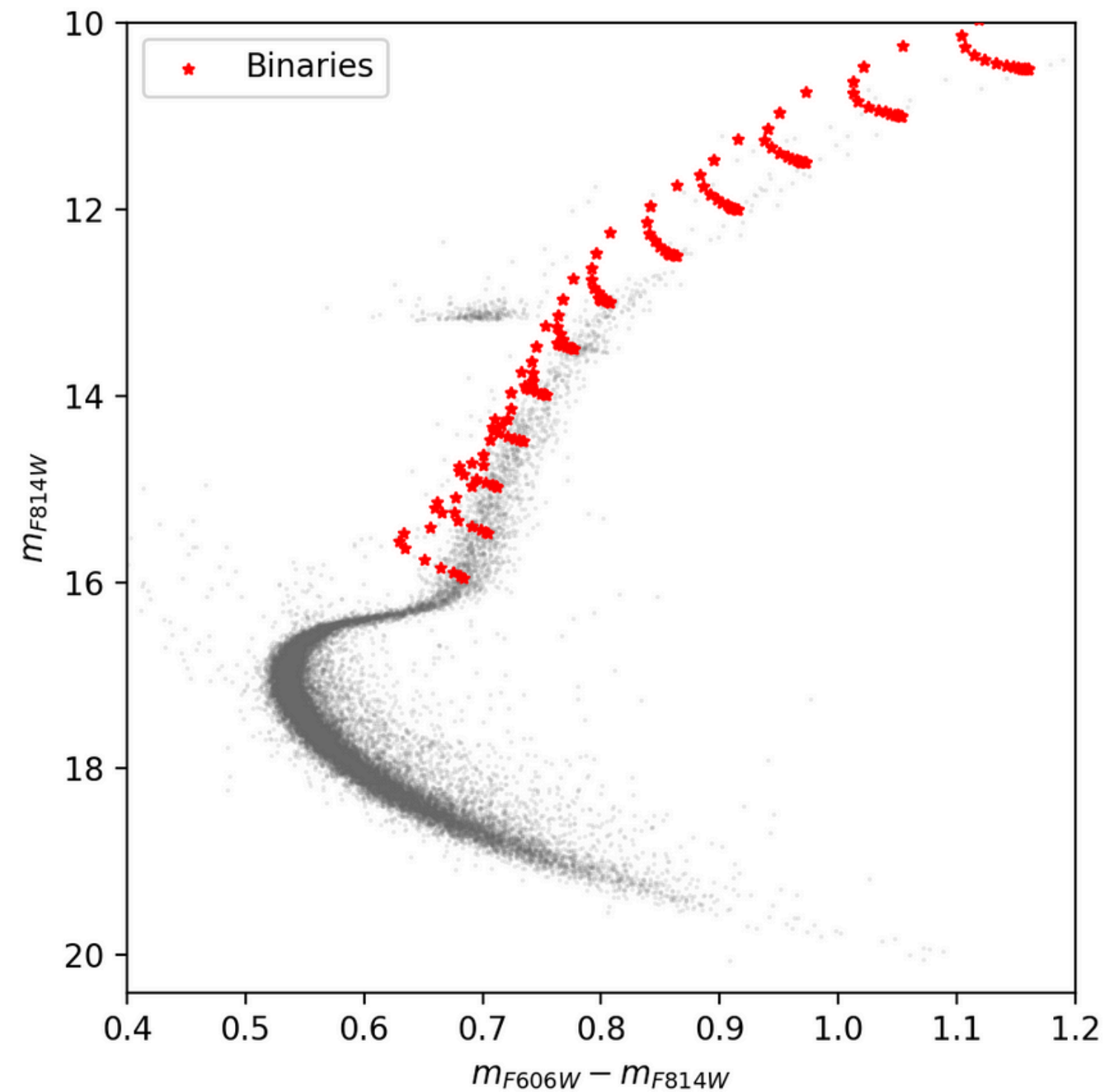
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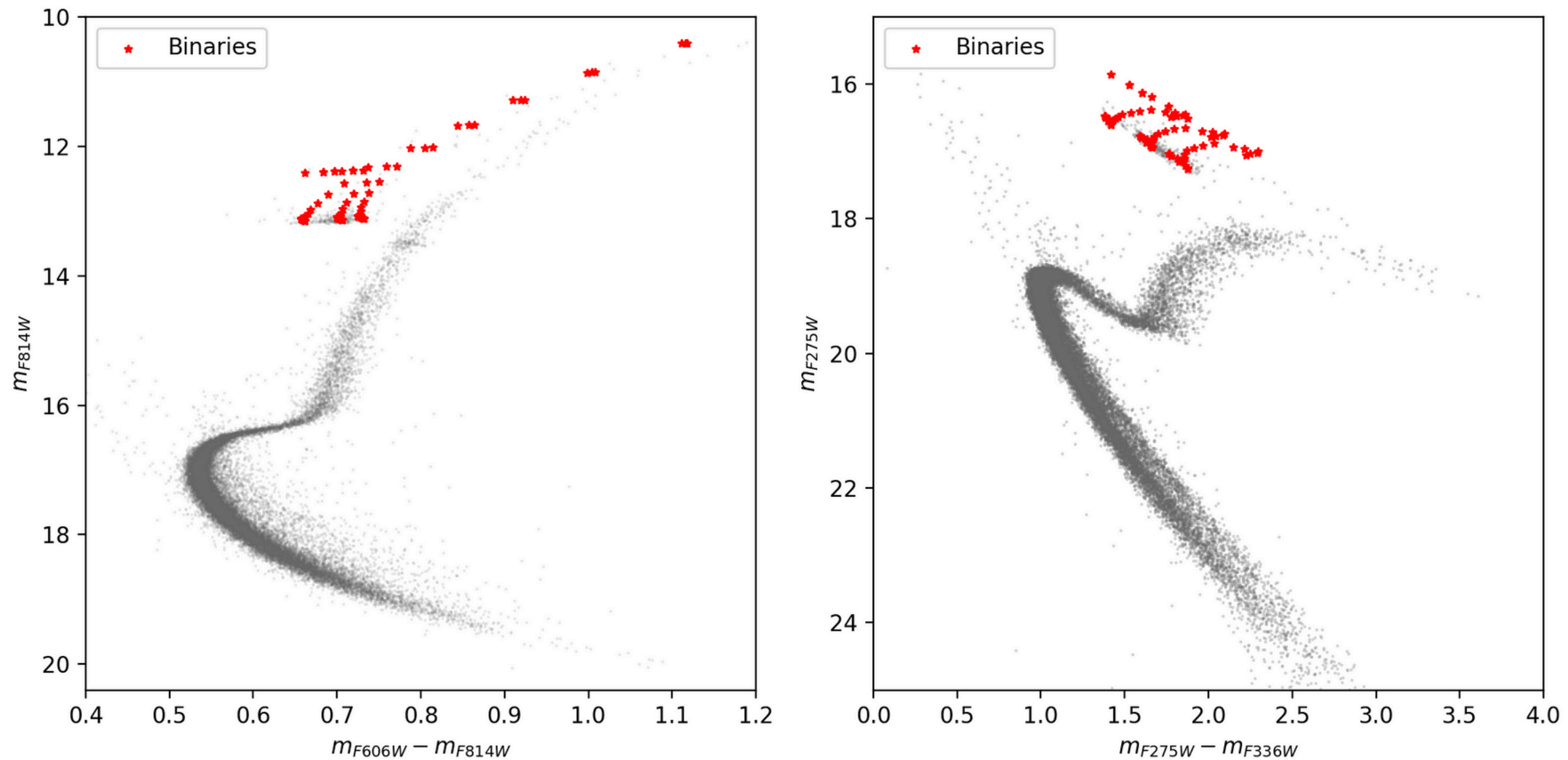
Synthetic Binaries

47 Tucanae – RGB binaries



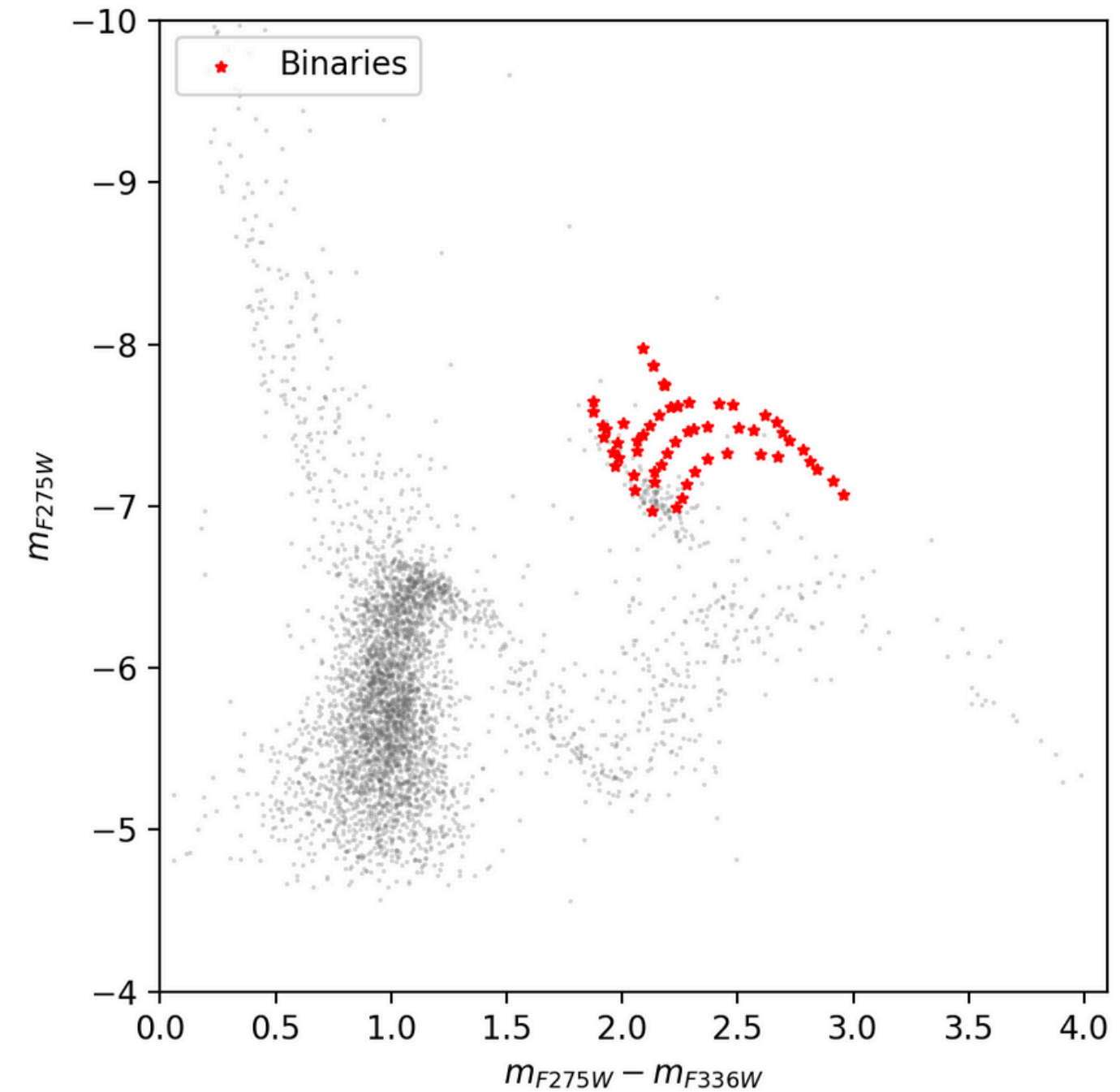
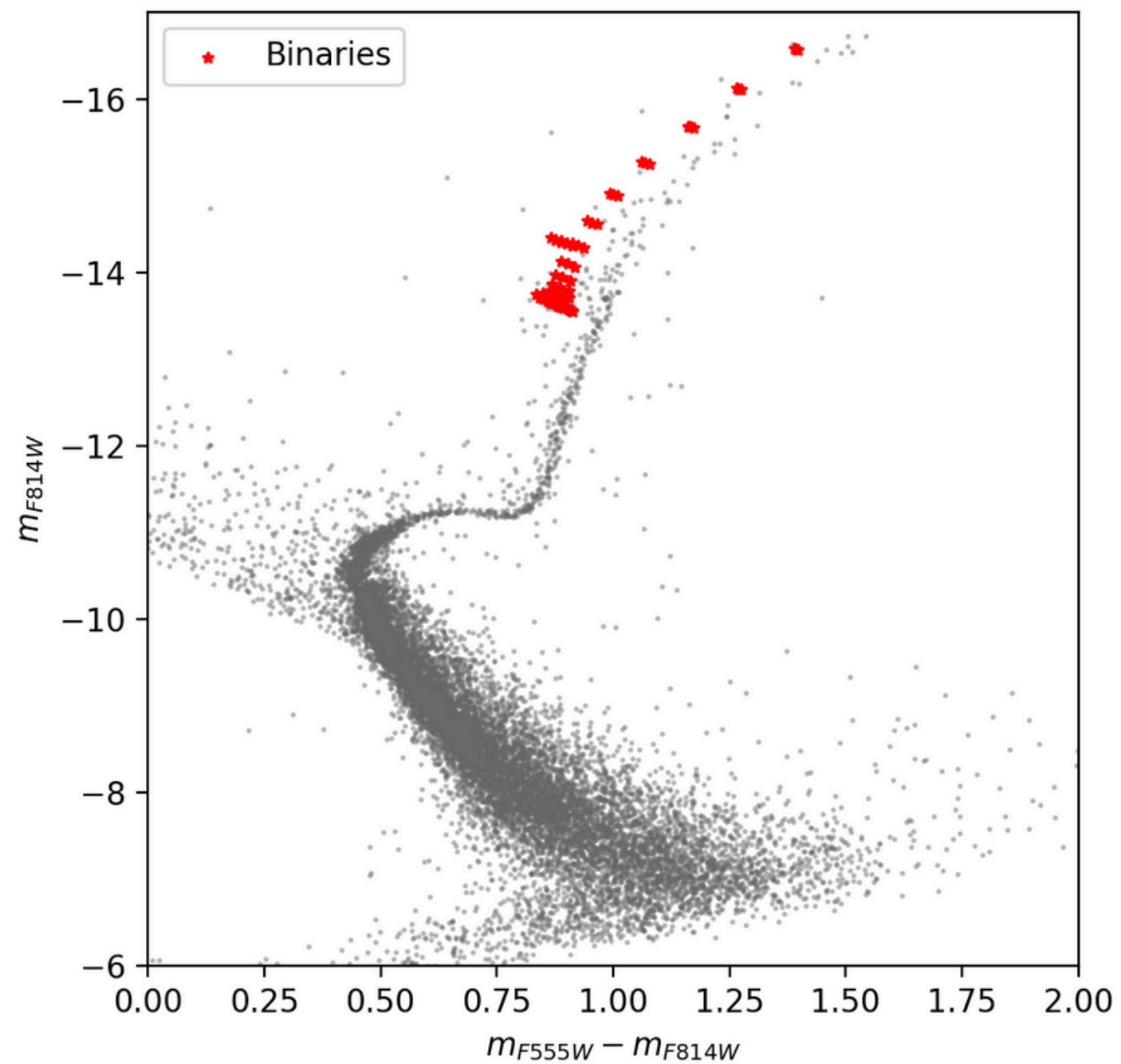
Binaries beyond the MS

47 Tucanae – HB binaries (data Nardiello et al. 2018)



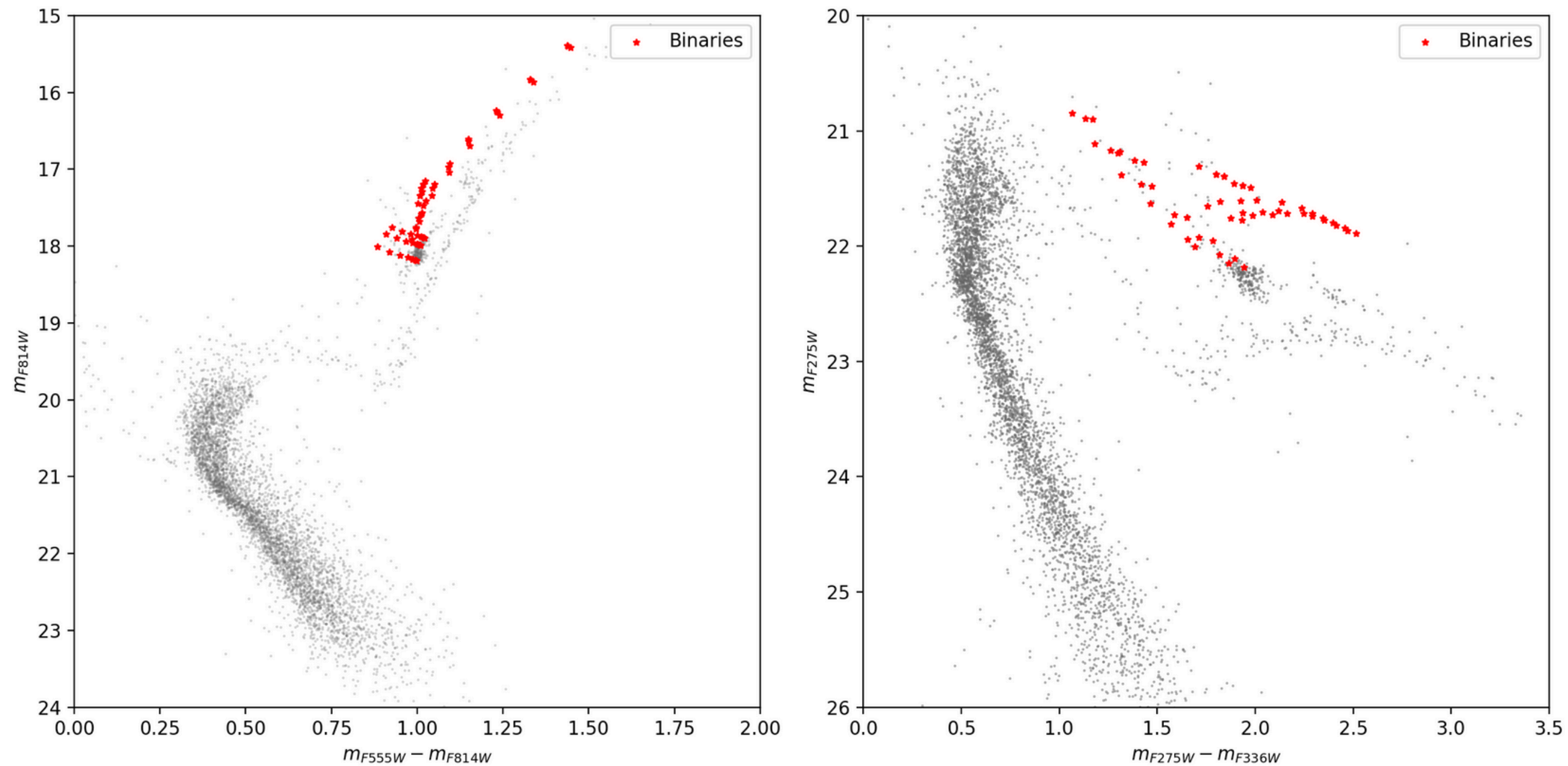
Binaries beyond the MS

BS 90 – HB binaries



Binaries beyond the MS

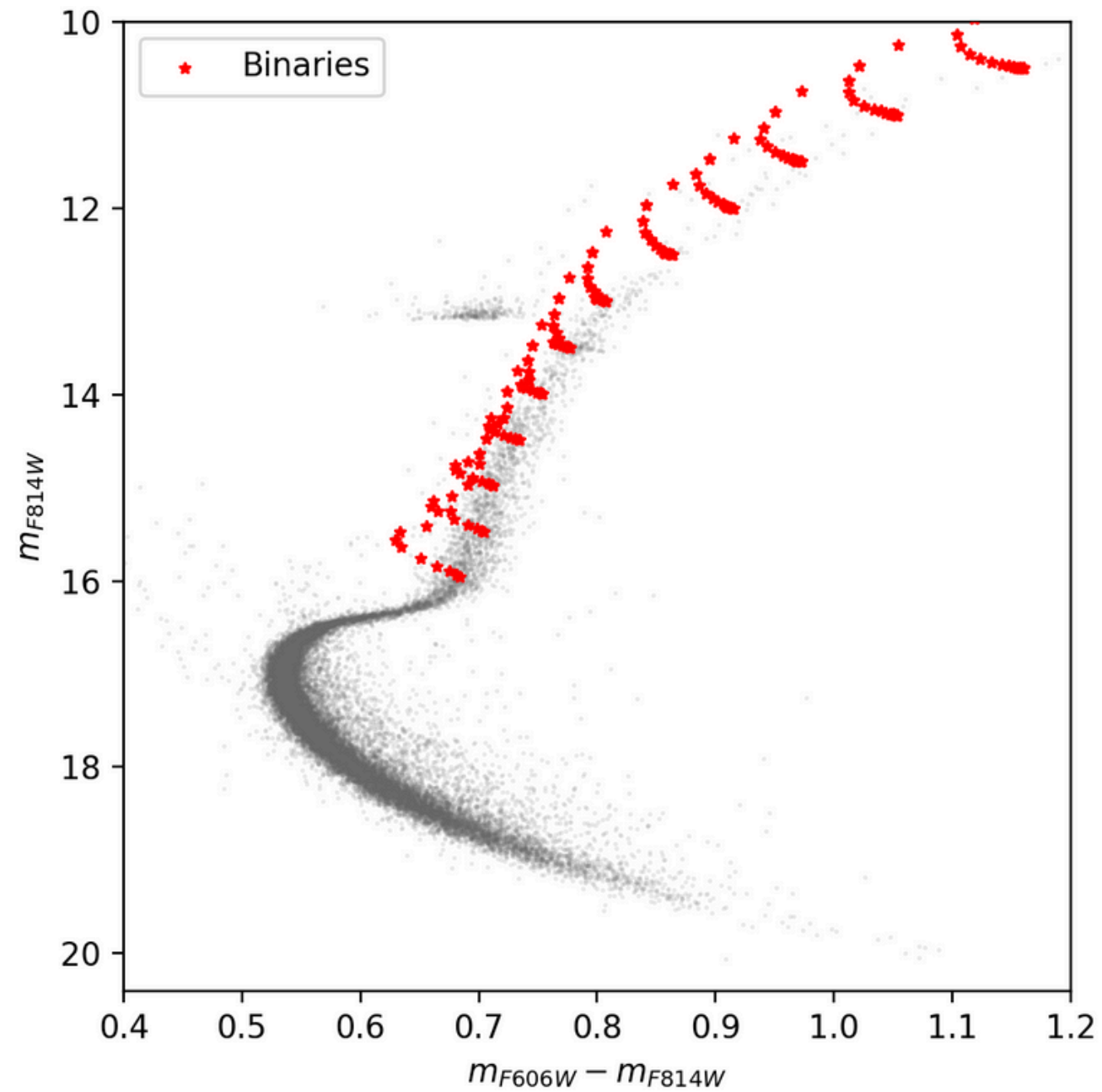
NGC 1783 – HB binaries (data Milone et al. 2023)



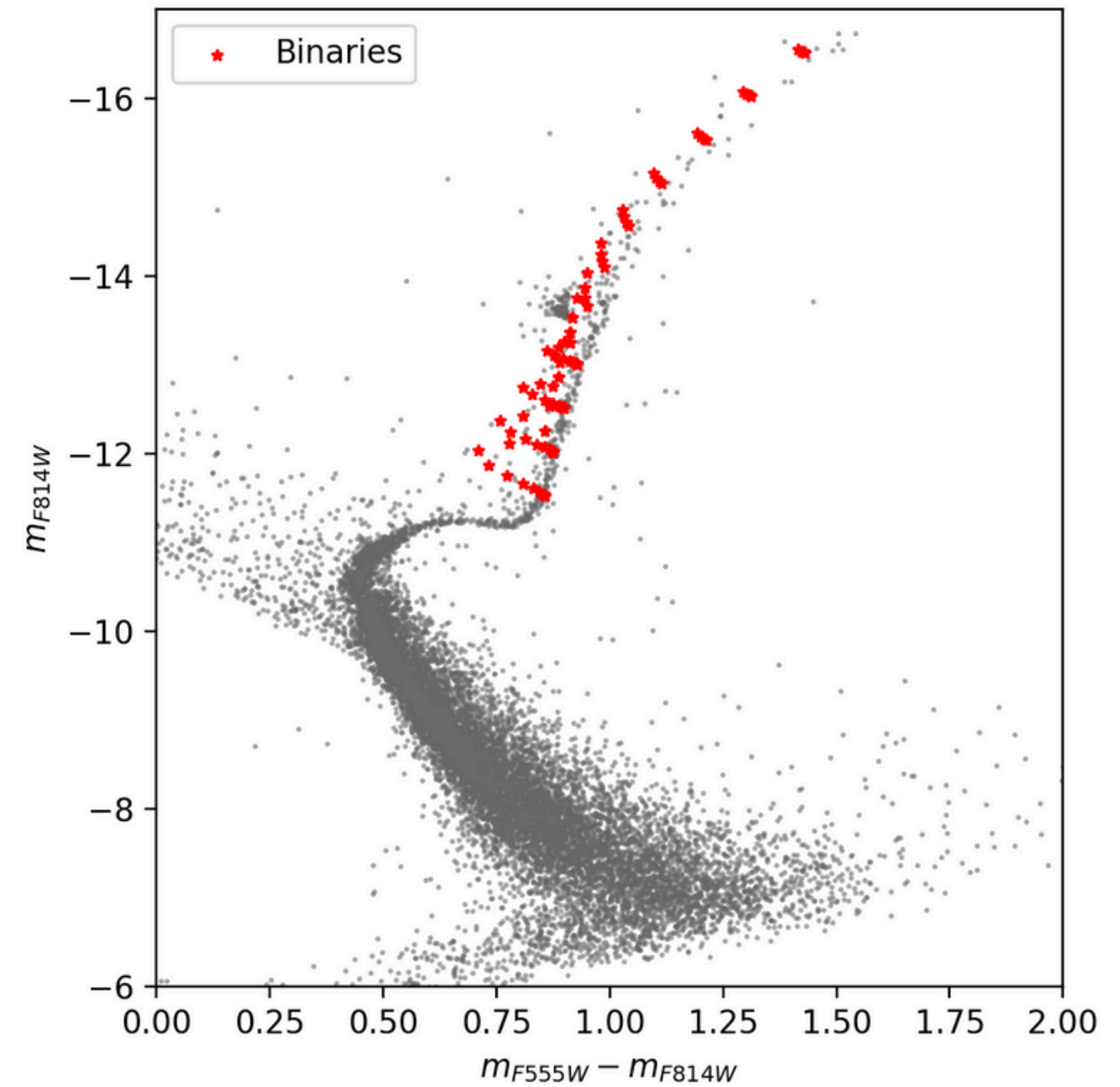
Binaries beyond the MS

RGB binaries

47 Tucanae



BS 90



Binaries beyond the MS

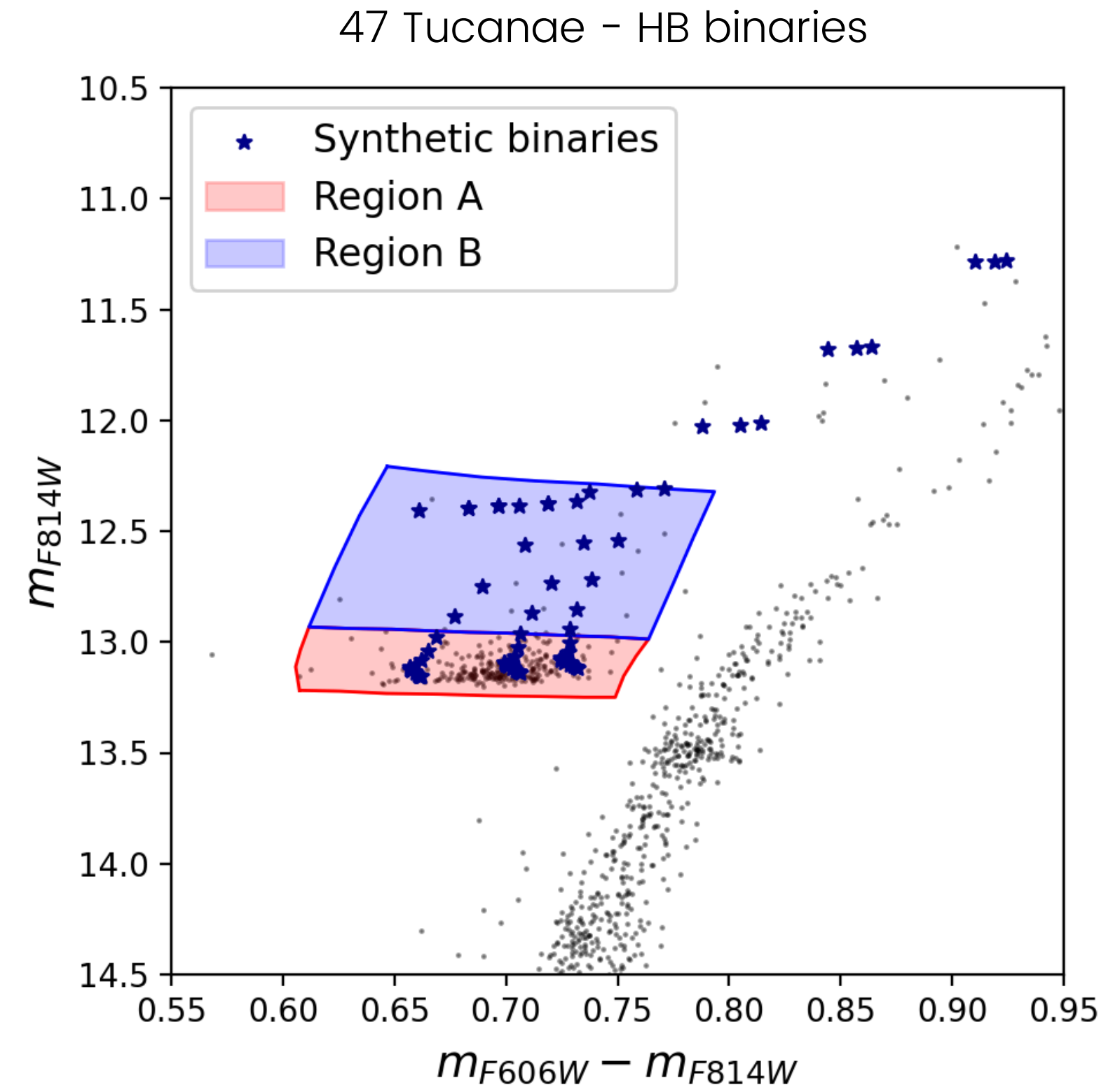
Computing the Binary Fraction

To estimate the binary fraction I select two regions:

- **Region A** – contains single stars
- **Region B** – contains binaries

Then the binary fraction is given by:

$$f_{\text{bin}} = \frac{N_B}{N_B + N_A}$$



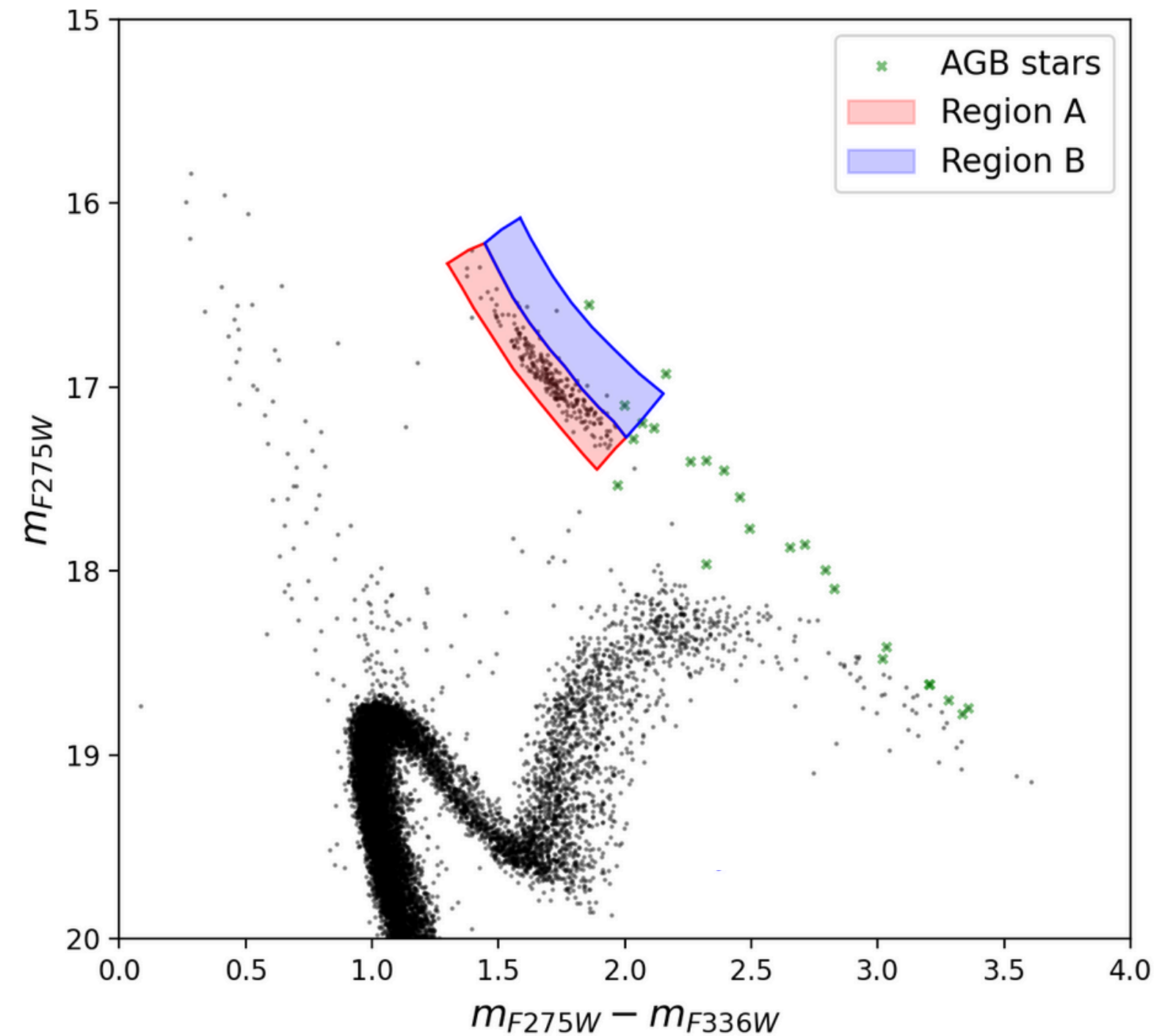
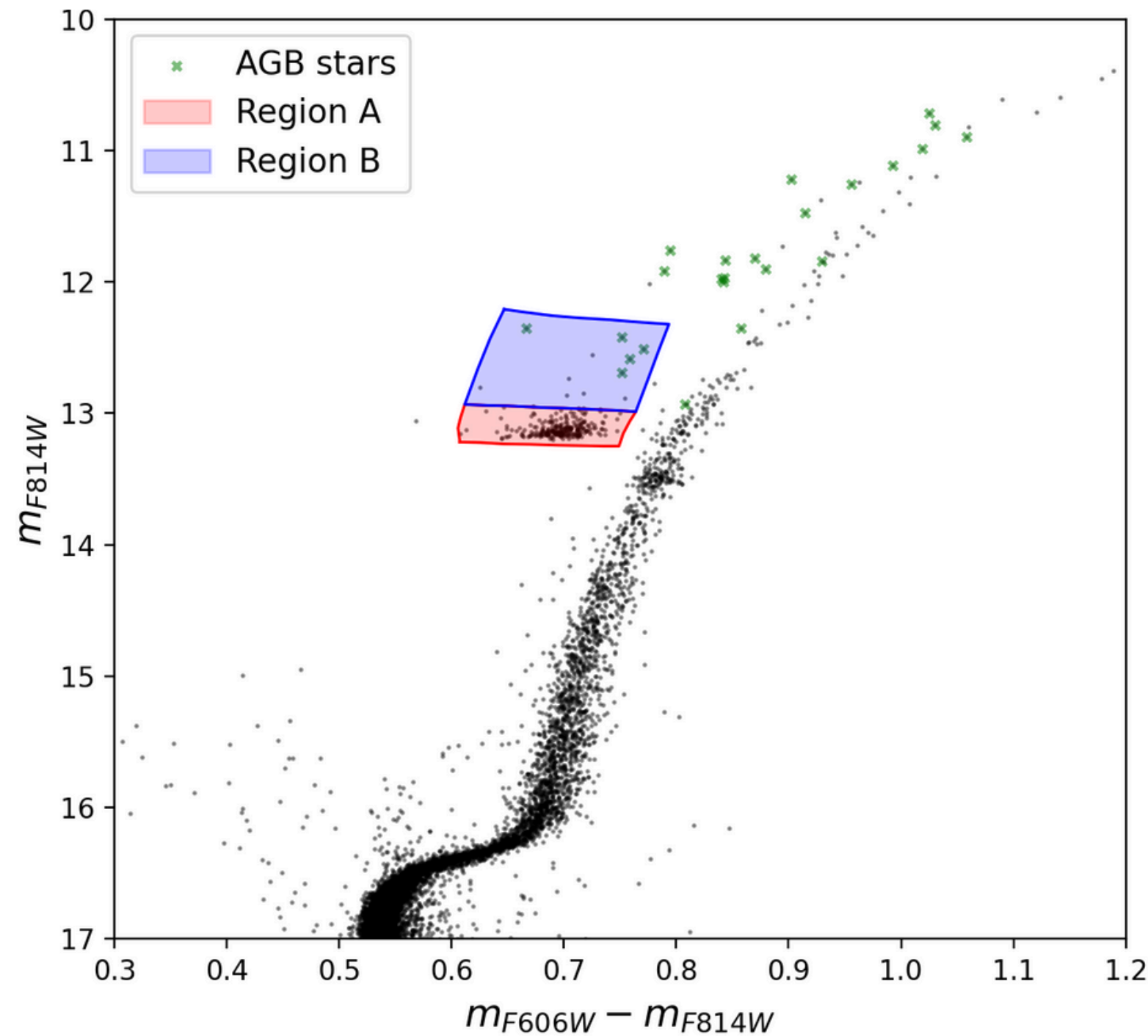
Binaries beyond the MS

Computing the Binary Fraction – HB

47 Tucanae

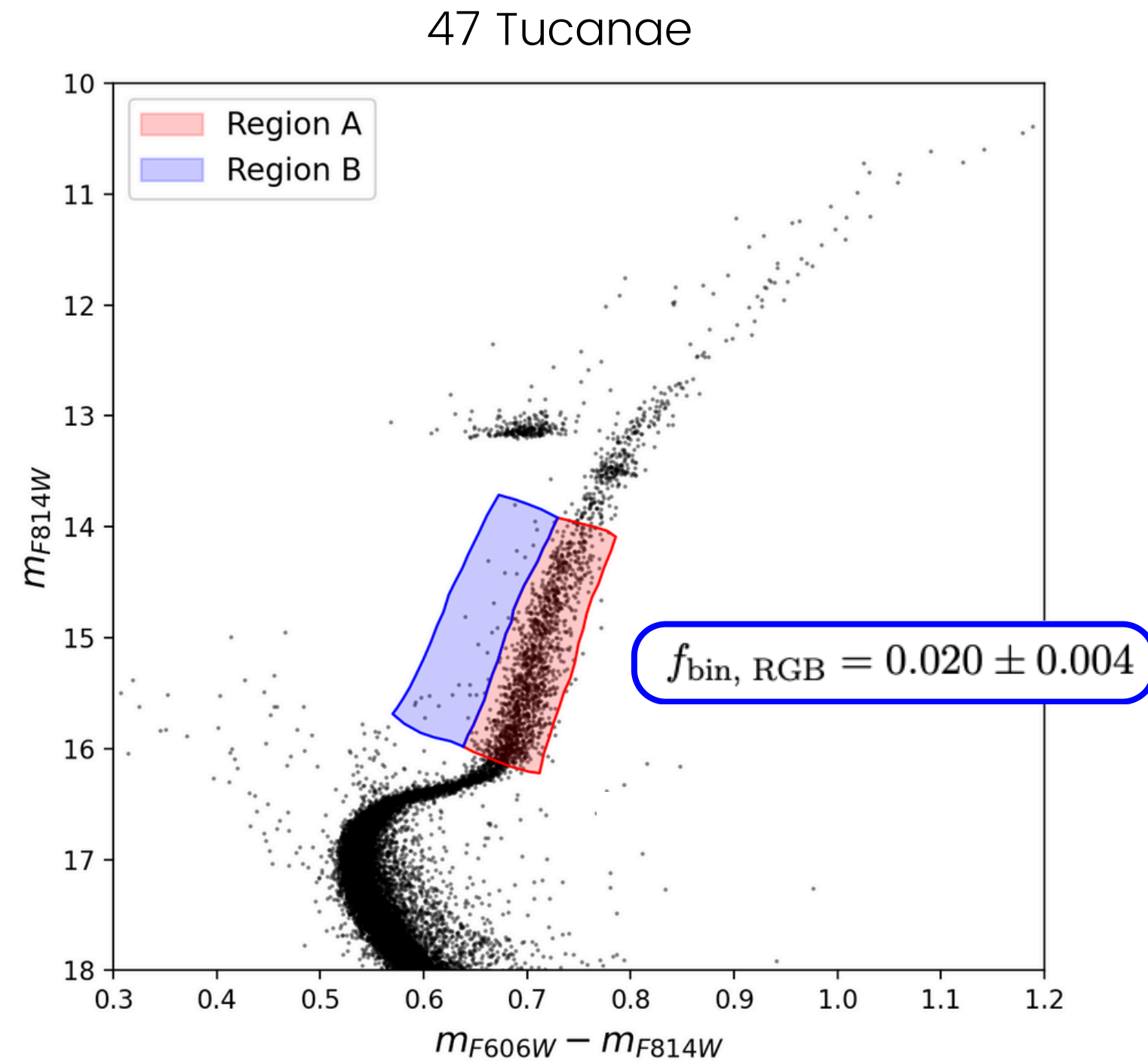
$$f_{\text{bin, HB}}^{\text{optical}} = 0.040 \pm 0.012$$

$$f_{\text{bin, HB}}^{\text{UV}} = 0.040 \pm 0.012$$



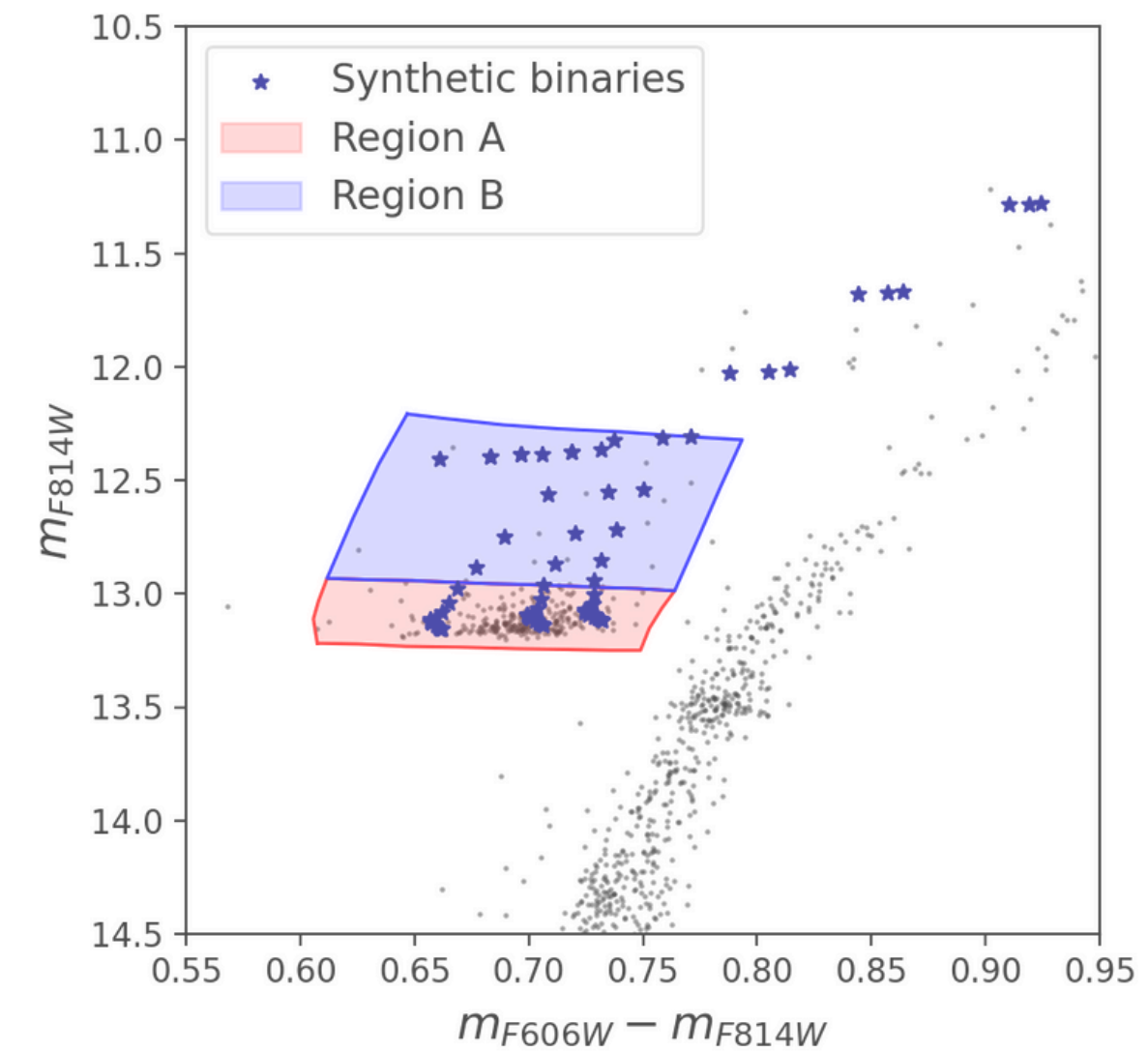
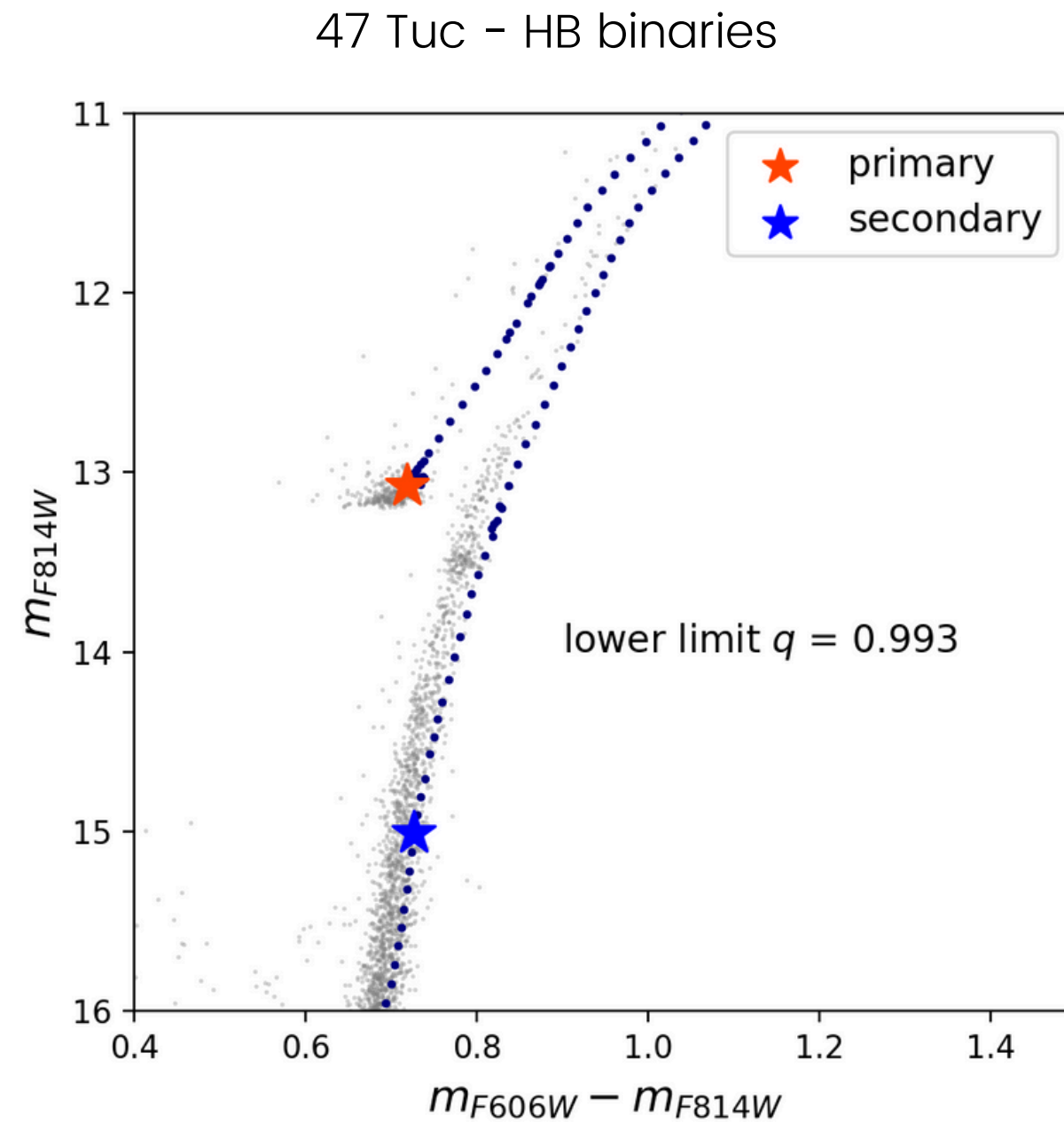
Binaries beyond the MS

Computing the Binary Fraction – RGB



Binaries beyond the MS

Computing the Mass Ratio



Binary's position is linked to the luminosity and mass of its member stars



Determination of the **minimum mass ratio** required for a binary system to be detectable.

Binaries beyond the MS

Summary of the Results



	HB			RGB	
	$f_{bin}^{optical}$	f_{bin}^{UV}	q_{min}	f_{bin}	q_{min}
47 Tuc	0.040 ± 0.012	0.040 ± 0.012	0.90	0.020 ± 0.004	0.90
BS 90	0.046 ± 0.024	0.048 ± 0.035	0.90	0.038 ± 0.022	0.82
NGC 1783	0.093 ± 0.021	0.094 ± 0.020	0.82	-	-

- The values are not compatible with the ones for the MS
- Extrapolating the total binary fraction assuming a flat mass distribution leads to a paradox

	MS	
	f_{bin}	q
47 Tuc	0.005 ± 0.003	0.7
NGC 1783	0.094 ± 0.007	0.7

(Milone et al. 2012, Mohandas et al. 2024)

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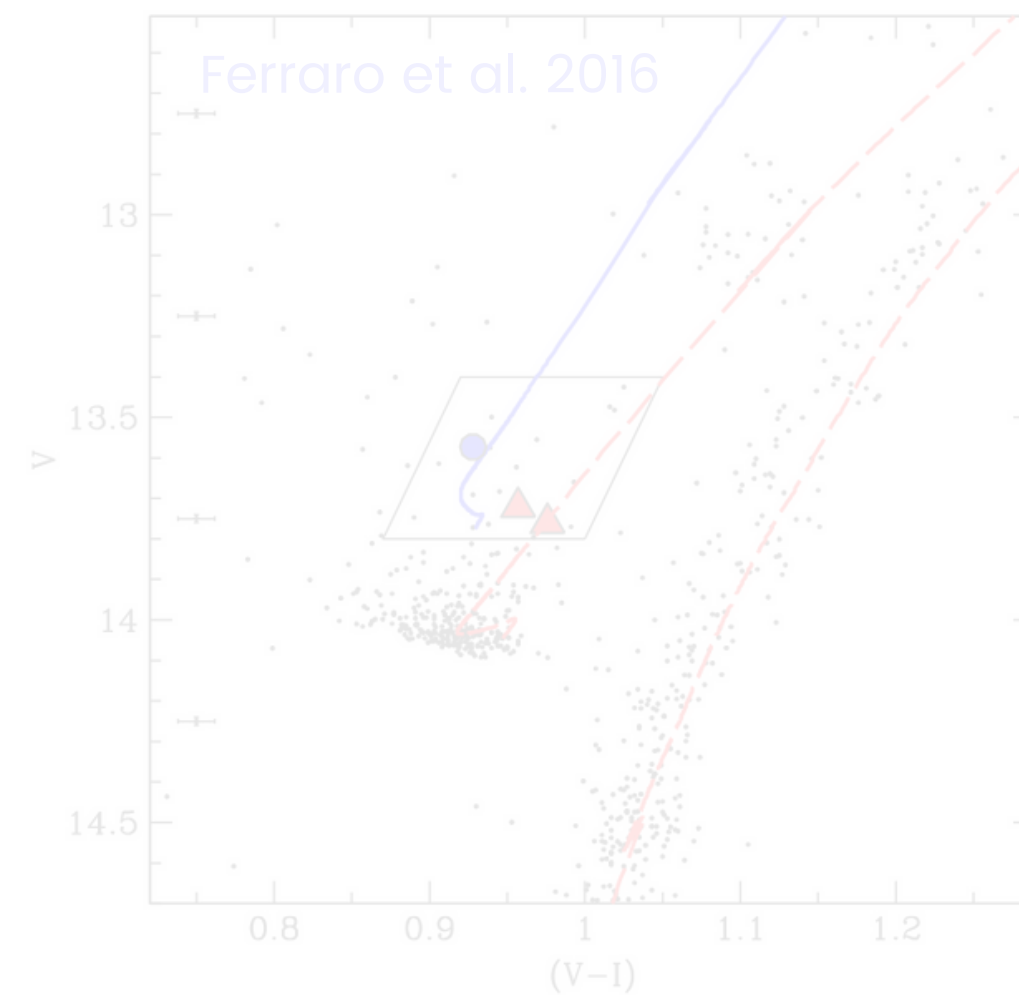
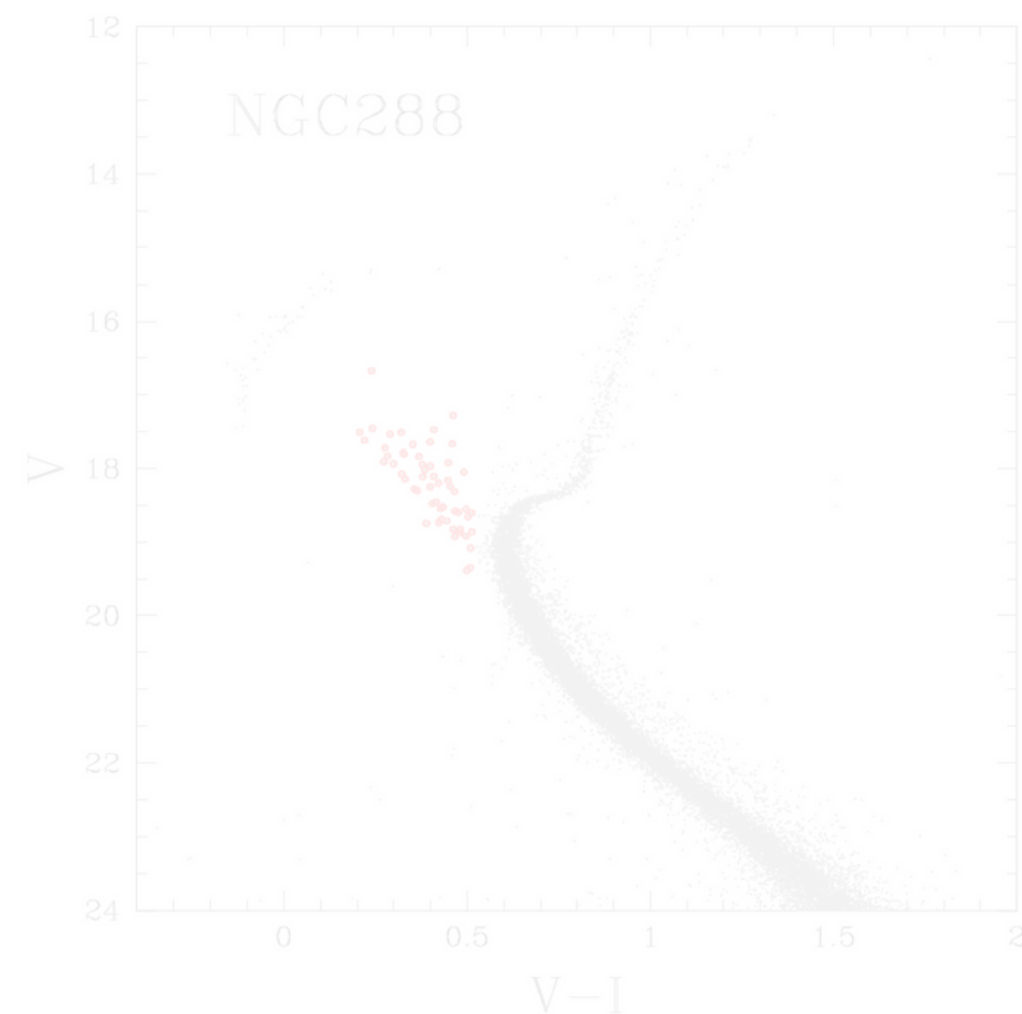
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Binaries beyond the MS

Speculations on the Results

- Mass distribution is **peaked around one** $\longrightarrow$ Makes the binary fractions compatible
- Presence of **Blue Stragglers** in both RGB and HB

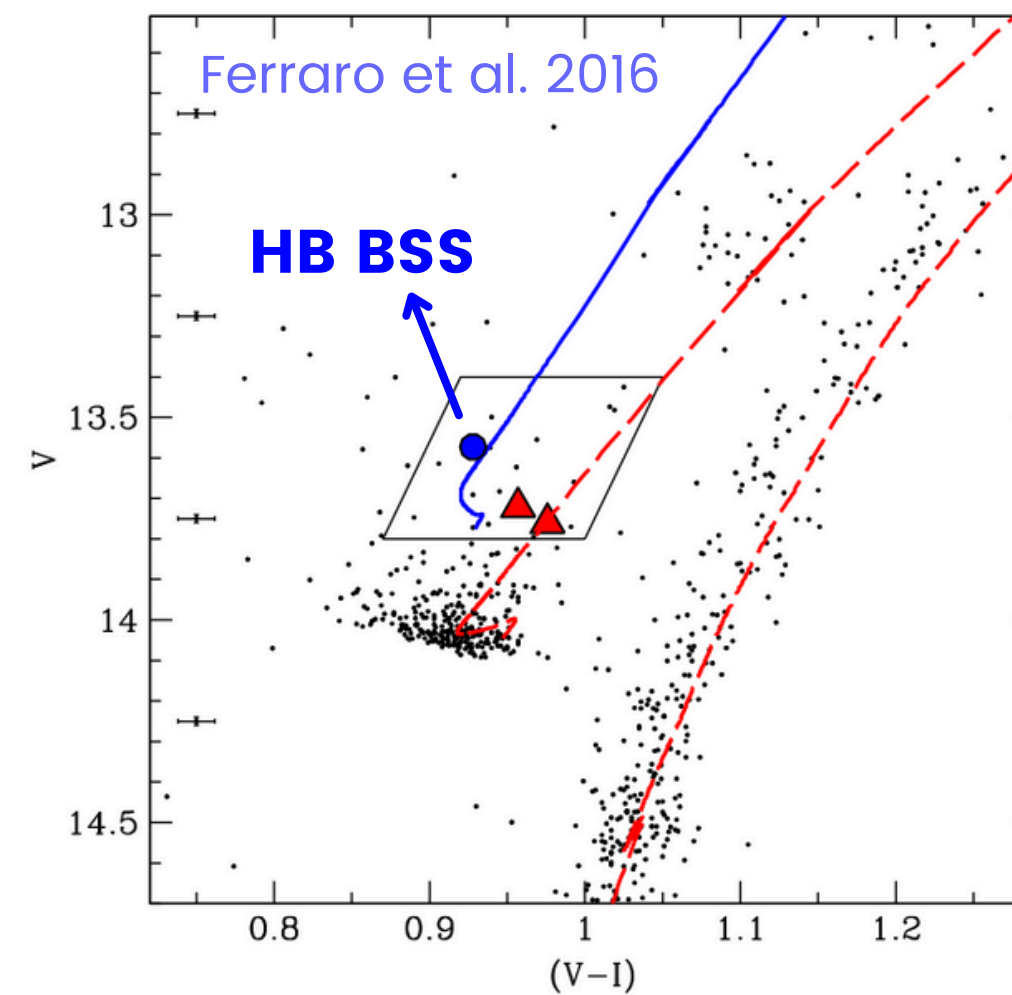
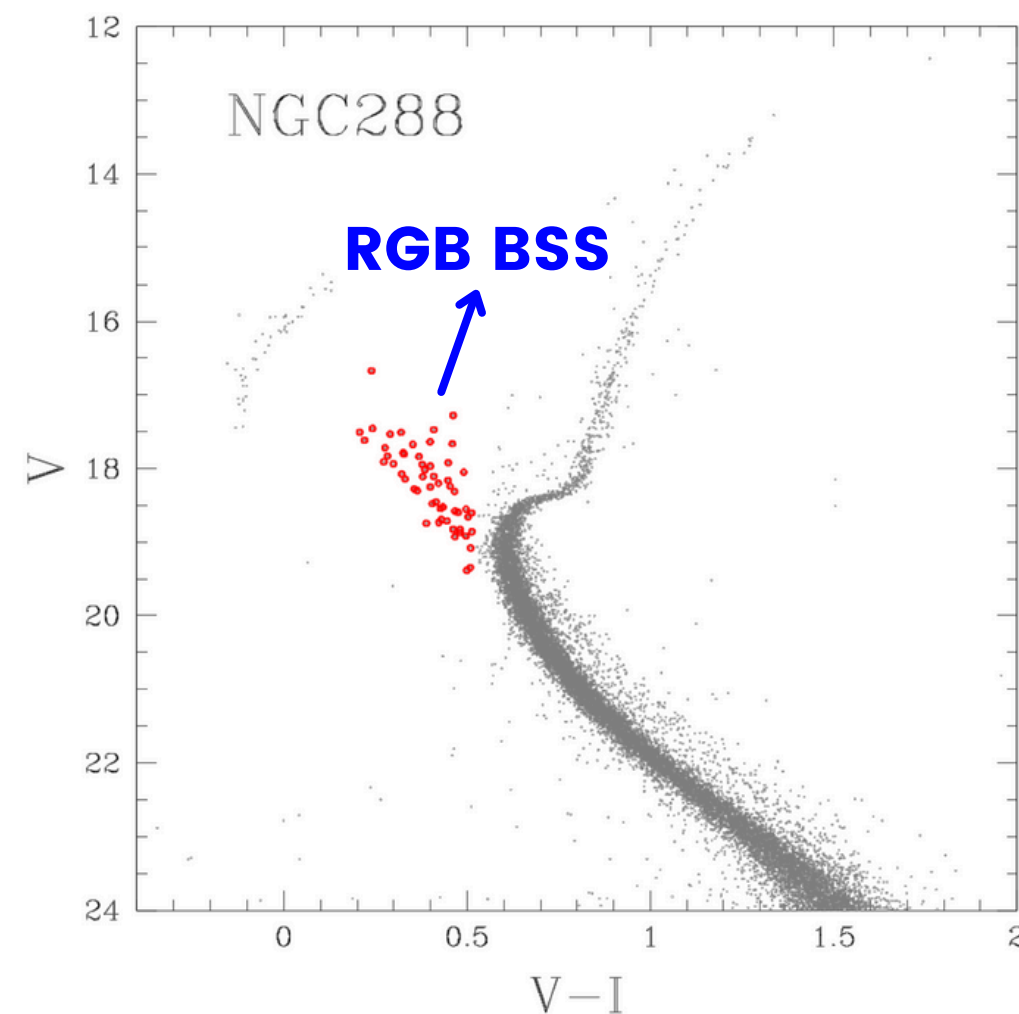


BSSs are located on the left of the RGB and above the HB

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Binaries beyond the MS

Speculations on the Results

- The mass distribution is **peaked around one** $\longrightarrow$ Makes the binary fractions compatible
- Presence of **Blue Stragglers** $\longrightarrow$ **BSSs** $\longrightarrow$ **HB**

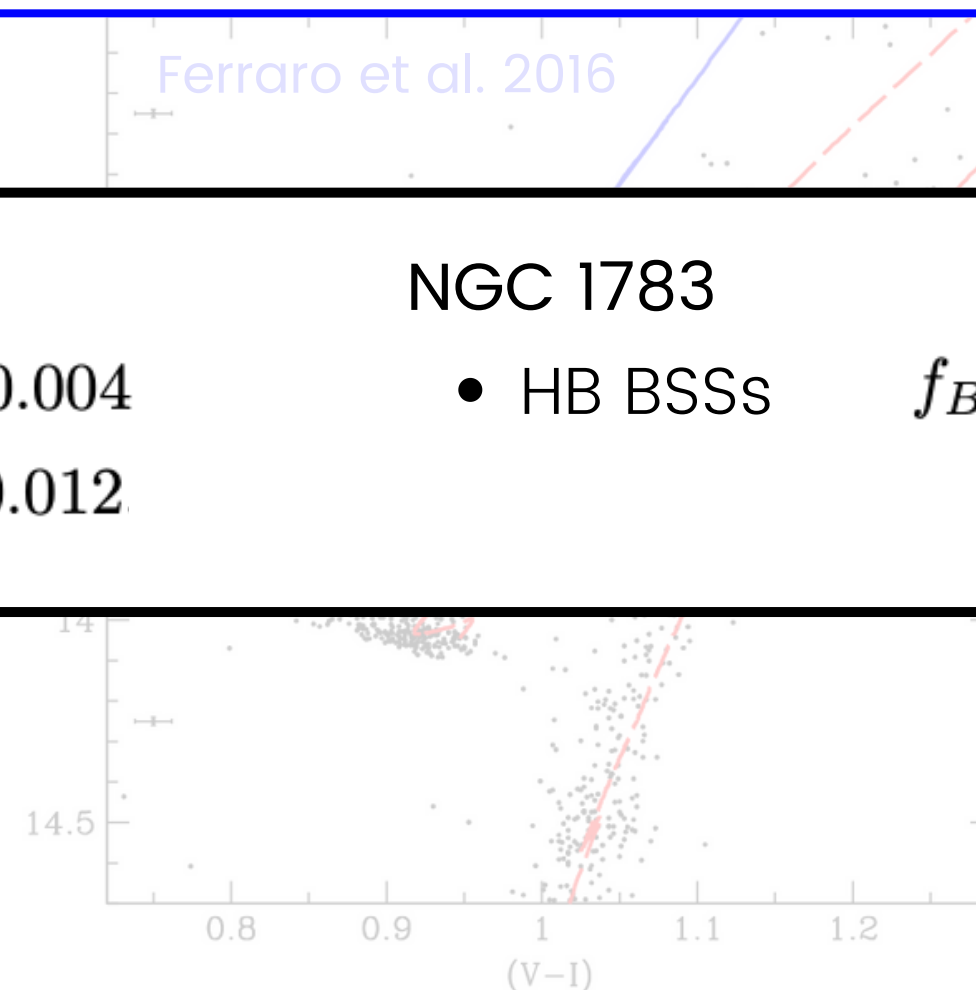
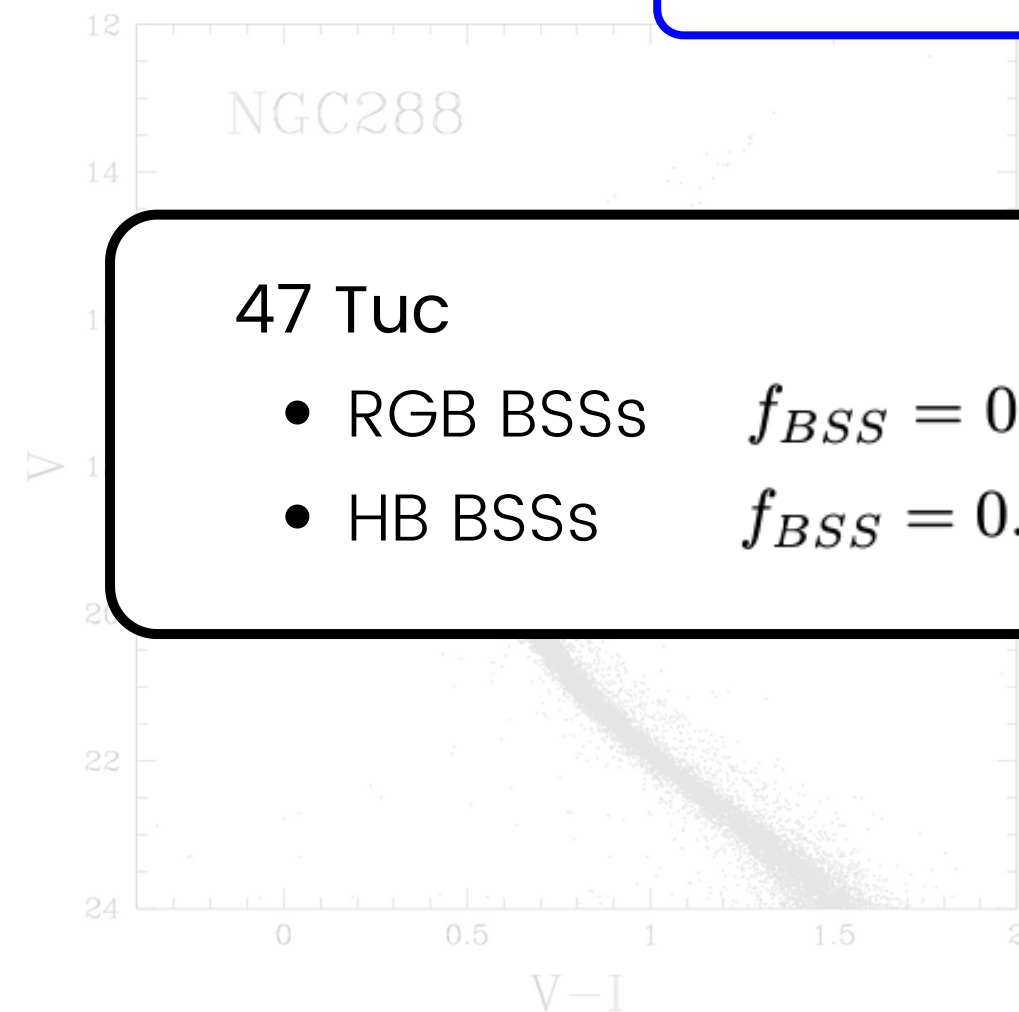
I can infer the BSSs fraction!

47 Tuc

- RGB BSSs $f_{BSS} = 0.018 \pm 0.004$
- HB BSSs $f_{BSS} = 0.032 \pm 0.012$

NGC 1783

- HB BSSs $f_{BSS} = 0.030 \pm 0.022$

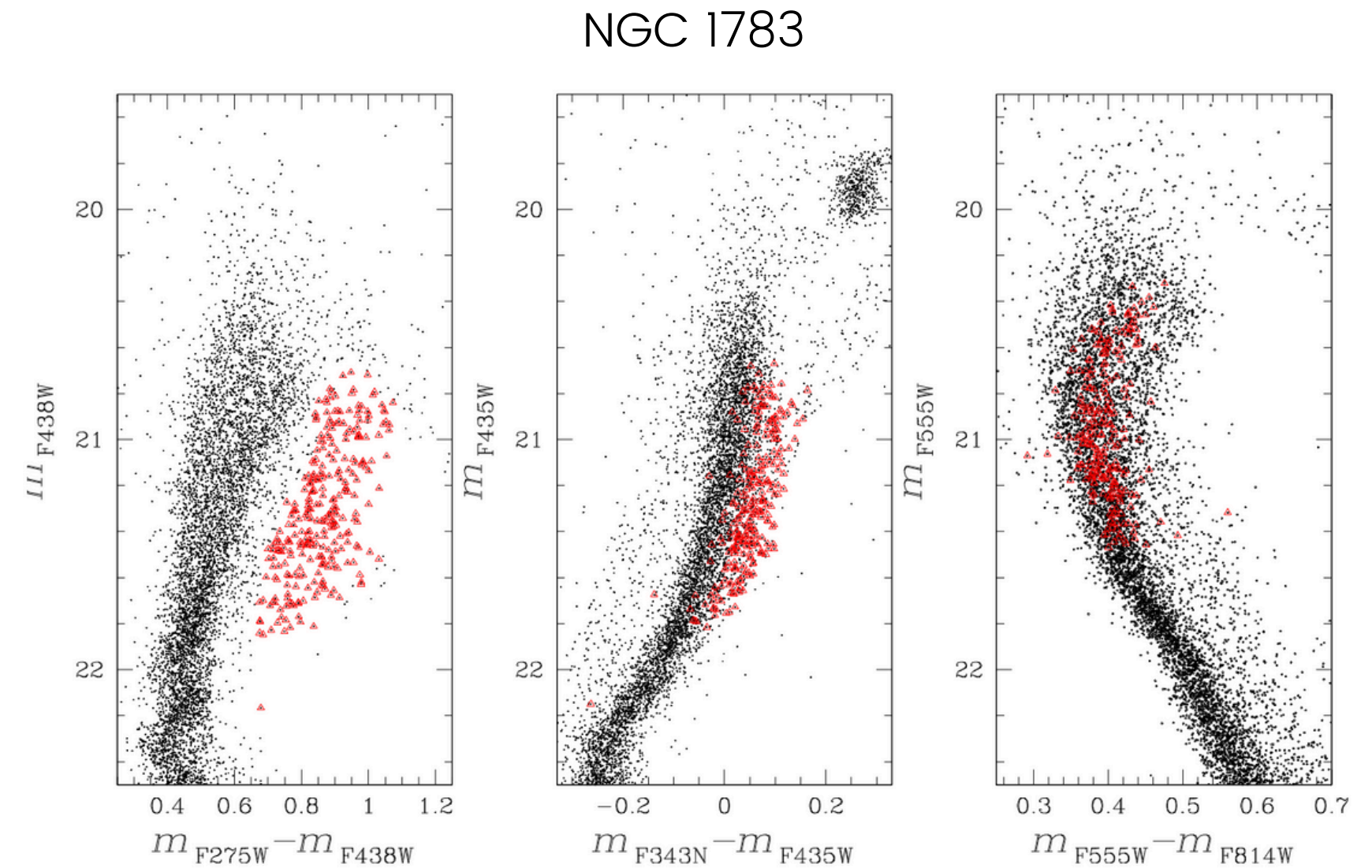


UV-dim stars

A new class of stars

UV-dim stars are stars with unexpectedly low UV flux, first discovered in NGC 1783 in 2022.

From Milone et al. (2023) they appear to be common in young clusters (~ 200 Myr).



Milone et al. 2023

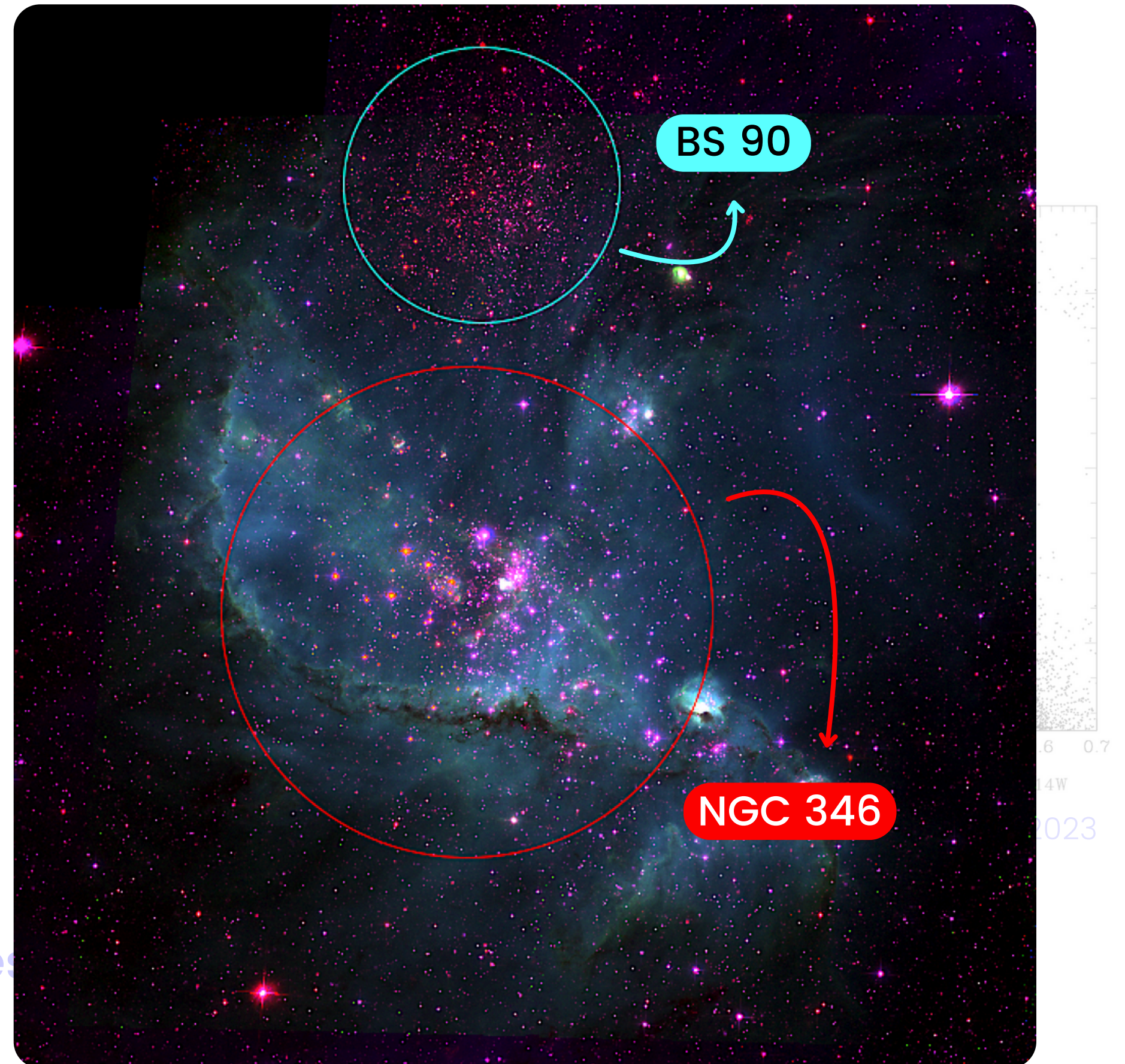
Are they also present in very young clusters?

UV-dim stars

A new class of stars

NGC 346 is an ideal candidate:

- same field as BS 90
- very young (~ 3 Myr)



Are they also pres

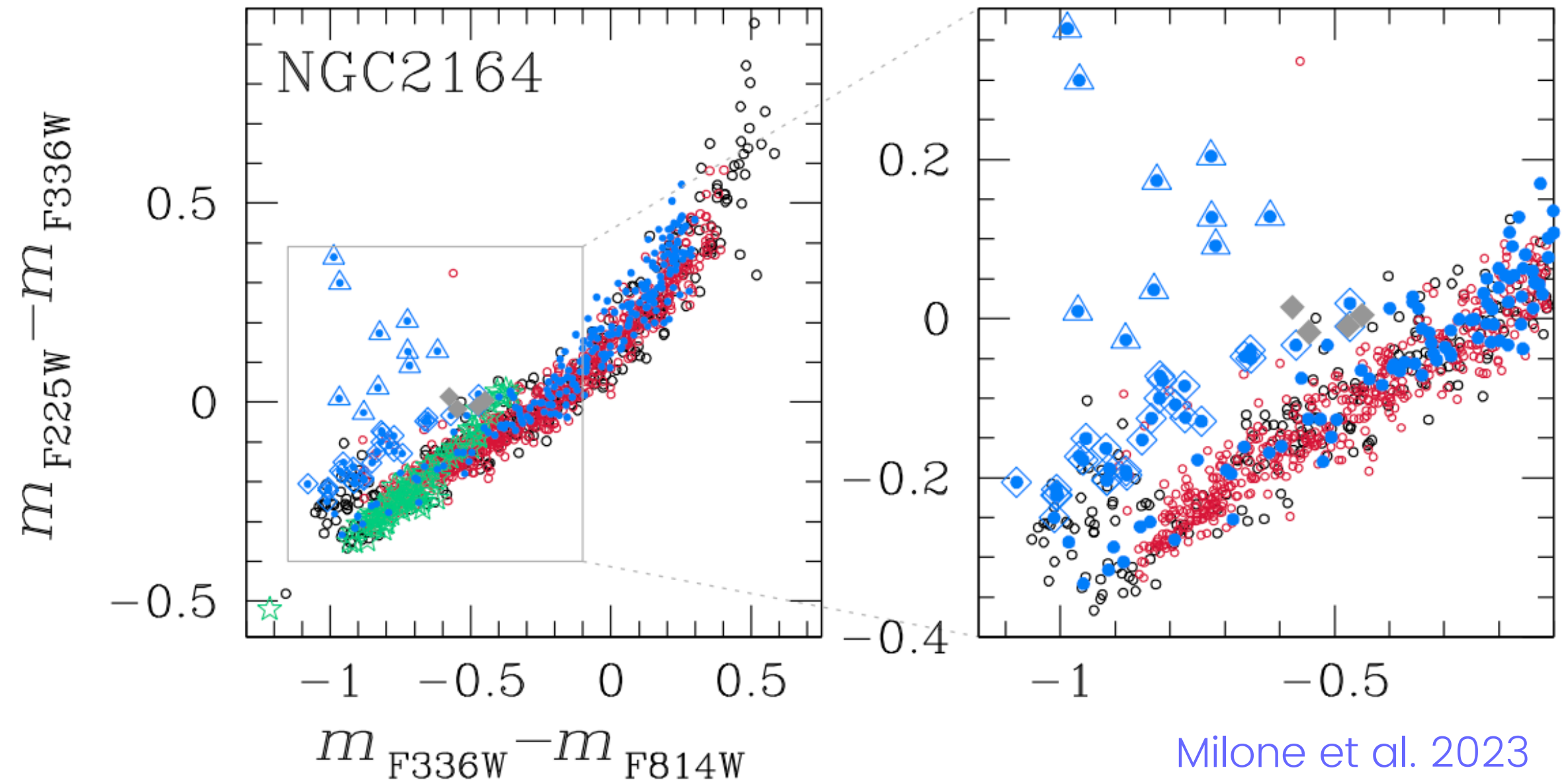
UV-dim stars

How to detect UV-dim stars?

A good diagnostic to detect UV-dim stars is a two-color diagram.

This diagram is not sensitive to binaries, they distribute as single stars.

▲ = UV-dim stars

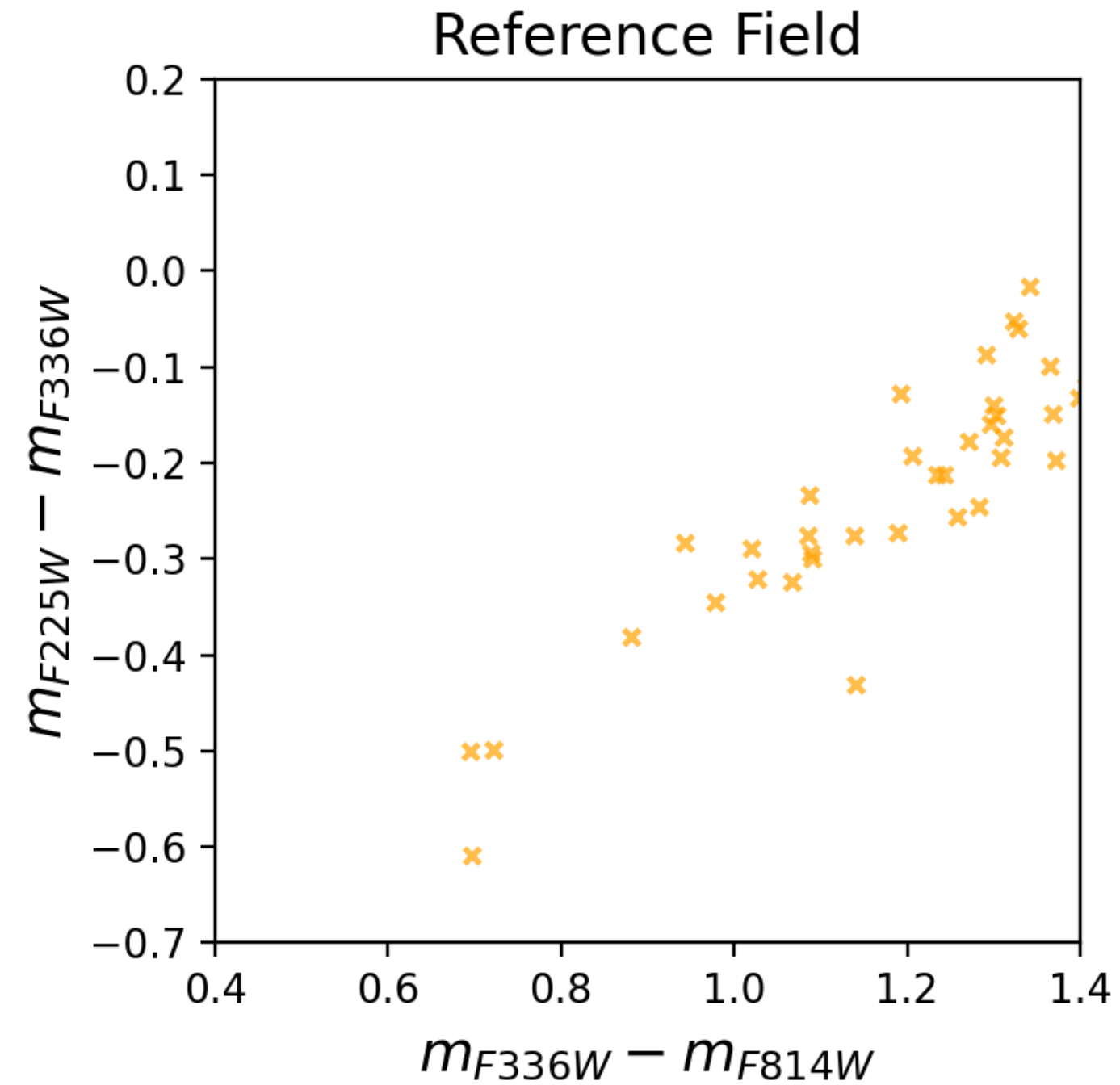
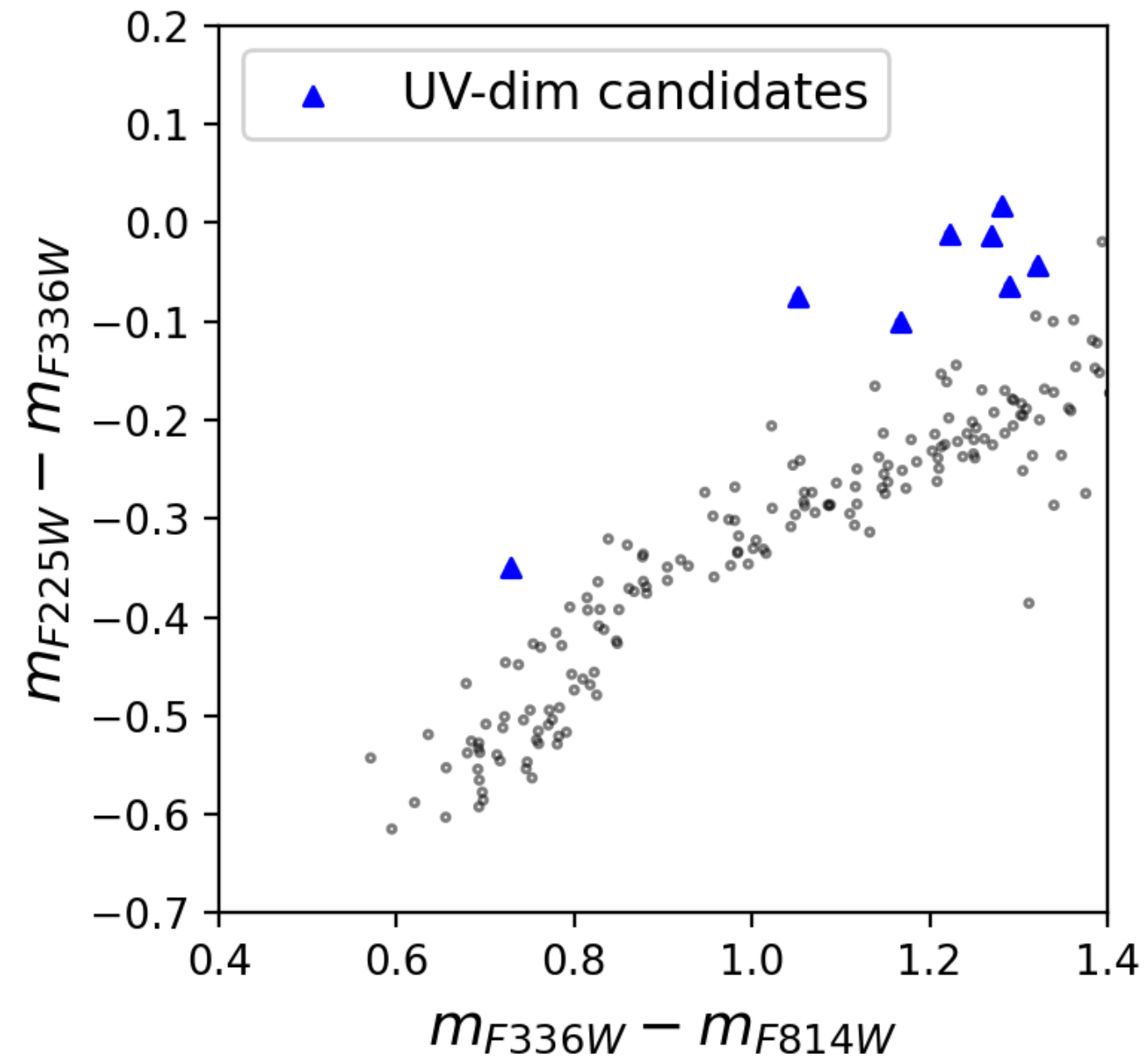
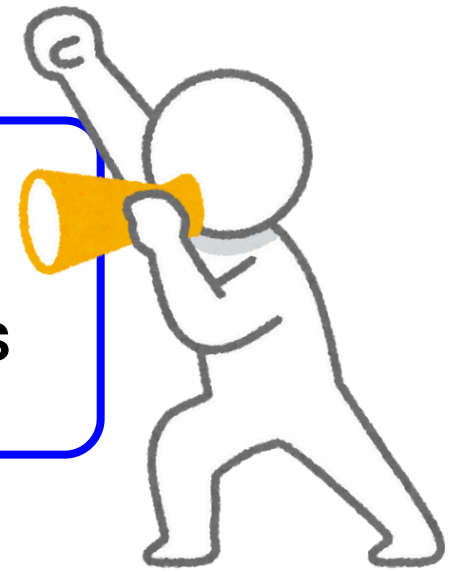


Milone et al. 2023

UV-dim stars

The young cluster NGC 346

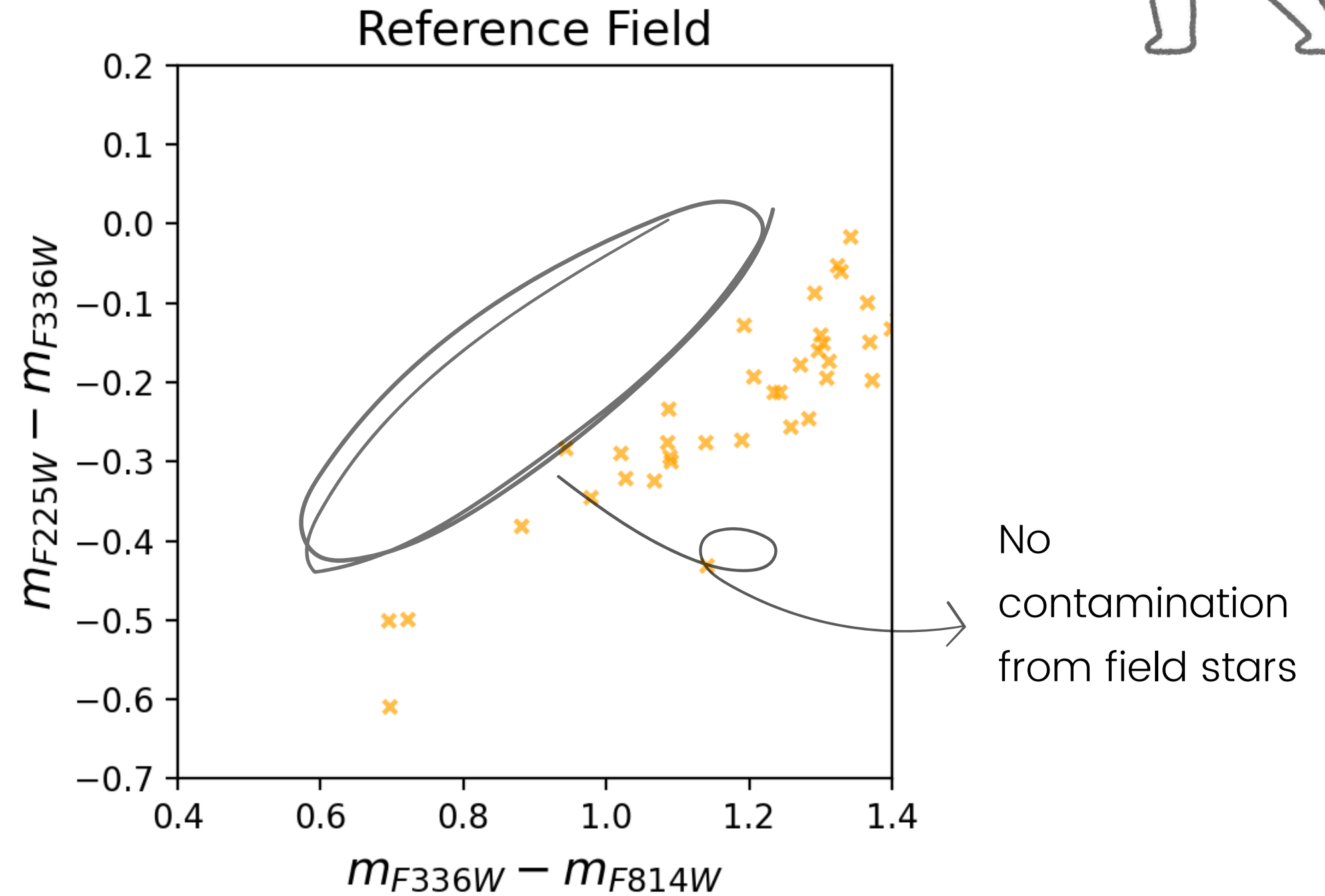
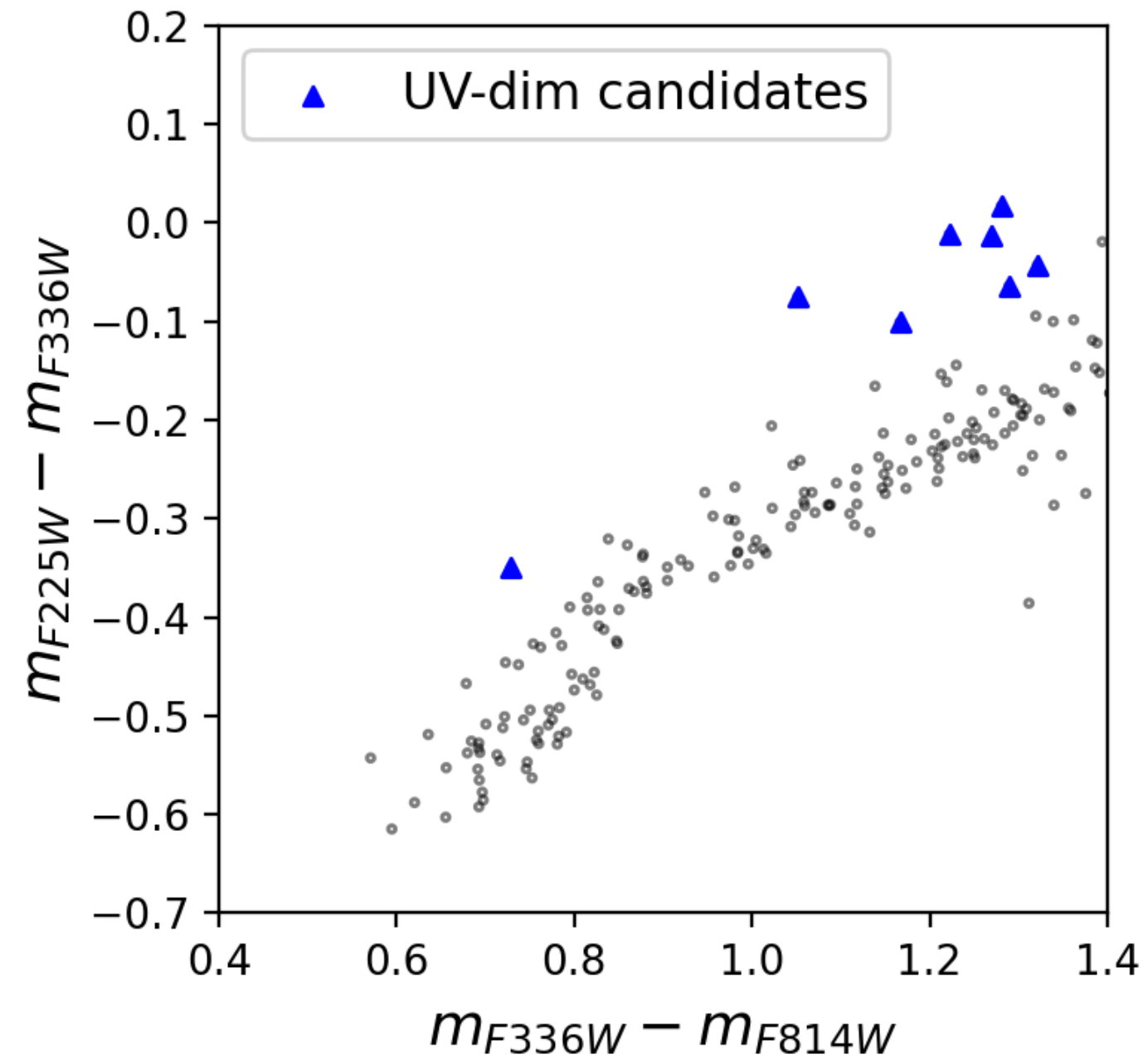
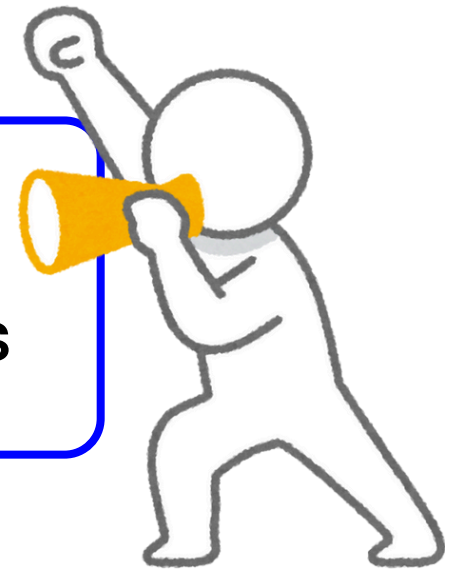
8 candidate
UV-dim stars



UV-dim stars

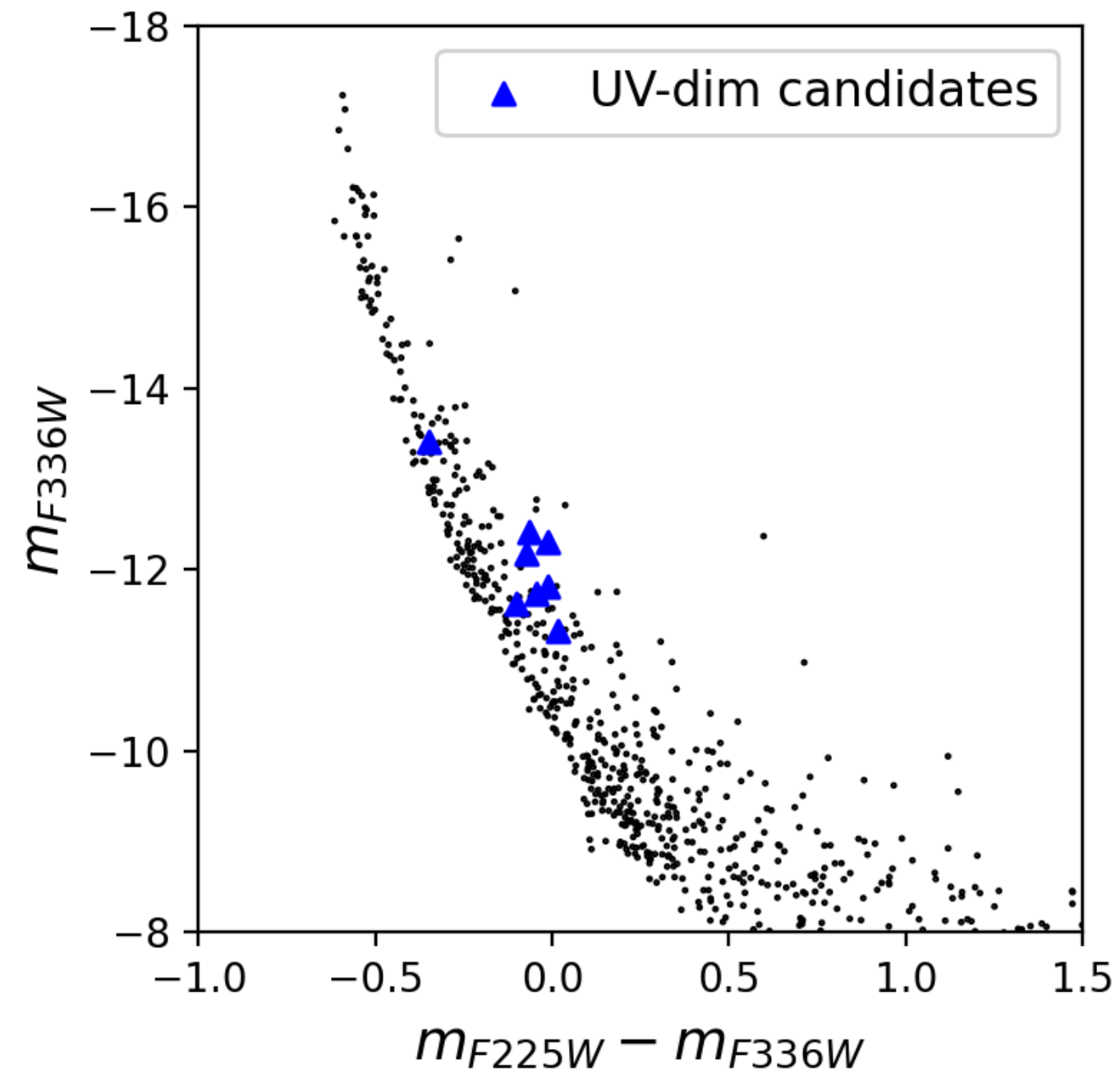
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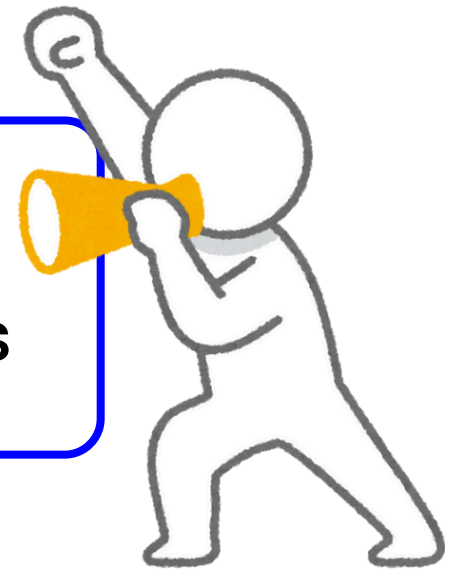


UV-dim stars

The young cluster NGC 346



8 candidate
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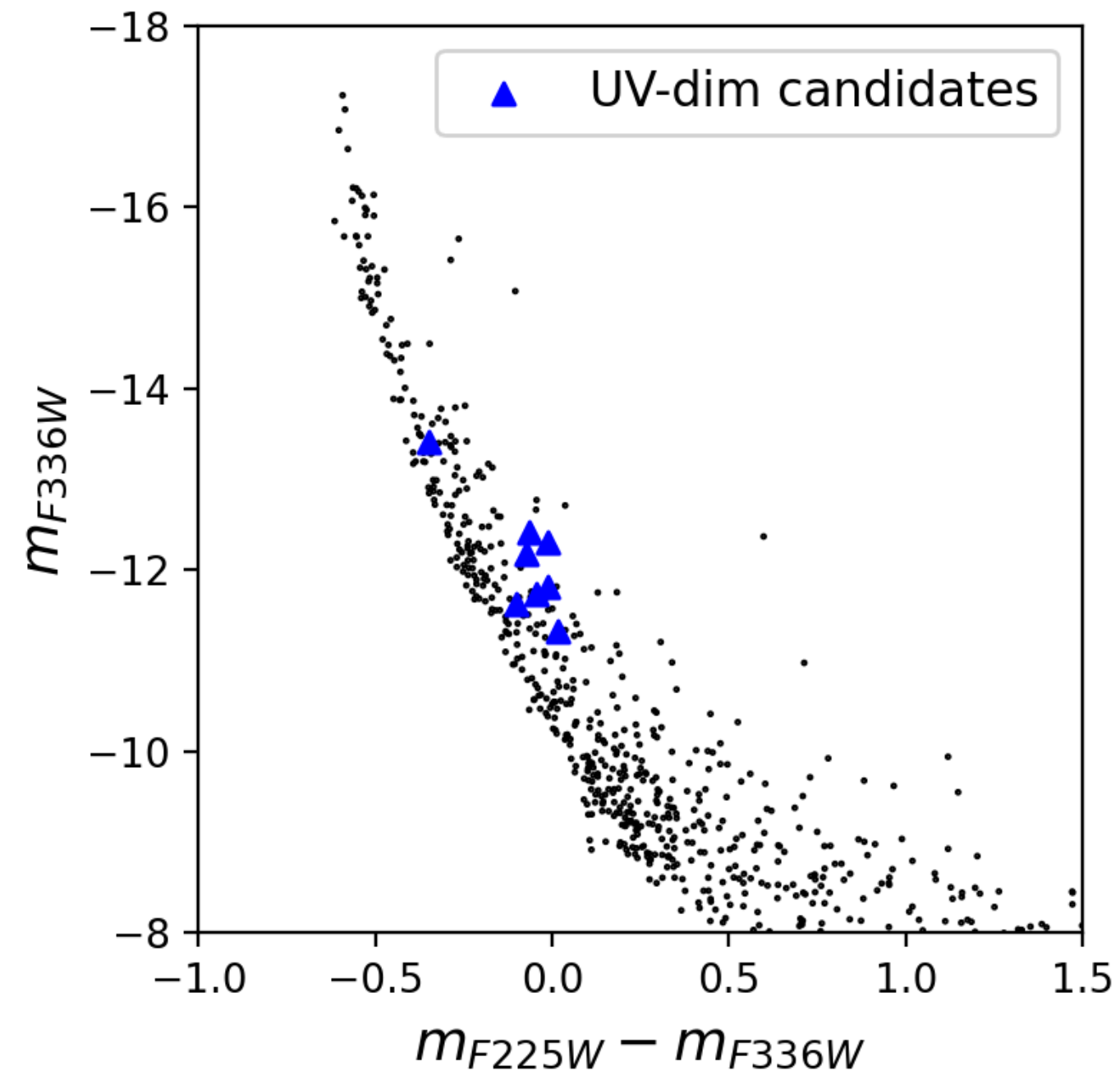


Origin of UV-dim stars:

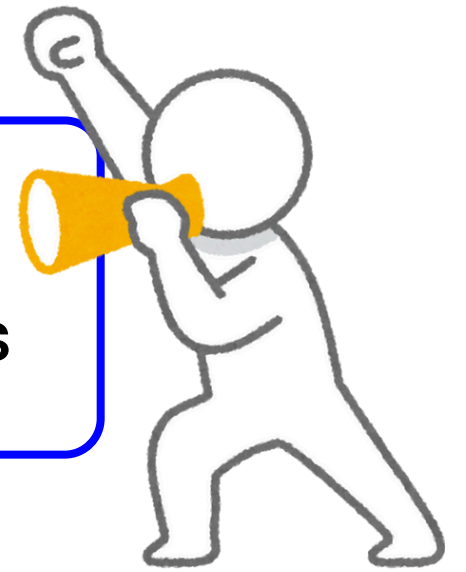
- Rapid rotators that became slow rotators due to a braking mechanism
- Circumstellar dust ring that absorbs part of the UV flux of the star

UV-dim stars

The young cluster NGC 346



8 candidate
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Origin of UV-dim stars:

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Conclusions

Binaries beyond the MS

Studies on binaries can be extended to extragalactic clusters.

Two scenarios that explain the binary fractions in the HB and RGB:

- New insights about the **mass ratio distribution**
- New method to determine the **BSSs fraction**

UV-dim stars

The discovery of UV-dim stars in the very young cluster NGC 346 is a novel result that could challenge existing theories about the nature of these peculiar stars.



Conclusions

Binaries beyond the MS

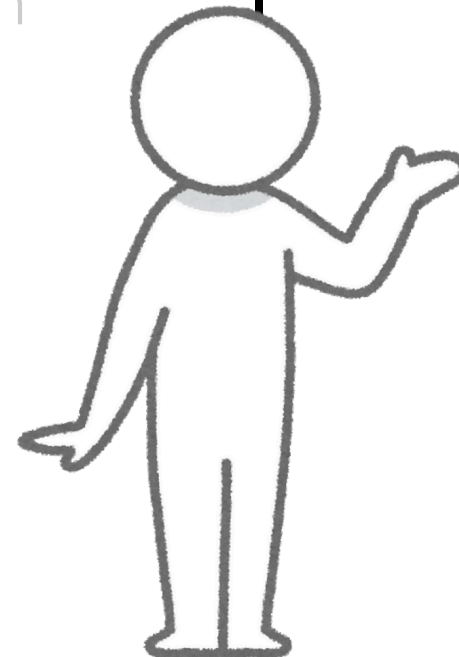
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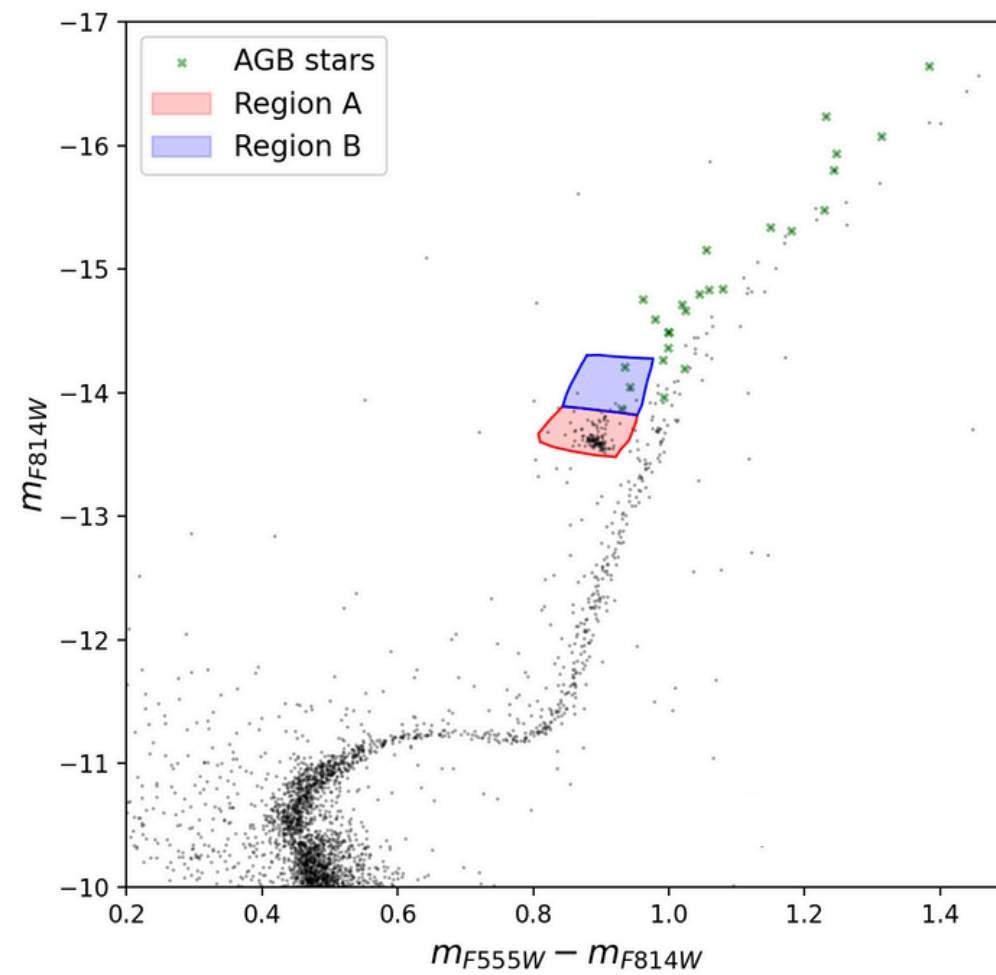
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Any questions?

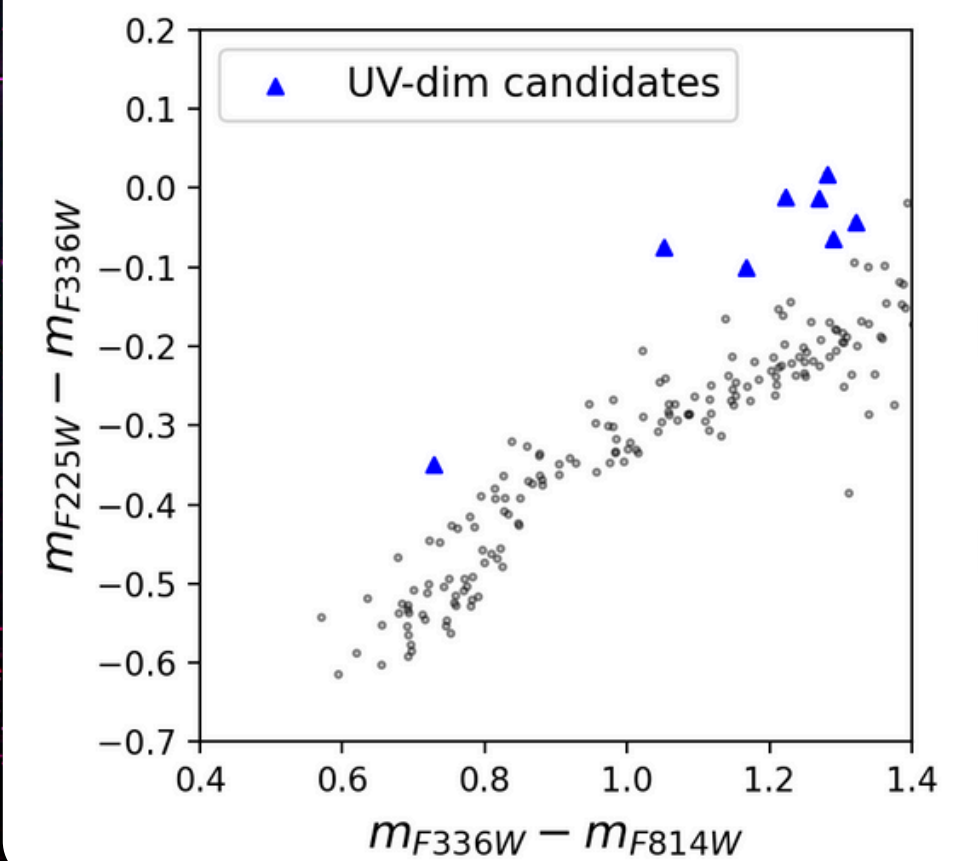
BS 90

Binary Stars
Beyond the MS



NGC 346

The Enigma of UV-
dim stars



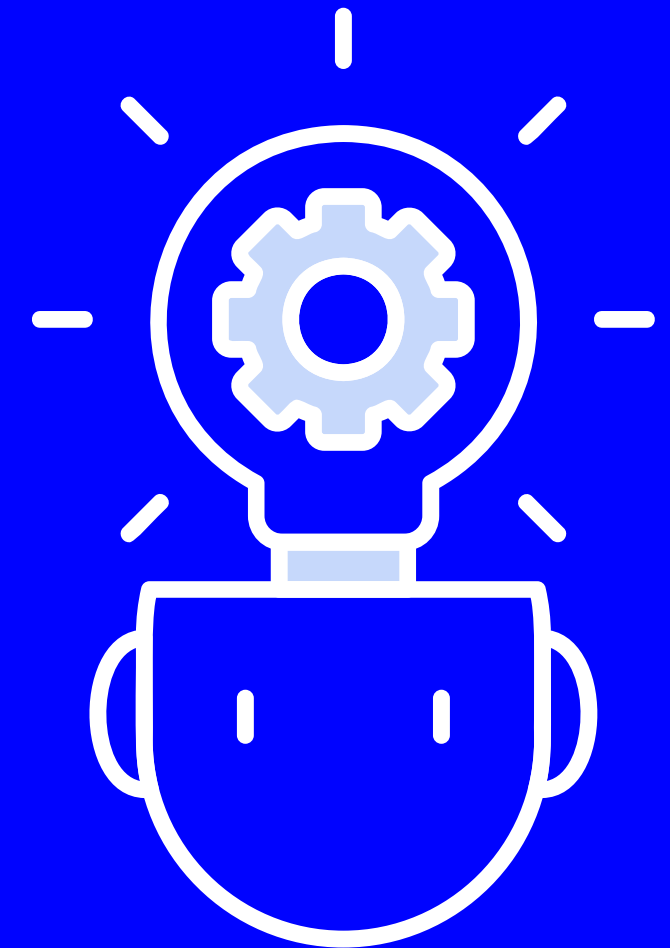


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Thank you!



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Binaries beyond the MS

Where are the binaries?

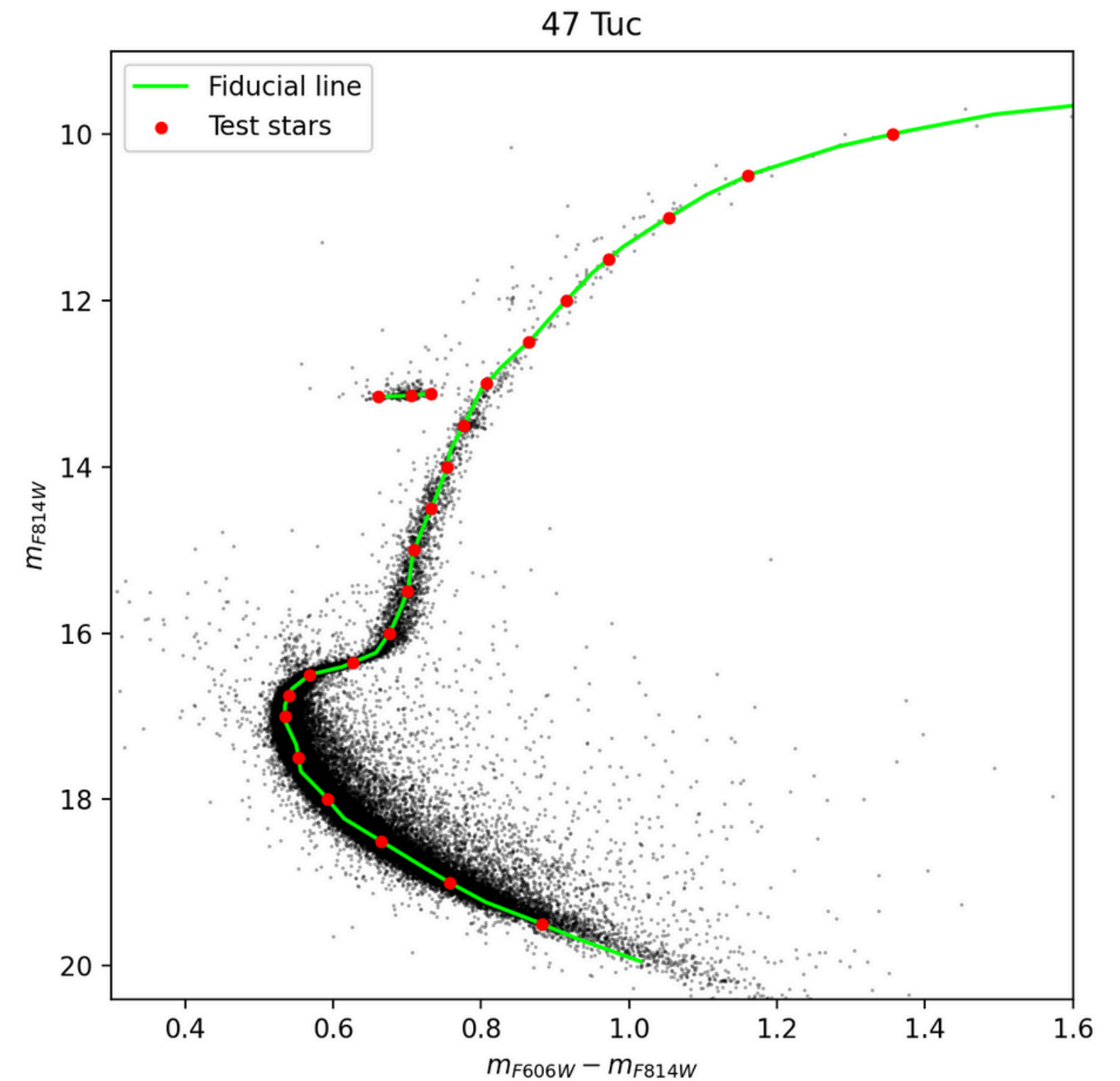
No studies
about binaries
outside the MS



Where are
these systems
located?

Procedure:

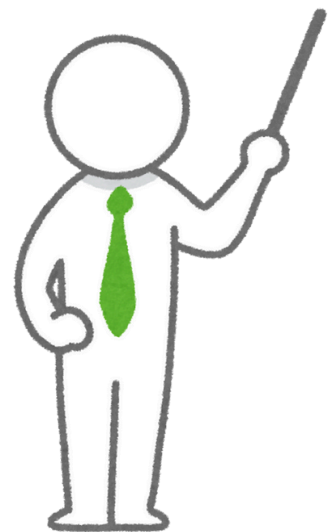
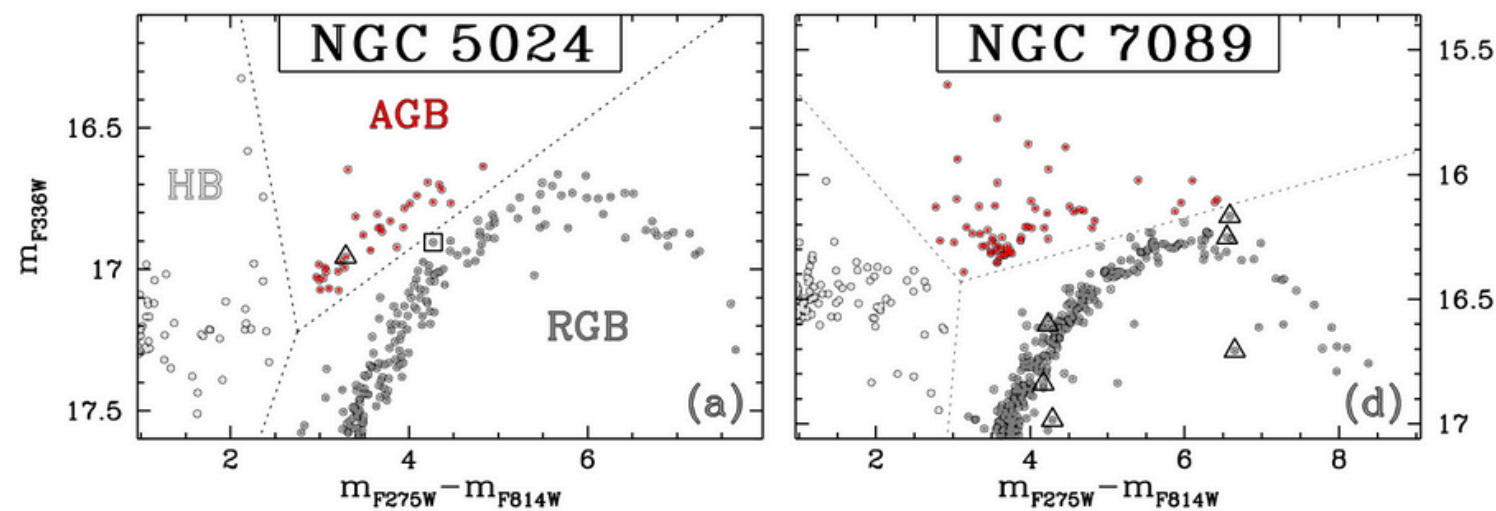
1. Select test stars along the CMD
2. Build synthetic binaries to see where they locate in the CMD



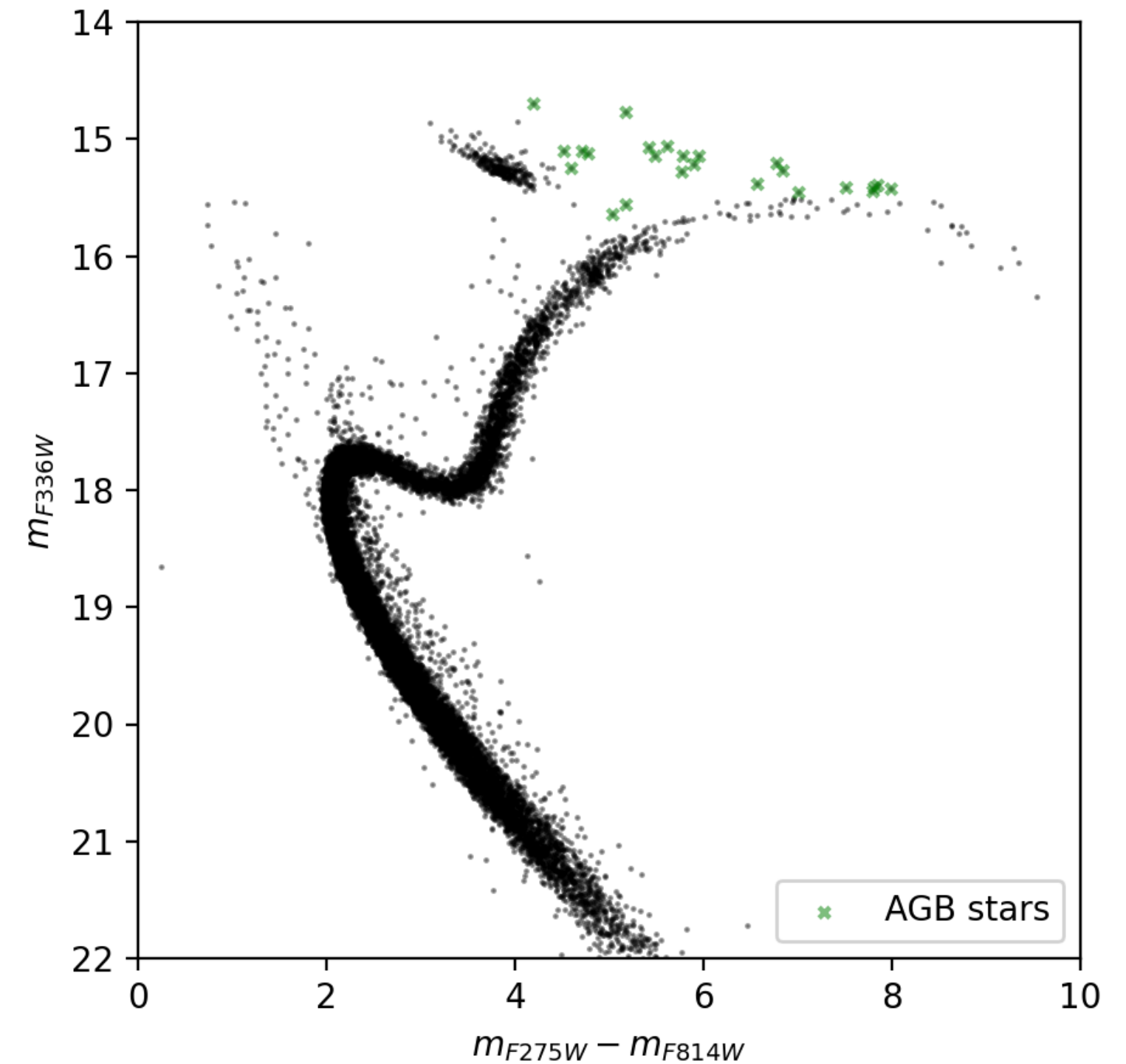
Binaries outside the MS

AGB stars selection

AGB stars were selected in a far-UV-optical CMD following the procedure described in Lagioia et al. (2021)



47 Tucanae - AGB stars



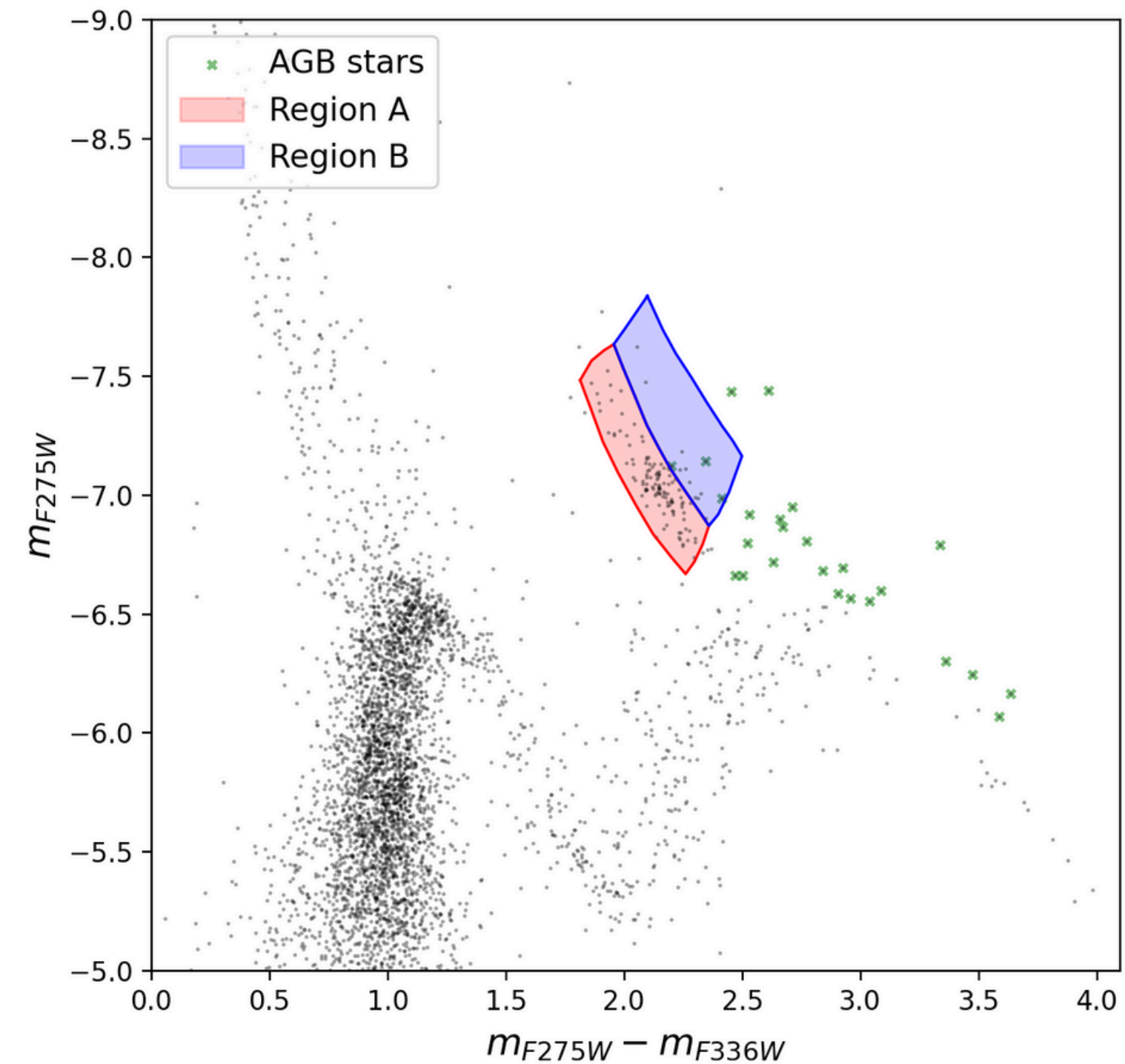
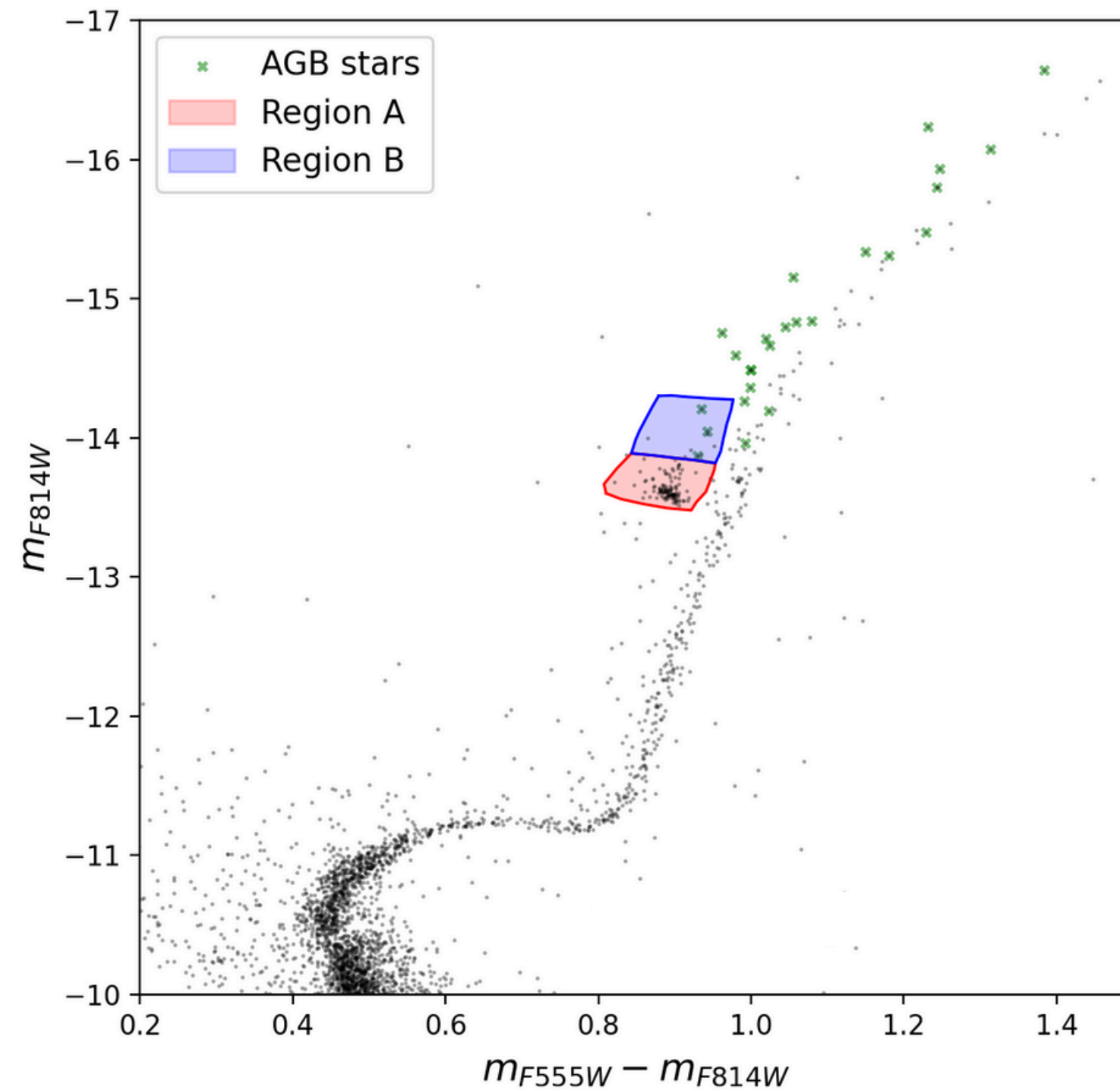
Binaries beyond the MS

Computing the Binary Fraction - HB

$$f_{\text{bin, HB}}^{\text{optical}} = 0.046 \pm 0.024$$

$$f_{\text{bin, HB}}^{\text{UV}} = 0.048 \pm 0.035$$

BS 90



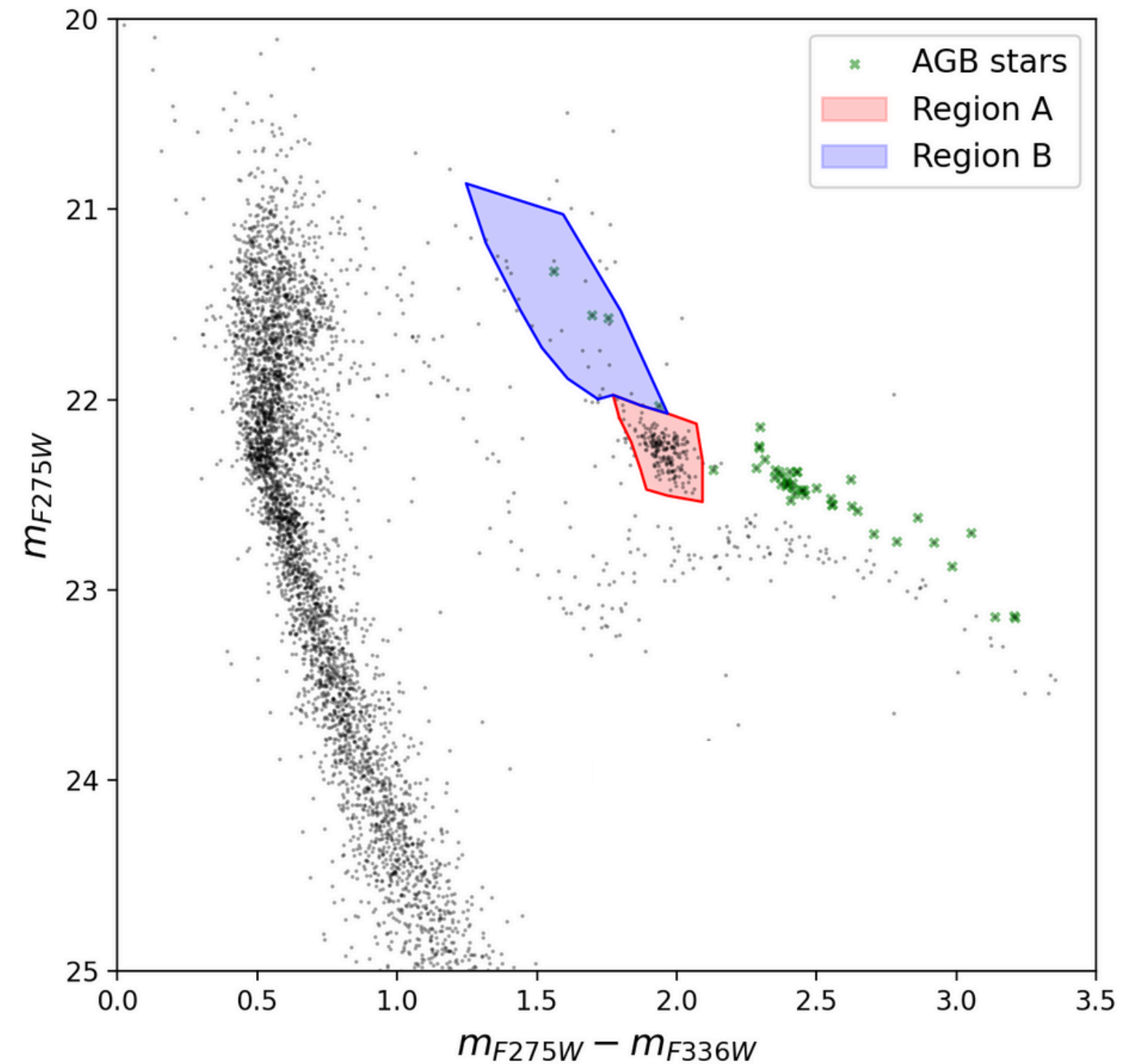
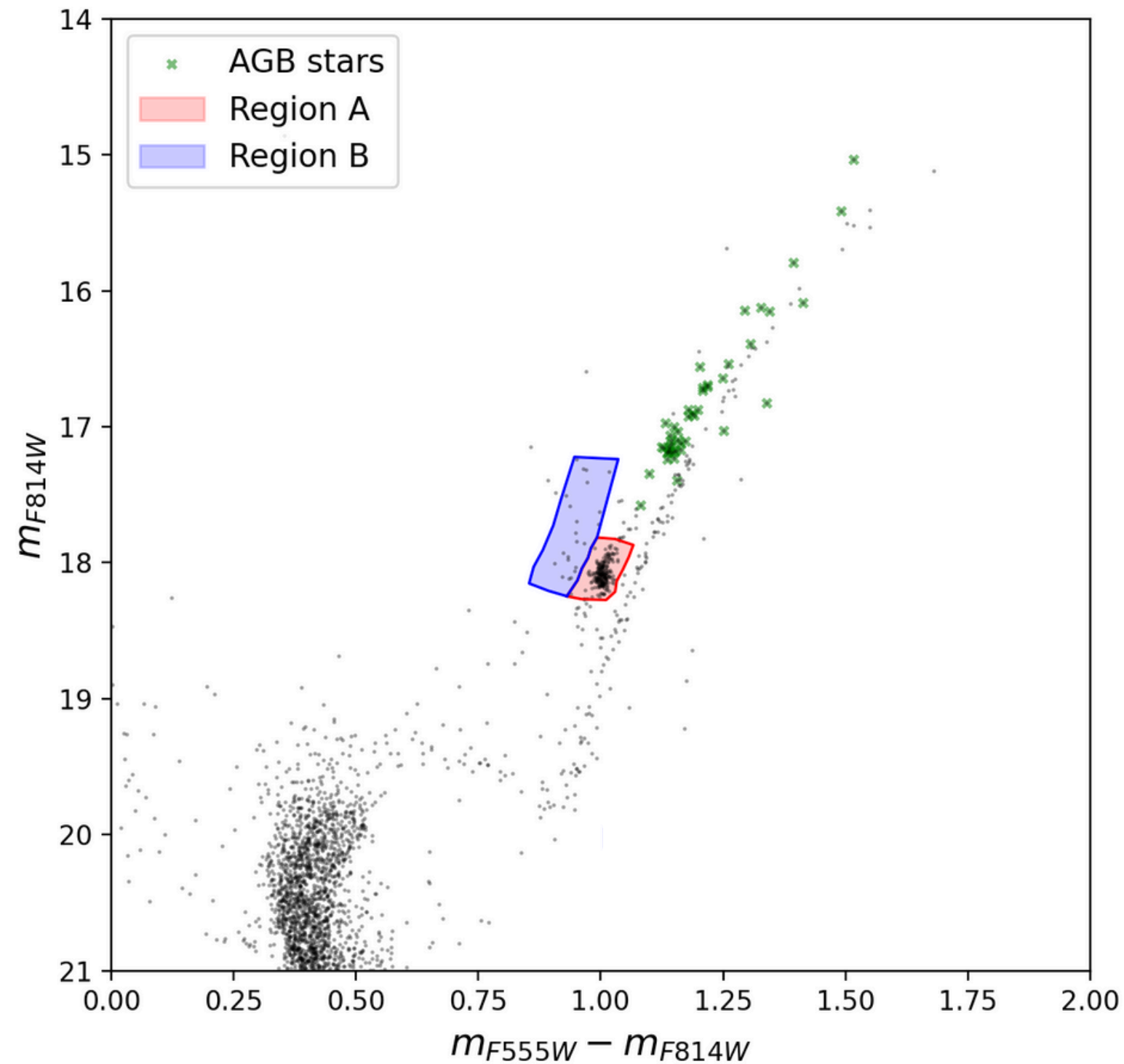
Binaries beyond the MS

Computing the Binary Fraction - HB

$$f_{\text{bin, HB}}^{\text{optical}} = 0.093 \pm 0.021$$

$$f_{\text{bin, HB}}^{\text{UV}} = 0.094 \pm 0.020$$

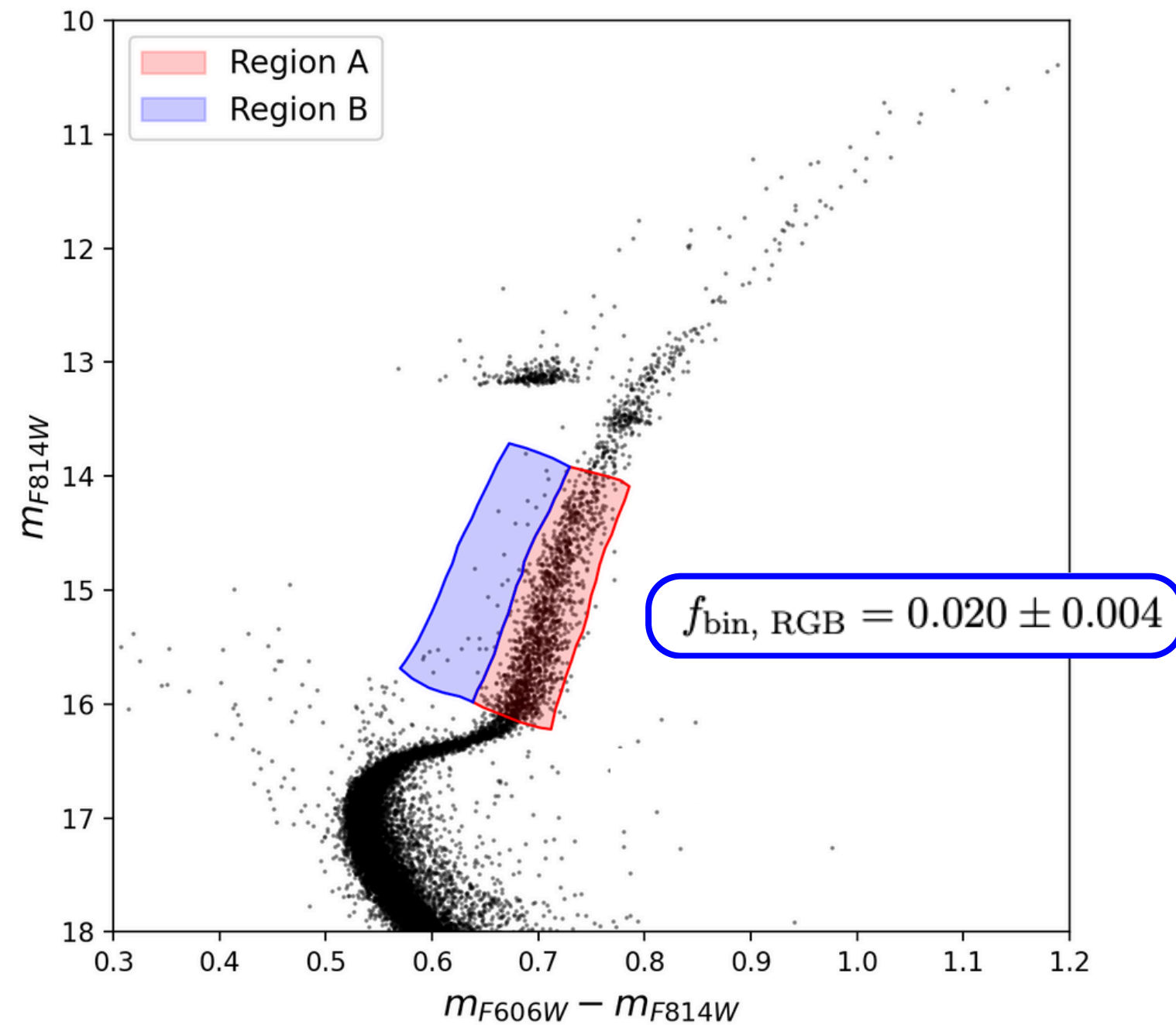
NGC 1783



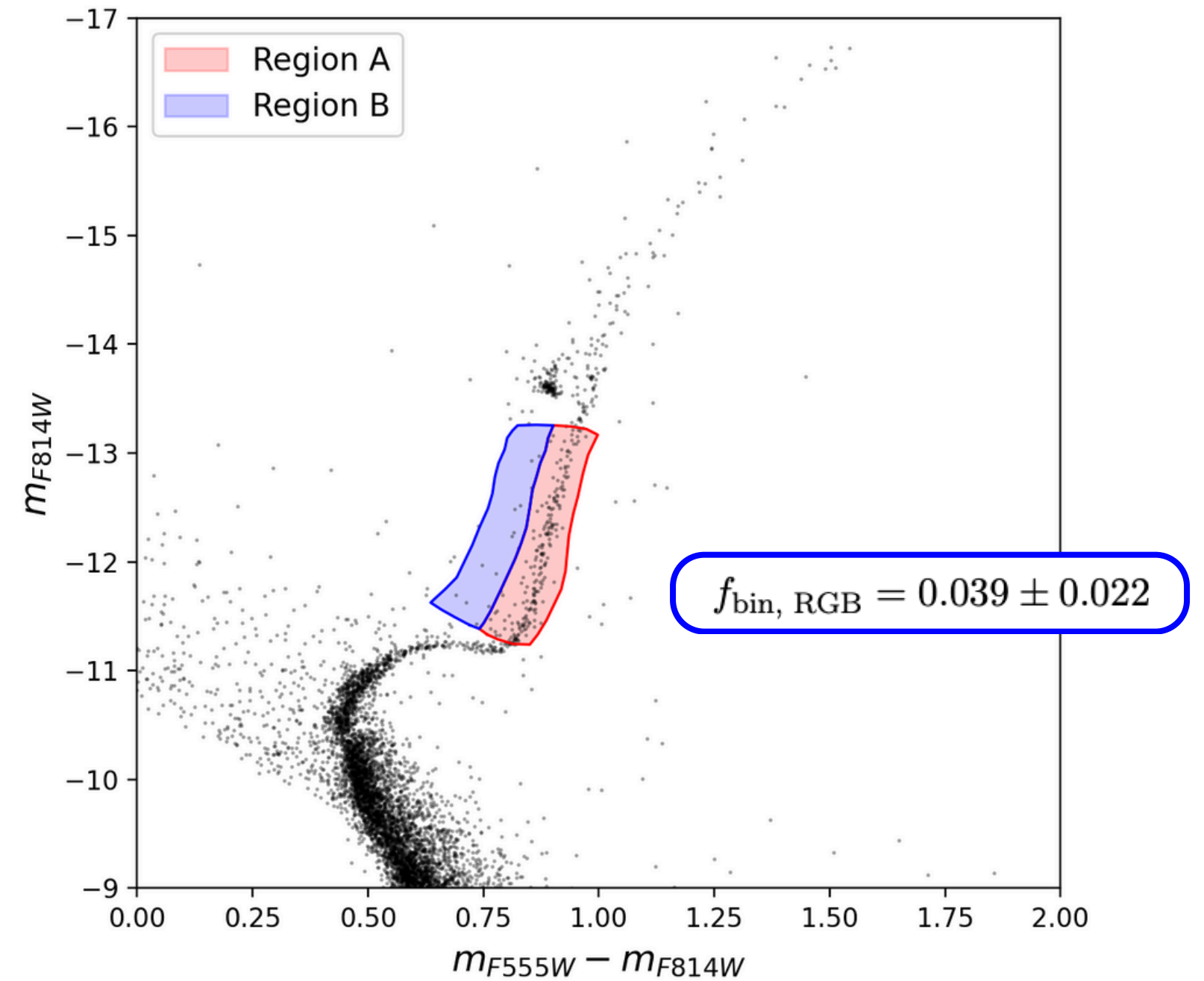
Binaries beyond the MS

Computing the Binary Fraction – RGB

47 Tucanae



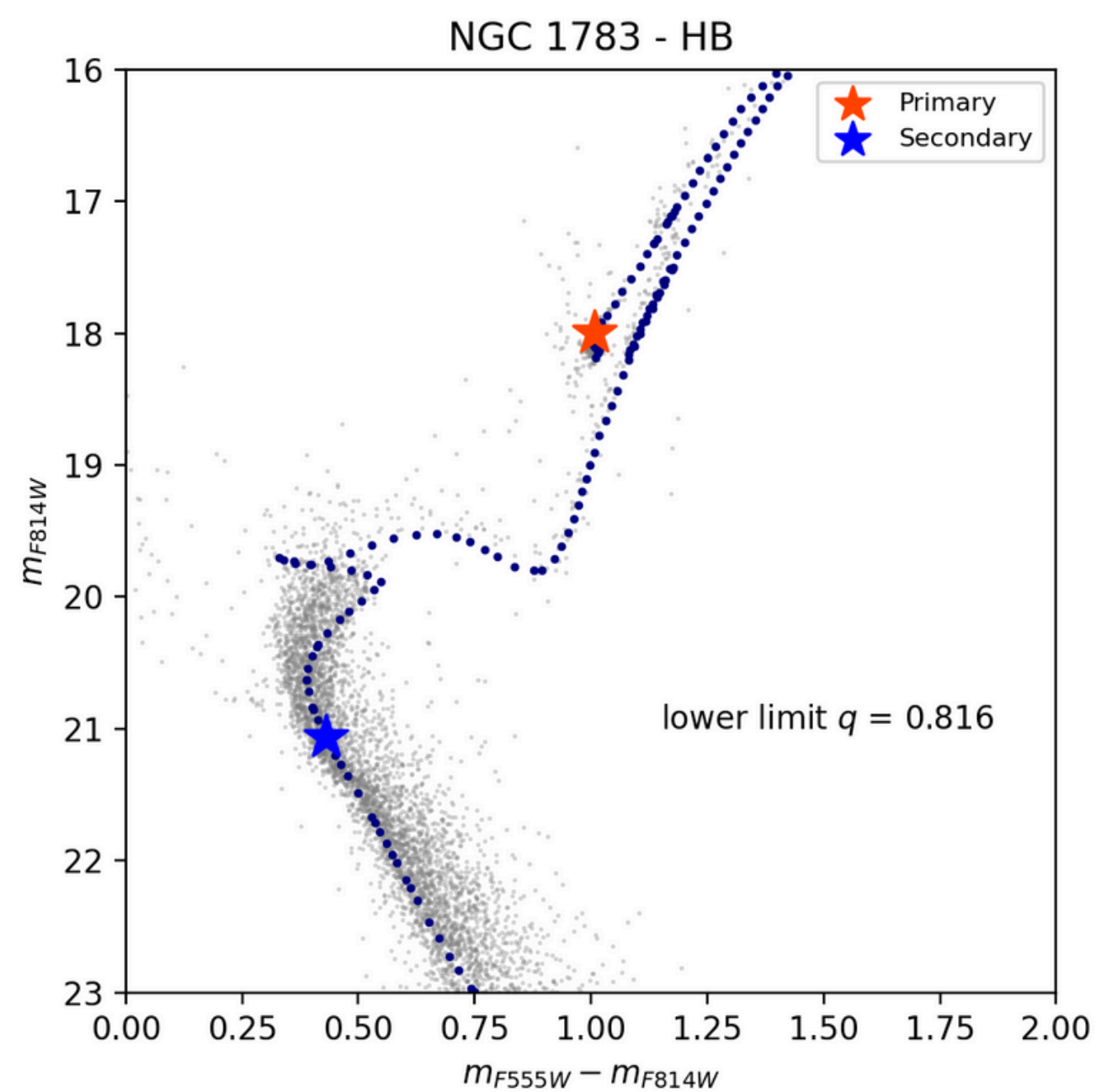
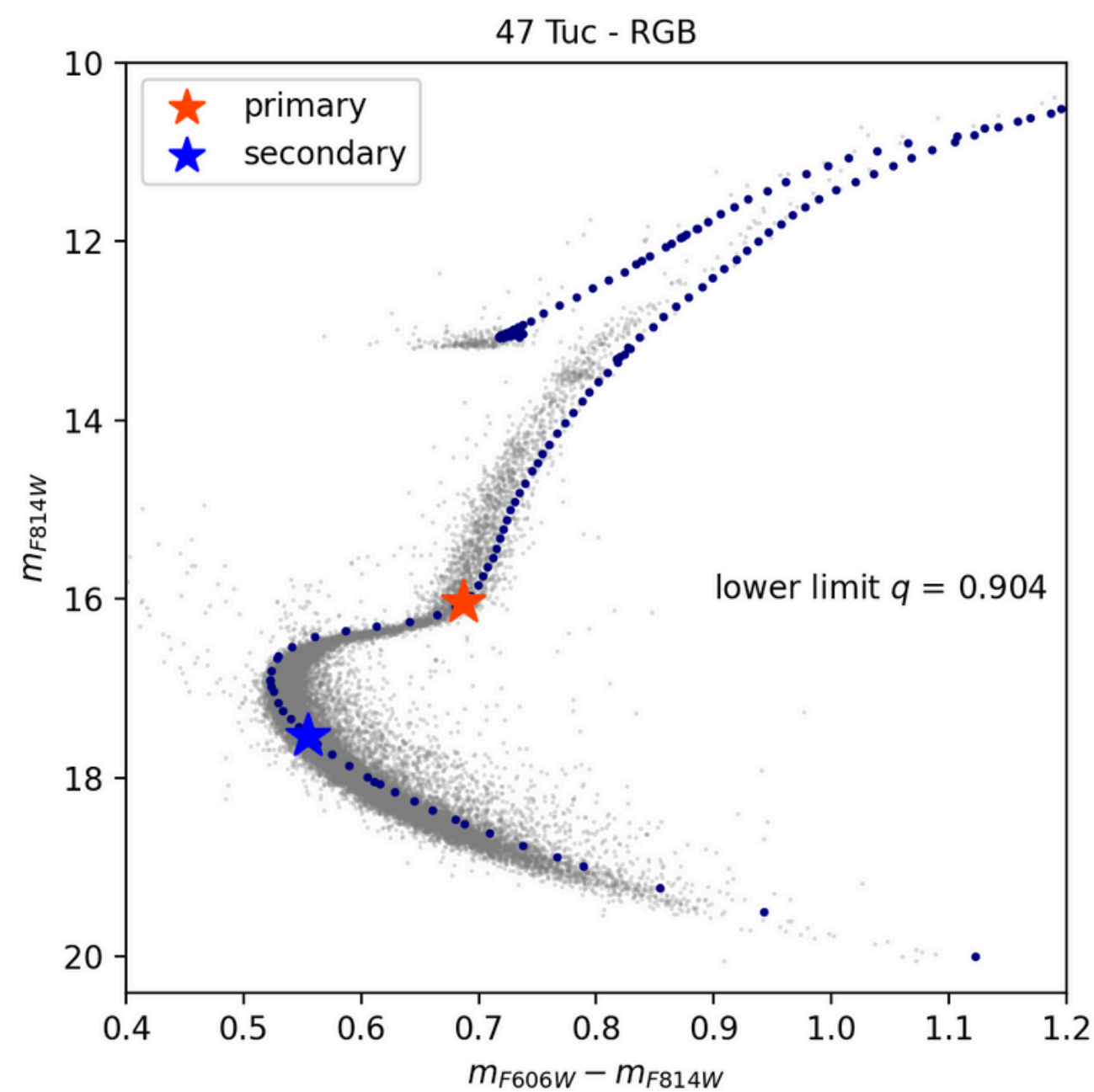
BS 90



Binaries outside the MS

Computing the Mass Ratio

	Age [Gyr]	$(m - M)_0$ [mag]	$E(B - V)$ [mag]	$[M/H]$ [dex]
47 Tuc	11.8	13.21	0.02	-0.45
BS 90	4.2	18.91	0.03	-0.7
NGC 1793	1.95	18.51	0.03	-0.4

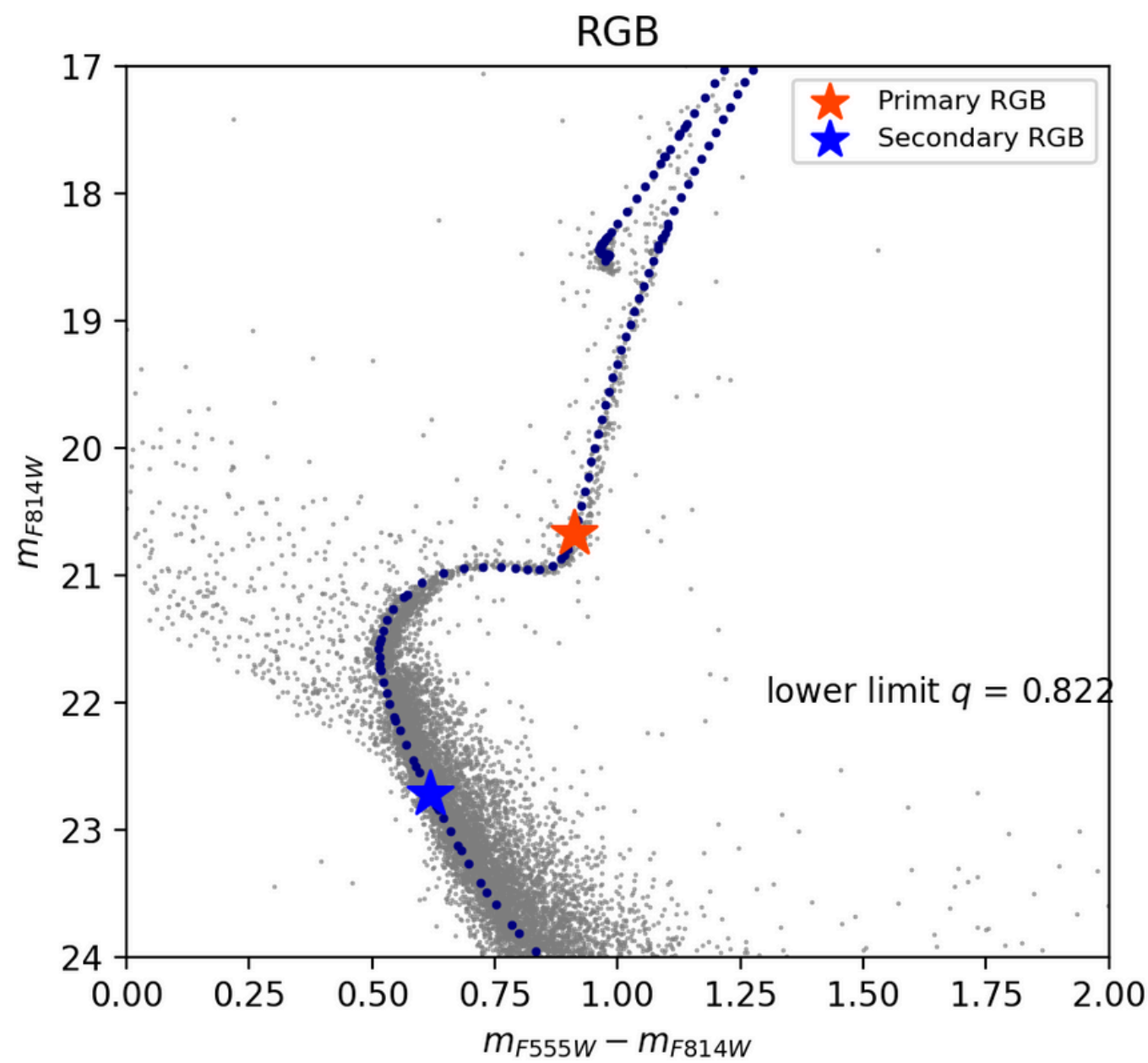
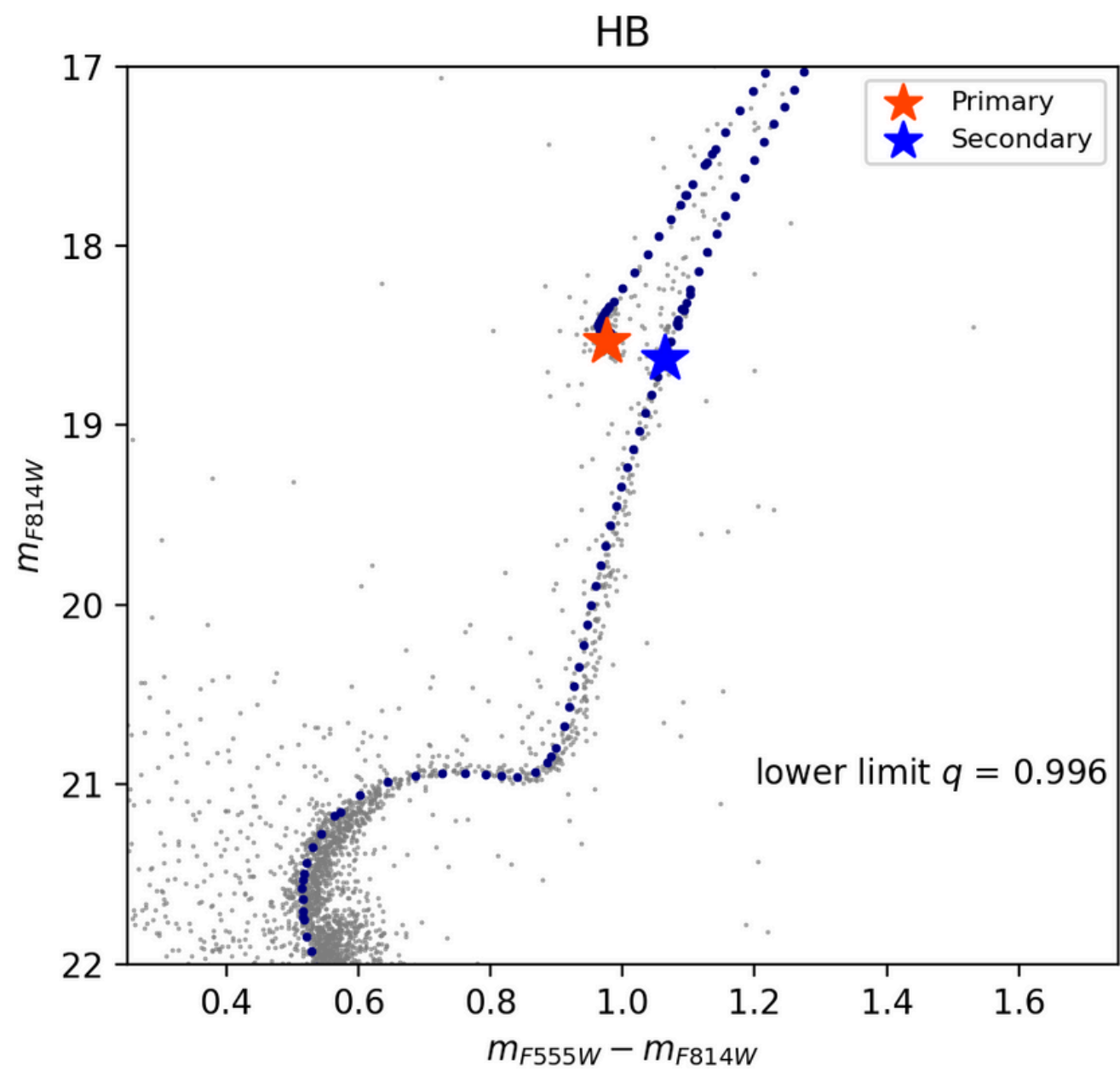


Binaries outside the MS

Computing the Mass Ratio

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BS 90	4.2	18.91	0.03	-0.7
NGC 1793	1.95	18.51	0.03	-0.4

BS 90



UV-dim stars

UV-dim selection

